An Executable Observer for Observer Patch Holography: Specification, Executed Components and Ordered Joins

Bernhard Mueller[1,2,*] Mario Poneder Jinwook Kim[3] Daniel Ari Friedman[4,5]

October 28, 2026

Paper release: r2055 **Released:** October 28, 2026

[1]Pragma Research Inc. [2]Information Physics Institute [3]Oraclizer Labs [4]Active Inference Institute [5]FractiAI
[*]Corresponding author: bernhard@floatingpragma.ai

Abstract

Observer Patch Holography (OPH) is made testable by its executable-model hypothesis: a fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose records have the observed quantum-field and gravitational descriptions. This paper is the base of that hypothesis. It specifies the twelve-port icosahedral carrier, the native C/E seam repair with its exactly classified family of quantum record laws, the finite observer algebra, the declared services and the scheduler, and reports the finite components executed on that specification. Uniform seam attempts have exact quantum gap $(5-\sqrt{5})/60$; a record-steered observer moves a final occupation from 1/6 to 35/264; nine source-executed measurements identify every coordinate of the record-law family. Overlapping observers read fields and event clocks from one continuing history, and one register joins quantum electric flux, charged fermions and a scalar with a feedback response certified nonzero at infinite occupation. On the refining spherical source one repair history defines a continuum operator of spectral dimension two governing Gaussian density histories read by native incident clocks; a declared radial extension gives a smooth $1+3$ free-scalar description of every fixed finite observer menu. Five conjunctive obligations form the open joins: a physical clock and energy normalization, an interacting chiral 3+1 field theory, native defects with identity and spectrum, realized Einstein premises and joint validation. The paper orders these joins with acceptance conditions and stopping tests and records four negative controls: the native double-load defect decays with survival $(5/6)^n$, the static radial product fails scalar null balance, passive repair strictly attenuates coherence and has no wave sector, and the tested mass laws miss the charged hierarchy.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Supplied coherence, protected codes, radial directions, fields, clocks or couplings are premises until that process realizes them. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1 + 3$ scalar description. A physical clock and energy normalization and an interacting chiral $3 + 1$ field description from native records are open joins.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. No native defect sector, maintenance-energy law or measured spectrum is established; the native double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join

needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[1]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a derived clock calibrates it. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

Finite C/E campaign. A versioned finite family tests both complete C and E instruments, their tagged classical mixtures, positive-probability identity holds and bounded classical memory with a positive uniform seam-exploration floor. On the existing 78-state carrier, the continuing observer retains a finite record buffer and explicitly exports displaced records or halts when full. Exact six-attempt executions preserve complete instrument probabilities and untouched-reference marginals. The transition kernel specifies every finite record history; the exponentially growing history tree is not materialized. Growing exports remain a resource, and preparation, randomization, access, storage and delivery services remain supplied.

The adaptive observer also has a constructed stationary reference. Its source/current-sign quotient has a unique recurrent class of 129 of 132 states. Lifting that exact law through the actual complete record instrument gives 128 positive record blocks for C and 192 for E, with 4,218 positive source/record diagonal entries in either case and exact stationary residual zero. This is the finite source/buffer marginal at repair probability one, not the growing history; faithfulness requires its stated support restriction. The delayed member needs its own reference construction.

Two exact obstructions sharpen the next joins. Each law's complete active branch $*$-algebra is the full matrix algebra on both the 78-state source and its 66-state binary shell. Thus a source-only Hermitian charge commuting with every branch is scalar; this does not exclude forward-invariant subspaces, general expectation-conserved observables, history-dependent charges or controlled lifetimes. Separately, the frozen real-operation grammar cannot implement a nontrivial modular action of the constructed nontracial reference on its fixed full support algebra: a specified input and dimensionless modular parameter give normalized trace-distance error at least $1/2$. The separating observable is a mathematical witness, not an admitted native detector. This is not an exclusion of every clock or encoding.

These results extend the component foundations and leave the integrated joins open. They sharpen four of them: a versioned extension and refinement class beyond the finite contract; a realization of the observer's declared services; accessible noncommuting algebras with implemented clock operations; and an explicitly specified defect sector or controlled lifetime outside the excluded commuting-charge class. The campaign adds no independent physical comparison and neither freezes nor excludes the full executable hypothesis.

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/COMPLETION_PROGRAM.md

1 This paper's hypothesis and place in the tower

This paper is the executed part of layers two to four of the tower: the carrier and its repair law, the executable observer, and the fields and clocks read from one process. Its hypothesis is the executable-model hypothesis stated above, taken at the level a finite execution can test. The carrier, the native C/E repair law and its memory, the finite observer algebra, the declared services and the scheduler are fixed before testing, and the observers, records, local fields and event clocks of every result are read from a process run under that one specification. Fields, particles and geometry enter as descriptions of what observers record and jointly interpret; no fundamental field or spacetime is reconstructed first [43, 49, 4].

The paper proves finite theorems and certifies finite computations on the declared objects: the seam rule from four requirements, the classification of admissible quantum record laws, the exact gap, memory and recovery of the native channels, record capacity, finite observers and field detectors with every history retained, the common interacting source and its occupation limits, overlapping-observer correspondence, local event clocks and matched screens. For one repair history on the refining spherical source it proves source limit theorems: a common continuum operator of spectral dimension two, Gaussian density paths, incident-clock parametrizations and their fluctuations. Conditional analytic theorems hold on supplied objects: a declared radial product with a chosen wave action, and the supplied interacting actions with their occupation limits.

None of these components is a physical clock, field theory, particle or gravitational response. The five conjunctive obligations of the executable hypothesis are open joins of this construction [31]: observer algebras with faithful references, executed modular clocks and a physical energy normalization; a nontrivial interacting chiral 3+1 field theory with controlled refinement and detector maps; native defects identified through charge, spin and statistics, propagation, interactions and energy, with the measured spectrum; the Einstein reconstruction premises realized on the same matter, state, modular flow, clocks and refining geometry; and joint validation of complete observation and intervention laws against independent measurements. Table 1 separates what this paper executes from what each obligation requires beyond it, and Section 11 orders the joins with their acceptance conditions and stopping tests.

A replayable failure of the specified law, state, access or readout contract rejects that candidate and its tested interpretation. A proof covering every realization of a frozen admissible class stops that class, and it rejects OPH only if the excluded choice is proved necessary. Section 11 records four such failures of the tested law: the native double-load defect decays, the static radial product cannot balance a scalar excitation, passive repair has no wave sector, and the tested mass laws miss the charged hierarchy.

Table 1: The five obligations of the executable hypothesis [31], the components this paper executes toward each, and the open join.

Obligation	**Executed here**	**Open join**
Clocks and energy normalization	Finite modular flow of actual observation states and the fixed-reference relation $t = -\beta s$ (Section 6.6); local, joint and active event clocks (Sections 3.8, 4.2); native incident clocks with source limits and fluctuation predictions, and faithful encoded-field states sharing one active clock (Section 7).	A reference selected by the process, regional compatibility, calibrated duration and the energy normalization. Passive repair is no nontrivial modular automorphism of an invariant observer algebra, and the uncapped native-return reference tends to the trace.

Table 1 continued

Obligation	Executed here	Open join
Interacting chiral 3+1 field theory	Free-scalar refinement transfer and two-site occupation limits (Section 5); an interacting feedback signal at infinite occupation (Section 6); the native continuum operator with density, clock and admitted-wave limits, a declared radial product with Lorentz-signature density, smoothing theorem and a quartic projection with an occupation limit at fixed band (Section 7).	The fields are free or two-mode; radial product, filters, preparations and inverse temperature are declared; no interacting refining family and no chiral matter. The native decay spectrum excludes a complete linear identification with a compact free scalar.
Native defects, identity and spectrum	A charged realization with branch charge covariance (Section 3.4); a flux-dressed fermion code with operator Gauss constraints (Section 6.1); the collision candidate tested (Section 11.2).	No localized protected native sector, maintenance-energy law or spectrum. The collision decays and the tested mass laws miss the charged hierarchy.
Realized Einstein premises	A rank-three response, native causal cones, the product's area and volume dictionary, one characteristic cone for scalar, spinor and Maxwell symbols, and a necessary spherical focusing equation (Sections 7, 8).	Responsive geometry, conserved stress of the native process, fixed-proper-volume generalized-entropy stationarity and a positive Newton normalization. The static radial product fails null balance.
Joint validation	A prospectively fixed count protocol for the record-law coherence and its sequential design; retrospective Bell and optical-clock replays; a retrospective coupling audit (Section 9).	One frozen physical dictionary compared with independent field, particle and gravity data.

The theorem library holds conditional theorems on consensus and repair [6], observers and records [7], gauge structure [8], spacetime and Einstein dynamics [9], and screen microphysics [10]; the flagship synthesis gives the informal account [3]. The labels A1, A2 and A3 used there name theorem premises: A1 is the carrier, federation and response layer of the specification, A2 the agreement condition with its endogeneity clause, and A3 the least-informative state selection that the no-outside rule imposes. A theorem that uses them states an implication, and Section 10 records, route by route, which antecedents the construction supplies on its own finite objects.

The sections follow the tower. Section 2 states the specification (layer two). Sections 3 and 4 give the executable observer on one carrier and overlapping observers, event clocks and matched screens on one native process (layer three). Sections 5 to 7 give the fields and clocks (layer four): field detectors on supplied actions, the common interacting source, and the geometry, continuum operator, clocks and fields of one repair history, followed by the common matter metric with its gravitational control (the interface to layer six). Section 9 gives the prediction protocols, Section 10 the premise map and Section 11 the ordered joins. Every number is copied from a retained receipt or a model document of the construction [11], cited at a pinned revision of the public repository [2].

Results carry the claim classes of Table 2. Actions, preparations, couplings, instruments, services and schedules are inputs of the specification, and apart from the retrospective replays of Section 9.5 no result is a comparison with independent observations. Numbered theorems and propositions are finite theorems unless they state a limit; items that quote a certified enclosure, table or replayed count are certified finite computations; the occupation limits of Section 6 are conditional analytic theorems on their supplied actions; the native continuum statements of Section 7 are source limit theorems with their domain, normalization and order of limits; Sections 9.5 and 9.3 are diagnostics, and the count protocol of Section 9.1 is a prospectively fixed conditional test.

Table 2: Claim classes used in this paper, from the flagship synthesis [3].

Class	Meaning
Finite theorem	A deductive result proved from displayed premises on a stated finite domain. An exhaustive computation counts as part of the proof only with a specified domain and a soundness argument.
Certified finite computation	An executable witness whose inputs, controls, and output are fixed by a reproducible certificate; execution alone does not promote its interpretation to a theorem.
Conditional analytic theorem	A theorem whose hypotheses include a supplied continuum action, state space or limit; its conclusion holds on that supplied object.
Source limit theorem	A theorem about the limit of a declared family of finite source executions along a stated subsequence, with its normalizations and order of limits displayed; the limit is derived, and its physical identification is a separate claim.
Reconstruction implication	An exact implication whose antecedents include displayed geometric or physical premises.
Unsupplied realization map	A required map from a finite object to a physical event, current, field, clock, scale, or continuum structure for which no construction is asserted.
Diagnostic	A certified computation compared against measured values that it consumes; never counted as a prediction.
Prospectively fixed conditional test	A diagnostic with a custody-bound target definition, kill band, precision floor, and decision rule fixed before comparison.

Seven claim records of the repository [2] state the scope of the executed components (Table 3). Further model records own the published-oscillator and anharmonic-transfer comparisons and the matter, common-field and static-product witnesses; they are cited where those results appear.

2 The specification

The specification is layer two of the tower. The carrier, the repair law and its record laws, the observer algebra, the services and the time parameters below are fixed before any test, and every later result runs on them.

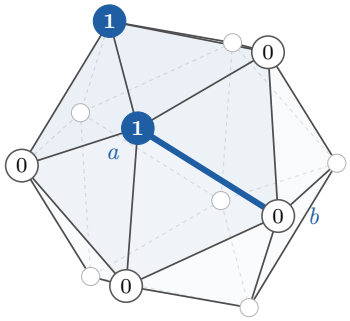
2.1 The carrier and its load configurations

The specification has one carrier with twelve primitive ports. Its incidence is the oriented boundary of an icosahedron, with thirty seams and twenty faces; five seams meet at every port, the neighbours of each port form a five-cycle, and every seam lies on two oppositely oriented faces [10, 3]. Write A for the adjacency matrix of the seam graph and $L = 5I - A$ for its Laplacian. The spectrum of L is 0, $5 - \sqrt{5}$ with multiplicity three, 6 with multiplicity five and $5 + \sqrt{5}$ with multiplicity three, so its characteristic polynomial is $x(x-6)^5(x^2-10x+20)^3$ [53, 94, 87].

A load configuration assigns a nonnegative integer x_p to each port p, with total load $\sum_p x_p = 2$. There are 78 configurations. The 66 with two distinct occupied ports form the binary shell; the other twelve are collisions $|2u\rangle$ with load two at one port u. We write $\mathcal{H}_{78} = \mathbb{C}^{78}$ for the quantum load space with basis $|x\rangle$, $\mathcal{H}_{66}$ for the span of the binary shell, and M_{78}, M_{66} for the corresponding full matrix algebras. For a binary configuration occupied at ports u and w we also write $|uw\rangle$ or $|\{u, w\}\rangle$.

Table 3: Executed components and the repository's claim records that state their scope.

Component	Sections	Claim record
Two-event native observer, classification of record laws, quantum mixing of the repair channels, recovery from complete records, phase protocol and its source-executed implementation	2.3, 2.4, 3.1, 3.5, 3.6, 9.1, 9.2	`OPH-MODEL-NATIVE-OBSERVER-EXECUTION`
Correctable record capacity and compact controller	3.3	`OPH-MODEL-OBSERVER-RECORD-CAPACITY`
P-dependent binary detector and retrospective coupling audit	3.4, 9.3	`OPH-MODEL-P-RESPONSE-EXECUTION`
Memory certificate at observer-local ticks for the reciprocal exchange law	3.8	`OPH-MODEL-N4-LOCAL-TICK-CERTIFICATE`
Scalar detector, two-read continuation and two-site source apparatus	5.1, 5.2, 5.3	`OPH-MODEL-SCALAR-RESPONSE-EXECUTION`
Common interacting source, occupation and readout certificates, smooth continuation, compiled native attachment and encoded memory, observer readouts, history repair, geometric and field readouts	6, 7.1	`OPH-MODEL-COMMON-SOURCE-EXECUTION`
Overlapping observers, local and active event clocks, record-law identification, federations and matched screens, the native continuum operator and its limits, the sequential protocol and public-record replays	7.2–7.6, 4, 9.4, 9.5	`OPH-MODEL-NATIVE-PROCESS-RECOVERY`



twelve ports, thirty seams, twenty faces; a binary load of total two (filled ports)

$d = x_a - x_b, \qquad j = \frac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big]$

d	$c=0$	$c=1$	effect of the attempt
2	1	1	balance
1	1	0	swap or wait
0	0	0	no move
-1	0	-1	wait or swap
-2	-1	-1	balance

The move j carries j units from a to b, and the fair coin c reads no load. At $|d| = 2$ the attempt lowers $V = \sum_i x_i^2$; at $|d| = 1$ it keeps V fixed, so the native process keeps moving. The seam drawn has $d = 1$.

Figure 1: The twelve-port carrier and its native repair. Left: the icosahedral incidence of the twelve ports, with thirty seams and twenty faces, in an orthographic view with back edges dashed; the filled ports carry a binary load of total two, and the drawn seam (a, b) has $d = 1$. Right: the seam rule for every mismatch the total-two carrier admits.

Ports are numbered 0 to 11 by the frame vectors u_p: $(-1, \varphi, 0)$, $(1, \varphi, 0)$, $(-1, -\varphi, 0)$, $(1, -\varphi, 0)$, $(0, -1, \varphi)$, $(0, 1, \varphi)$, $(0, -1, -\varphi)$, $(0, 1, -\varphi)$, $(\varphi, 0, -1)$, $(\varphi, 0, 1)$, $(-\varphi, 0, -1)$, $(-\varphi, 0, 1)$, with $\varphi = (1 + \sqrt{5})/2$. The thirty seams are the pairs with $|u_a - u_b|^2 = 4$, each written (a, b) with $a < b$, and that order is the orientation used by the repair law below [87, 88].

Premise A1 of the theorem library is the layer of carriers, their federation through typed seams and coherent triple overlaps, the oriented spherical support and the complete reversible port response [5]. The construction realizes one carrier of that layer: its ports, seams and faces, its load configurations and the repair events on its seams. Coherent quantum operations run through a

six-level processor whose twelve source generators, one per port, exponentiate to every executed gate on two- and four-dimensional proper codes through declared load and export services [74, 65]. Spherical-screen executions join many carriers under explicit access and refinement contracts [12]; they identify no physical current for the complete response algebra and do not discharge the full A1 attachment.

2.2 Native repair and its quantum record laws

Fix an oriented seam $e = (a, b)$ and a fair placement coin $c \in \{0, 1\}$. The native repair law is

$$d = x_a - x_b, \qquad j = \tfrac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big], \qquad y = x + j\,(\mathbf{e}_b - \mathbf{e}_a), \tag{1}$$

where $\mathbf{e}_p$ is the unit load vector of port p and y is the repaired configuration. The two endpoint loads become the floor and the ceiling of their mean; the coin assigns the ceiling to a when $c = 1$; all other loads are unchanged.

Proposition 2.1 (The classical seam rule)**.** *Let a stochastic rule on one seam (i) conserve $x_a + x_b$, (ii) end at a configuration minimizing $x_a^2 + x_b^2$, (iii) break any remaining tie by a random choice independent of the loads, and (iv) be invariant under exchange of the two endpoints. Then it is* (1) *with a fair coin.*

Proof. By (i) and (ii) the outputs are the floor and the ceiling of the mean. An odd total leaves two outputs, and by (iii) endpoint a receives the ceiling with a constant probability q. Exchange of the endpoints maps q to $1 - q$, so (iv) gives $q = \frac{1}{2}$. □

Without (iii), the rule that leaves the extra unit with the endpoint holding more satisfies the other three requirements. It fixes every binary configuration, so the binary shell carries no motion and the records carry no geometry. Requirement (iii) applies the no-outside rule to the tie: nothing recovered prefers one endpoint.

The record is $r = (c, j)$, the coin and the signed current. Compatibility on a seam is specified independently of the update; it requires the two endpoint loads to differ by at most one. On the quantum load space the record defines the current instrument C with Kraus operators and record-discarded channel

$$L^e_{(c,j)} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j} |y(x,c)\rangle\langle x|, \qquad \mathsf{C}_e(\rho) = \sum_r L^e_r\, \rho\, L^{e\dagger}_r. \tag{2}$$

Both maps are written in Kraus form [115]. The contrast-record variant E keeps $r = (c, d)$ instead, with Kraus operators $M^e_{(c,d)} = 2^{-1/2} \sum_{x:\, x_a - x_b = d} |y(x,c)\rangle\langle x|$ and record-discarded channel E_e. Uniform independent seam attempts form the baseline schedule; on diagonal binary states the repair is then the lazy symmetric exclusion process with two particles, with generator $\frac{1}{60}\sum_e (S_e - I)$ per attempt, S_e exchanging the endpoint loads [46, 53].

Proposition 2.2 (Native repair instrument)**.** *Let $e = (a, b)$ be an oriented seam of the twelve-port carrier, let the placement coin be fair, and let C_e and E_e act on $\mathcal{H}_{78}$. Then:*

(i) For every record value the branch map $x \mapsto y(x, c)$ is an injective translation of its support. Consequently $\sum_r L^{e\dagger}_r L^e_r = I$, and C_e is completely positive and trace preserving (CPTP), also when tensored with an untouched reference system. The same holds for E_e.

(ii) Every branch conserves the total load, leaves spectator ports unchanged, never increases $\sum_p x_p^2$, and maps the binary shell into itself.

(iii) *The current labels satisfy* $c \in \{0,1\}$ *and* $j \in \{-1,0,1\}$*; all six labels occur on* $\mathcal{H}_{78}$ *and four occur on the binary shell.*

(iv) C *and* E *induce the same classical transition on every configuration, and*

$$\mathsf{E}_e^2 = \mathsf{E}_e, \qquad \mathsf{E}_e\mathsf{C}_e = \mathsf{C}_e\mathsf{E}_e = \mathsf{E}_e, \qquad \mathsf{C}_e^n = \mathsf{E}_e + 2^{1-n}(\mathsf{C}_e - \mathsf{E}_e) \quad (n \geq 1).$$

On the binary shell E_e *is a trace-preserving conditional expectation.*

(v) *With the seam-total parity* $p = (x_a + x_b) \bmod 2$ *one has* $d = 2j + (2c-1)p$*, and* E *equals* C *followed by a binary record of* p *and this relabelling, as an identity of complete unnormalized branches.*

Proof sketch. The inverse of a fixed-current branch adds j at a and subtracts it at b; every basis vector has one branch for each coin, so each column of the joint record/source isometry has two entries $2^{-1/2}$ with disjoint supports [51]. For fixed contrast, the unchanged spectators and the endpoint sum reconstruct the input, which gives (i) for E. Items (ii) and (iv) are the single-seam relations of the candidate register [46]. Item (v) follows from $j = \lceil d/2 \rceil$ for $c = 0$ and $j = \lfloor d/2 \rfloor$ for $c = 1$; the parity-refined Kraus operator $\Pi_p L_{(c,j)}$ has exactly the contrast group as support [68]. The executable checks all 312 outputs of the observer's two seams against an independent floor and ceiling rule [51]. □

The current instrument is no classical coarse-graining of the contrast instrument: grouping the labels of E sums density operators and cannot restore the coherent Kraus sums of C. Both belong to the family of quantum record laws classified in Section 2.3. One further law of the candidate register, the reciprocal exchange N4 that couples the native current to private six-level response factors, enters only the local-tick certificate of Section 3.8 [46].

2.3 Admissible record laws

The classical rule fixes the diagonal of a record law and leaves its coherences free. For the seam $e = (a,b)$ write $\kappa(x) = (x_a, x_b)$ for the seam key and $D = \mathbf{e}_b - \mathbf{e}_a$. A channel Φ on $\mathcal{H}_{78}$ is an admissible record law for e if it acts as the identity on the ten spectator ports, reproduces the classical repair distribution on every $|x\rangle\langle x|$, commutes with $U_t = \exp(i \sum_p t_p N_p)$ for every $t \in \mathbb{R}^{12}$, and is invariant under the carrier rotation that exchanges a and b with the current reversed. The third condition makes every local load phase a symmetry of the unread channel. It is a premise of the classification, a local charge-superselection rule for the repair channel, and it is what excludes a recorder of a pre-repair endpoint load. It guarantees that a definite-current Kraus representation exists; it does not force every recorded implementation of the channel to commit a definite current, since a change of record basis before commitment mixes the current labels and leaves the channel unchanged [45]. The classification below concerns unread channels, and statements about current histories use the compatible definite-current instrument.

Theorem 2.3 (Coherence tables)**.** *A channel* Φ *is an admissible record law if and only if*

$$\Phi(|x\rangle\langle x'|) = \sum_j G_j\big(\kappa(x), \kappa(x')\big)\, |x + jD\rangle\langle x' + jD|$$

for endpoint-symmetric positive semidefinite matrices G_j *on the seam keys that can carry the current* j*, with* $G_j(\kappa,\kappa) = \Pr(j \mid \kappa)$*. On* $\mathcal{H}_{78}$ *the free entries are five multipliers:* k *between the empty and the full seam,* g *between the empty seam and a waiting single load,* h *between the full seam and a*

waiting single load, a real m between the two waiting orientations, and n between a moving single load and a moving collision. Positivity gives $|g|, |h|, |n| \le 2^{-1/2}$ and $|m| \le \frac{1}{2}$, and $k = 1$ forces $g = h$. Repairing one seam with (k, g, h, m, n) and then with (k', g', h', m', n') gives $(kk', gg', hh', mm', n\bar{h}')$. Exactly two admissible laws are idempotent, those with $g = h = m = n = 0$ and $k \in \{0, 1\}$, and every law with $k = 1$ converges under repetition to the idempotent law with $k = 1$.

Proof sketch. Covariance places $\Phi(|x\rangle\langle x'|)$ in the span of units $|y\rangle\langle y'|$ with $y - y' = x - x'$. Complete positivity confines nonzero Choi entries to pairs with nonzero diagonal, which the classical law makes $(x, x + jD)$, so both sides carry the same current and the Choi matrix is the direct sum of the G_j. Trace preservation and the classical law fix the diagonals, and spectator invariance makes each G_j a function of the seam keys. Under repetition every coherence keeps its seam key, so each multiplier applies again; idempotence requires $k, g, h, m \in \{0, 1\}$ and $n = n\bar{h}$, and the bounds leave $g = h = m = n = 0$. Exact arithmetic in $\mathbb{Q}(\sqrt{2})(i)$ checks the composition law on every pair of seam keys for 64 ordered pairs of tables and on all 6,084 matrix units for two pairs [44]. □

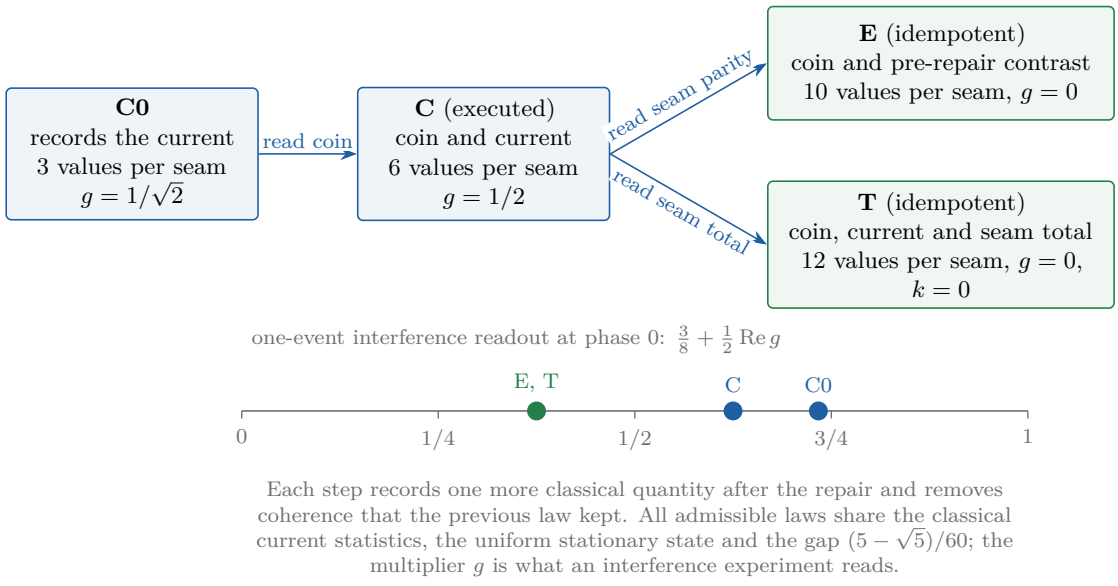


Figure 2: The record-law ladder of Theorem 2.3 and Table 4, and the one-event interference readout $\frac{3}{8} + \frac{1}{2}\,\mathrm{Re}\,g$ at phase zero that separates its members.

Four members form a ladder (Table 4 and Figure 2): C is C_0 followed by a diagonal readout of the coin, E is C followed by a readout of the seam parity as in Proposition 2.2(v), and T is C followed by a readout of the seam total. The completion of C in Proposition 2.2(iv) is the case $g = h = n = \frac{1}{2}$, $m = 0$, and E is the idempotent law with $k = 1$. A recorder of an endpoint's pre-repair load keeps the classical law and fails covariance, because one of its branches carries two currents. C_0 records only what conservation requires, C records the tie-break as an event, E is the coarsest law for which one repair completes the seam, and T records the full seam occupation; none of these principles follows from the theorem premises. The construction executes C as its convention and compares it with E. Under a common definite-current record interface all members assign the same probabilities to every classical current observation; Section 9 reads g in one coherent event, and Section 4.3 identifies all nine real coordinates. Every admissible table, including complex phases and different laws on different seams, has the uniform binary density as its unique stationary source state and the same spectral gap $(5 - \sqrt{5})/60$, so equilibrium selects none of the coordinates;

a finite unread warm-up of 221 attempts on the binary shell, or 318 on the full carrier, brings an arbitrary initial density within 1/1000 of that state, uniformly over untouched references [44].

Table 4: Four distinguished admissible record laws [44].

Law	Each event records	Values per seam	k	$g = h$	Idempotent
C_0	the current	3	1	$2^{-1/2}$	no
C	coin and current	6	1	1/2	no
E	coin and pre-repair contrast	10	1	0	yes
T	coin and both endpoint loads	12	0	0	yes

2.4 The finite observer algebra

An observer of length n on the carrier consists of a preparation on $\mathcal{H}_{78}$, a record tape $R^{\otimes n}$, seam policies s_k that map committed records and classical randomness to seams or to supplied detector outcomes, the coherent writes W_{s_k}, optional supplied instruments whose committed bits may enter s_k, and a halting rule after n writes; a policy reads only committed records. Definition 2.4 is the instance $n = 2$, and Sections 3.4, 5 and 6 add declared instruments to it.

Definition 2.4 (Two-event native observer)**.** Let $R = \mathbb{C}^6$ carry one basis vector per current label (c, j). The observer carrier is

$$\mathcal{H}_{\mathrm{cap}} = \mathcal{H}_{78} \oplus (R \otimes \mathcal{H}_{78}) \oplus (R \otimes R \otimes \mathcal{H}_{78}),$$

with sectors of dimensions 78, 468 and 2808, total 3354; the sector index is the phase, the number of filled record slots.

(a) *Preparation.* The scored preparation is uniform on the 66 binary configurations; the full input family uses all 78.

(b) *Measurement and record.* At phase zero the coherent write $W_e = \sum_r |r\rangle \otimes L_r^e$ acts on the seam $e = (0, 1)$.

(c) *Record-only feedback.* At phase one the seam is $s(r) = (0, 5)$ if the stored current is positive and $(0, 1)$ otherwise, and $V_1(|r\rangle \otimes \psi) = |r\rangle \otimes W_{s(r)}\psi$.

(d) *Forecast.* Before the second event the controller reads its stored letter and issues a supplied forecast $q(r)$ of a positive second current.

(e) *Halting.* At phase two the observer acts as the identity.

The controller reads only the committed first record; it has no access to source loads or future outcomes.

Proposition 2.5 (One CPTP map with classical-prefix preservation)**.** *Let $V_0 = W_{(0,1)}$ and V_1 be extended by zero to the other sectors, giving A_0 and A_1, and let P_2 project onto the full sector. Then:*

(i) V_0 and V_1 are isometries on every nonfull sector, including unreachable combinations of letters and loads, and $T(\rho) = A_0\rho A_0^\dagger + A_1\rho A_1^\dagger + P_2\rho P_2$ is CPTP;

(ii) *on the inhabited subspace $K = \mathcal{H}_1 \oplus \mathcal{H}_2$ every first-record projector Q_r obeys $T^*(Q_r) = Q_r$, so a committed first record is reread with probability one;*

(iii) *for the blank input with occupations $(1, 1, 0, \ldots, 0)$, the first-record projector of the letter $(0, 0)$, extended by zero to phase zero, has expectation 0 before the tick and $1/2$ after it, so writing a record is a transition distinct from preserving it;*

(iv) *for the equal superposition of $\{0, 2\}$ and $\{1, 2\}$, the reduced-density entry of the first-record pair $((0, 1), (0, 0))$ equals $1/4$ after the first write and $1/8$ after the second while the classical diagonals are unchanged, so coherence between different record values need not survive feedback.*

Proof sketch. Orthogonal retained labels and $W_s^\dagger W_s = I$ give $V_0^\dagger V_0 = V_1^\dagger V_1 = I$, hence $A_0^\dagger A_0 + A_1^\dagger A_1 + P_2 = I$. Every subsequent map intertwines Q_r on K. In (iv) the working state attached to the entry has occupations $\{1, 2\}$, on which the isometries of $(0, 5)$ and $(0, 1)$ overlap by $1/2$ [51]. The checkpoint adapter verifies all $78 + 468 + 2808$ columns and both the preservation identity on the 3276-dimensional checkpoint space and the blank-sector control [74, 100]. □

Lemma 2.6 (Feedback requires classical record preservation)**.** *Let Λ be a CPTP map from $R \otimes W$ to $R \otimes W'$ with $\Lambda^*(B \otimes I) = B \otimes I$ for every B in the full matrix algebra of R. Then $\Lambda^*(I \otimes O) = I \otimes \Phi^*(O)$ for a unital completely positive map Φ^*, so changing only the record preparation of a product input changes no output statistic on W'.*

Proof sketch. The operators $B \otimes I$ lie in the multiplicative domain of the unital completely positive map Λ^*, because the Schwarz inequality is an equality for them [107]. Then $\Lambda^*(I \otimes O)$ commutes with every $B \otimes I$ and has the stated form [51]. □

Lemma 2.6 fixes the shape of the observer contract. A record that steers the working state is preserved as a classical record, with its projectors fixed, while coherence between different record values may change.

2.5 Declared services

Each component names the services it assumes; they belong to the specification and enter the resource counts where a component counts resources. The services are coherent proper-code load and export, with complementary failure outcomes retained by a complete channel definition; blank initialization, reset, storage and routing of quantum registers and classical records; Born readout of pointers and fair coins; and analogue precision of source pulses. Particular components add a calibrated external flux and probe (Section 3.4), incident-event recognition and bare-occupation readout (Section 3.8), coherent phase preparation and readout (Section 9), and, for the native process results, independent uniform seam scheduling, uniform fixed-number preparation, full initial and current readback, coherent wave processing, the return-length sampler and the response controller (Sections 7, 4 and 9). The coherent services are not products of the native grammar: committed C/E repair with diagonal accessible inputs and fresh independent diagonal ancillas cannot prepare the declared coherent reference to arbitrary accuracy, even with adaptive heralding. This coherence floor of the native interface makes the coherent services a necessary resource of the specification. The sharp joint instrument of Section 6.1 uses native integer repair as a declared primitive, and Section 6.5 attaches the compiled 78-dimensional C/E channels, routing and encoded recovery to the continuing work history. The processor algebra is M_6; its twelve generators are the exact source generators of the compiler, one per port [65, 74]; a proper code is a two- or

four-dimensional subspace of $\mathbb{C}^6$ with its load and export channels and the complementary failure outcome; and the service error model is the compositional contract of Section 9.2. Exact certificates exclude numerical execution error and physical service error unless a component states otherwise.

2.6 The closure inputs P and N

The pixel ratio P is the area of one screen cell in units of $\ell_\star^2$, and N is a record capacity in nats [7, 59]. The construction accepts both as closure-derived inputs and fits neither to an output. P enters through a declared coupling map, the source-only structured-running branch of the fine-structure closure (one-loop running and matching conventions with the tree-level m_Z closure), defined in the P-derivation code of the repository [2] and consumed unchanged; the map is a declared selection, not a result of this construction. At the pinned closure value $P = 1.630972095858897\ldots$, the midpoint of the pinned forward-closure interval, the map gives $\alpha^{-1}(P) \approx 136.9948351774$ and $g(P) = \sqrt{4\pi\alpha(P)} \approx 0.302867613117$ in Heaviside–Lorentz natural units, numerical evaluations of the executed map [57, 96]. N enters only as a bound, and no component consumes its closure value. For a declared capacity carrier of dimension D, the record capacity $N_{\mathrm{record}} = \ln D$ bounds every correctable classical record code by $\ln M_0 \le N_{\mathrm{record}}$ [43, 52]. The cosmological relation $N = 3\pi/(\Lambda\ell_\star^2)$ additionally requires a physical identification of capacity with horizon area, which the construction does not make.

2.7 Time parameters

The construction keeps the time parameters of Table 5 distinct [49]. Conversions between them need explicit maps. The construction uses three: the fixed-reference relation $t = -\beta s$ of Section 6.6 between a model coordinate and a modular parameter, the instrument-valued conversion between two local event clocks of Section 4.2, which keeps their whole count law, and the rescaling of native attempt count along the refining screens of Section 7. A calibration of native calls against a monotonic host counter is an engineered software correspondence. No preferred universal physical clock is required.

3 The finite observer

This section is layer three on one carrier: an observer that prepares, records, retains and acts on native records, with its capacity, memory and recovery limits.

3.1 The native observer

The native observer executes Definition 2.4 on the scored preparation [50, 51]. A trial is one fresh preparation followed by two repair events and a halt; running many trials repeats the preparation and does not extend one observer's memory. Event order is the time parameter. The tape, prior, forecast table, seam selector and halting rule are designed apparatus, and the forecast is a fixed table without learning.

Proposition 3.1 (Statistics of the native observer)**.** *Prepare the 66 binary configurations uniformly, use fair coins and the controller of Definition 2.4, and let Y indicate a positive second current. Then:*

(i) the 264 histories, two coins for each of the 66 configurations at each of the two events, each have weight $1/264$*;*

Table 5: Time parameters and where they occur.

Parameter	**Meaning**	**Occurrence**
Host time	Runtime of a software execution; a resource cost	A software calibration of native calls; never a physical quantity
Solver counter	Position in a search or repair schedule; the attempt count of a declared schedule	History repair (Section 6.7); attempt counts in the observer theorems
Causal order	Precedence of recorded events; counts of native operations	Observers, detectors, phase protocol, local ticks
Local event count	Attempted repairs incident to an observer region, or occupation changes at a port	Overlapping observers, local and active event clocks, matched screens (Section 4)
Scaled attempt count	Native attempt count n at level l rescaled to $\tau = n/(1260 \cdot 16^l)$ on the refining screens	Native density limits and the controlled repair response (Sections 7.3, 9)
Coherent wave parameter	Parameter of the admitted coherent continuation $\cos(t\sqrt{A})$	Native waves and the return reference (Sections 7.2, 7.7)
Model coordinate	Parameter of a declared Hamiltonian or action	Scalar detectors, two-site apparatus, common source
Modular parameter	Parameter of the modular automorphisms of an algebra-state pair	Modular clock and fixed-reference relation (Section 6.6)
Operational clock	Observer-accessible reading with calibrated duration	Supplied by no component

(ii) the first record has masses $14/33$, $14/33$, $5/66$, $5/66$ *for* $(c, j) = (0,0), (1,0), (0,1), (1,-1)$, *the conditional probabilities* $q(r) = \Pr(Y = 1 \mid r)$ *are* 0, $5/56$, 0, $1/2$ *in the same order, and* $\Pr(Y = 1) = 5/66$*;*

(iii) the Brier risk of forecasting Y *is* $395/7392$ *with* $q(r)$, $140/2013$ *with the best forecast based on the chosen seam alone, and* $12875/150304$ *when* q *is evaluated on an independent record resampled within the same seam;*

(iv) the mean final load at port 0 *is* $35/264$ *under record feedback and* $1/6$ *when the selector is disconnected or replaced by an independent selector with the same seam frequencies; the feedback effect is* $-3/88$.

Proof sketch. For (iii), $\mathbb{E}[(Y - q(R))^2] = \mathbb{E}[\mathrm{Var}(Y \mid R)]$. The chosen seam S is a function of R; the best S-only forecast has excess risk $\mathbb{E}[\mathrm{Var}(q(R) \mid S)]$, and an independent resample within the seam adds twice that amount. For (iv), a positive first current occurs on ten binary inputs with coin zero, mass $5/66$. On that branch $x_0 = 0$, $x_1 = 1$ and $\mathbb{E}[x_5] = 1/10$; a second repair on $(0,1)$ gives port-0 mean $1/2$ and one on $(0,5)$ gives $1/20$. Hence the change is $(5/66)(1/20 - 1/2) = -3/88$ and the mean is $1/6 - 3/88 = 35/264$. An independent selector leaves the uniform shell uniform under either seam [51]. □

The other two current letters occur only on the full carrier, where the forecast abstains. The exact receipt retains all 264 histories [91], and a seeded execution of 20,000 fresh trials matches every trial against its exact path reference, with sample final port-0 loads 0.1327 with feedback, 0.16645 disconnected and 0.1668 with the independent selector [92]. Since the selector is a design choice and the forecast table is the exact conditional law, (iii) and (iv) are consistency checks: the Bayes forecast beats coarser forecasts and the record-wired selector moves the occupation.

3.2 Continuing memory and the actual stationary reference

The finite campaign [2] extends the two-event observer with an explicitly declared memory law. It keeps the same source and complete C/E instruments. A classical tag chooses C with probability λ and E otherwise; a repair occurs with probability r, with an identity hold otherwise. A controller chooses a preferred seam from its retained records. The actual seam distribution is

$$p(e \mid m) = \frac{\epsilon}{30} + (1-\epsilon)\pi(e \mid m), \qquad r, \epsilon \geq \frac{1}{16}, \quad 0 \leq \lambda \leq 1.$$

The rational parameter bound and permitted controller tables are part of the campaign, not measured physical constants. The executed controller proposes $(0, 5)$ after a positive current and $(0, 1)$ otherwise. It does not read the hidden source load. A proposal intervention preserves the uniform exploration floor. Holds are recorded with current zero.

For each classical memory input, every source branch is $K_b = \sqrt{w_b} M_b$, with M_b a native partial injection or identity and $\sum_b K_b^* K_b = I_{78}$ exactly. Tensoring with the identity preserves an arbitrary untouched reference. The tests independently reconstruct the branch supports by endpoint floor/ceiling arithmetic, including their off-diagonal action. Composition therefore defines every finite history with its unnormalized quantum output and actual probability. Exact prefix replay and retained six-attempt source/reference/last-record marginals verify C, E and the delayed tagged mixture; the complete history tree remains implicit in that channel kernel.

One to eight complete records occupy the declared buffer. Under the executed one-slot export policy, n attempts leave one record and export $n-1$ records for $n \geq 1$; expected repair and hold counts are nr and $n(1-r)$. Concatenating exports and the remaining buffer reconstructs the history. The halt-on-full control instead preserves its source and records with an explicit timeout outcome. These are different resource contracts. Bounded working memory with growing exports is not closed finite-resource operation for unlimited duration. No new native source-word implementation of the supplied initialization, selection, write, export or delivery services is claimed.

The stationary reference must include this controller. For $r = 1$ and $\epsilon = 1/2$, a uniform binary source with blank memory becomes nonuniform after two attempts: 30 source probabilities change, with maximum deviation $1/960$ from $1/66$. Exact rational solution of the actual source/current-sign kernel instead gives one stationary law on a 129-state recurrent subset of the 132 source/bit states. Applying the complete C/E observer instrument to that law constructs the full-record reference. Its projection recovers the bit law, and one further actual observer step preserves the entire full-record density exactly. The resulting 128 C or 192 E record blocks have 4,218 positive diagonal entries overall. Positivity establishes faithfulness only on the explicitly compressed support of each block. The growing exported tape is not a stationary object, and the delayed member's hold updates require a separate stationary solve.

This reference also makes a clock obstruction precise. In an actual nontracial block choose positive weights with ratio $q = p_i/p_j > 1$ and the dimensionless modular parameter $s_* = \pi/(2 \log q)$. The state-dual modular action takes the real pair probe $(|i\rangle + |j\rangle)/\sqrt{2}$ to a density with off-diagonal entry $-i/2$. Every admitted history, with the declared real joint ancillas and classical memory, preserves real density matrices. A norm-one imaginary pair observable separates the target from every such normalized output, giving trace-distance error at least $1/2$. This witness is not a native detector, and the probe is an all-input map test, not an asserted native preparation. On the accessible commutative record algebra the modular action is trivial. Other algebras, encodings or coherent services have different contracts; no general physical-clock exclusion follows. The exact reference, implementation bound and full qualifications are retained with the executable model [2].

3.3 Record capacity

The retained records are evaluated with the correctable-public-record construction of the theorem library [7, 52]. There M_0 is the largest correctable set of reachable public records, the independence number of the compound checkpoint confusability graph; the record capacity is $N_{\mathrm{record}} = \ln M_0$ nats; and $M_0 \le D$ for a declared carrier of dimension D. The counts below describe the reachable record interface; program, source, probe and service memory are additional.

Proposition 3.2 (Record capacity of the native observer)**.** *Take halt and exact classical reread as the declared terminal continuations of the observer of Definition 2.4. Then the compound graph has no edges, and:*

(i) $M_0 = 11$ *on the binary preparation (four reachable first records, 264 histories) and* $M_0 = 17$ *on the full family of 78 inputs (six first records, 312 histories);*

(ii) each terminal word projector has rank 78 in the 3354-dimensional carrier, distinct projectors are orthogonal, and the eleven reachable ones have total rank 858, so the bound is unsaturated with slack $\ln(3354/11)$*;*

(iii) the binary preparation reaches $1{+}4{+}11 = 16$ *classical prefixes, so four bits encode the complete controller history; a compact controller that reads only this code reproduces the phase, action, forecast, source loads, record and halt of all 264 histories; the full family reaches 24 prefixes and needs five bits;*

(iv) coarser retained readouts give correctable codes of size 4 for the first letter alone, 4 for the second letter alone, 3 for the forecast alone, 2 for the chosen seam alone, 4 for forecast and seam, and 1 after complete erasure.

Proof sketch. Halt and reread leave the tape unchanged, so the joint kernel is the identity and the existing evaluator returns the number of reachable terminal words. The terminal projector of word w occupies the basis coordinates $546 + 78w$ to $546 + 78w + 77$ of the carrier [52, 93]. □

The terminal capacity is $\ln 11$ nats, about 3.459 bits, on the binary preparation and $\ln 17$ nats on the full family; the Shannon entropy of the benchmark record law, about 2.023 nats, is a different quantity. The library's global proposal adds closure across the complete terminal fiber, $N_\star = \ln M_0(U_{N_\star})$, and a de Sitter horizon identification giving $\Lambda\ell_\star^2 = 3\pi/N_\star$ [7, 52]; the native records supply an actual finite input to that evaluator, and the two further steps are not taken here.

3.4 The P-dependent detector

The detector carries the closure input P through the electromagnetic coupling into a recorded binary observation that controls the next repair [57, 58]. The charge assignment, probe, calibrated flux, readout and controller are specified apparatus (Section 9.3).

Choose the charged realization $Q_v = g(P)N_v$ of the native occupation N_v. A current record j on the seam (a, b) then transfers the charge $\Delta Q = g(P)\, j\, (\mathbf{e}_b - \mathbf{e}_a)$, and total charge is conserved. For port gauge parameters λ_v and $U_\lambda|x\rangle = e^{ig\sum_v \lambda_v x_v}|x\rangle$, every current-grouped Kraus operator obeys $U_\lambda L_{(c,j)} U_\lambda^\dagger = e^{igj(\lambda_b-\lambda_a)} L_{(c,j)}$. The probe is a unit-charge path qubit prepared in $|{+}X\rangle$. The first native current routes it around a calibrated external loop with signed exposure $j\chi$, where $\chi = \oint A{\cdot}dx$ is prepared independently of g, so the relative phase is $\varphi_j = g(P)\, j\, \chi$, the Aharonov-Bohm phase [103]. A Y measurement commits $y = \pm 1$ with amplitudes $m_y(j) = \frac{1}{2}[e^{-i\varphi_j/2} - iy\, e^{i\varphi_j/2}]$ and effects

$E_y = \sum_r \frac{1}{2}[1 + y \sin \varphi_{j(r)}]|r\rangle\langle r|$ on the old current register. The actuator receives only y. It repairs $(0, 5)$ for $y = +1$ and $(0, 1)$ for $y = -1$, keeps y unchanged, and the protocol halts.

Proposition 3.3 (P-dependent response)**.** *Prepare the 66 binary configurations uniformly, run the protocol above, and put $s = \sin(g(P)\chi)$. Then:*

(i) $\{E_y\}$ *is a binary instrument on every input, including coherent records; each* $M_y = \sum_r m_y(j(r))|r\rangle\langle r|$ *commutes with the old record projectors, and the composed experiment is one CPTP map that preserves the committed bit;*

(ii) $\Pr(y \mid j_1, P, \chi) = \frac{1}{2}[1 + y \sin(g(P) j_1 \chi)]$*;*

(iii) at $\chi = 1$ *all 528 state, coin, detector and coin histories have positive probability, and*

$$\mathbb{E}[y] = 0, \quad \mathbb{E}[y j_1] = \tfrac{5}{33} s, \quad \mathbb{E}[j_2] = \tfrac{5}{264} s, \quad \mathbb{E}[g(P) j_2] = \tfrac{5g(P)}{264} s;$$

(iv) for weak exposure, $\mathbb{E}[g\, j_2] = \frac{5\pi}{66}\alpha(P)\chi + O(\chi^3)$*;*

(v) disconnecting the probe, or replacing y by an independent fair bit, gives $\mathbb{E}[j_2] = 0$ *with the same detector and seam frequencies.*

Proof sketch. The eigenvalues of E_y lie in $[0, 1]$ and sum to one, which gives (i). For (iii), after the first repair let the source be z; averaging the second coin gives $\mathbb{E}[j_2 \mid z, j] = \frac{1}{4}(2z_0 - z_1 - z_5) + \frac{sj}{4}(z_1 - z_5)$. The first term averages to zero because the first repair preserves the uniform shell. Currents ± 1 each have mass $5/66$, and on either branch the other load is uniform over ten ports, so $\mathbb{E}[j(z_1 - z_5)] = 5/66$. An independent floor and ceiling enumerator reproduces every coefficient over all $66 \cdot 2 \cdot 2 \cdot 2 = 528$ paths [58, 60]. □

At the computed P and $\chi = 1$ the plus probability is 0.649129261647 for $j = +1$, 0.350870738353 for $j = -1$ and $1/2$ for $j = 0$; the mean second current is 0.00564883566843, the mean transferred charge 0.00171084937579 in natural units, and the final port-0 load 0.161017830998235 against $1/6$ for the controls. Off-closure controls at $0.999P$ and $1.001P$ change the detector and feedback probabilities at fixed χ, and a seeded run of 200,000 trials gives mean second current 0.006390 with standard error 0.00085715 [57, 96]. The audit record (r_1, y, r_2) has 24 terminal words and $M_0 = 24$ [59]. If a cut enclosing a retained record obeys the area-capacity law with area K chart cells, then $N_{\mathrm{cut}} = KP/4$ and $\ln M_0 \le N_{\mathrm{cut}}$ give $K \ge 4 \ln M_0 / P$; with the pinned P one detector bit needs $K \ge 2$ ($4 \ln 2 / P \approx 1.6999608573$) and the full audit $K \ge 8$ ($4 \ln 24 / P \approx 7.7942567832$).

3.5 Quantum mixing and memory

Discarding every seam, coin and repair record after each event turns the repairs into channels Φ_{C} and Φ_{E} on the source. Under independent seam attempts these channels have an exact spectral theory [53]. Theorems 3.4 and 3.5 concern this record-discarded channel under independent uniform attempts; they do not describe the retained tape or the feedback controller.

Theorem 3.4 (Signed-difference reduction and quantum gap)**.** *Let G be a finite connected graph with $n \ge 2$ vertices, positive edge weights w_{ab} and $W = \sum w_{ab}$. Let each event choose the edge ab with probability w_{ab}/W and a fresh fair coin, and discard its record. On the binary space $\mathcal{H}_k$ of load k, $1 \le k \le n - 1$, with $D = \binom{n}{k}$:*

(i) for a matrix unit $|x\rangle\langle y|$ the signed difference $z = x - y$ is invariant; if z has r positive entries and $Z = \{v : z_v = 0\}$, then on the sector of z

$$\Phi_{\mathsf{L}}\big|_z = I - \frac{L_{\mathrm{excl}}(G[Z], k-r) + K_{\mathsf{L}}(z)\, I}{2W}, \qquad \mathsf{L} \in \{\mathsf{C}, \mathsf{E}\},$$

with the exclusion Laplacian L_{excl} of the induced graph, $K_{\mathsf{C}}(z) = \sum w_{ab}|z_a - z_b|$ and $K_{\mathsf{E}}(z) = 2\sum w_{ab}\mathbf{1}[z_a \neq z_b]$;

(ii) both channels are unital and Hilbert-Schmidt self-adjoint with spectrum in $[0, 1]$, their fixed algebra consists of the scalars, and both have gap $\gamma = \lambda_1(G)/(2W)$, where $\lambda_1(G)$ is the first positive Laplacian eigenvalue; and $\|\Phi_{\mathsf{L}}^t - \mathrm{Dep}_D\|_{2\to 2} = (1-\gamma)^t$;

(iii) for every input and every untouched reference R, $\mathrm{dist}((\Phi_{\mathsf{L}}^t \otimes \mathrm{id}_R)(\rho), \tau \otimes \rho_R) \le \min\{1, \frac{D}{2}(1-\gamma)^t\}$, with $\tau = I_D/D$ and $\mathrm{dist}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$.

For the twelve-port carrier at load two ($n = 12$, $W = 30$, $\lambda_1 = 5 - \sqrt{5}$, $D = 66$) the gap is $\gamma = (5 - \sqrt{5})/60 \approx 0.046066$, and the slowest off-diagonal sectors contract by exactly $r_{\mathsf{C}} = 5/6$ and $r_{\mathsf{E}} = 7/10$ per attempt. On the full 78-state carrier the gap is the same, the full channel is block triangular with a diagonal collision block whose entries lie in $[0, 5/6]$, and the unique stationary density is the uniform binary density, of rank 66 on dimension 78.

Proof sketch. A matrix unit survives a coin branch exactly when its two records agree, and then both configurations translate by the same current; comparing the coin branches gives (i). The diagonal sector is the symmetric k-particle exclusion chain, whose gap equals $\lambda_1(G)$ by the theorem of Caputo, Liggett and Richthammer [106]. For a nonzero sector, $K_{\mathsf{E}} \ge K_{\mathsf{C}} \ge \frac{1}{2}\sum w_{ab}(z_a - z_b)^2 \ge r\lambda_1 \ge \lambda_1$, so no off-diagonal mode is slower. Item (iii) uses Schmidt rank at most D and $\|Y\|_1 \le D\|Y\|_2$. On the carrier, $r = 1$ gives $K_{\mathsf{C}} = 10$ and $K_{\mathsf{E}} = 2(10 - \mathbf{1}[uv \text{ edge}])$, while $r = 2$ gives $K_{\mathsf{C}} \ge 16$ and $K_{\mathsf{E}} \ge 24$. The collision block has largest entry $1 - d_{\min}/W = 5/6 \le 1 - \gamma$ [53]. The executable checks the reduction against the native primitive on all 4,356 binary matrix units and the transient structure on all 6,084 full matrix units [94]. □

Choose adjacent ports u, v and the two mixtures $\rho_\pm = \frac{1}{20}\sum_{w \notin \{u,v\}}(|uw\rangle \pm |vw\rangle)(\langle uw| \pm \langle vw|)$. They have identical load populations and orthogonal supports, and

$$\mathrm{dist}(\Phi_{\mathsf{L}}^t(\rho_+), \Phi_{\mathsf{L}}^t(\rho_-)) = r_{\mathsf{L}}^t, \qquad \mathrm{dist}(\Phi_{\mathsf{C}}^t(\rho_+), \Phi_{\mathsf{E}}^t(\rho_+)) = \tfrac{1}{2}(r_{\mathsf{C}}^t - r_{\mathsf{E}}^t).$$

The second distance is $1/15$ after one attempt and peaks at four attempts at $12259/101250$; the binary effect $\frac{1}{2}(I + A_{uv} + A_{uv}^\dagger)$ with $A_{uv} = \sum_{w \notin \{u,v\}}|uw\rangle\langle vw|$ attains it. Load populations, classical load histories and diagonal geometric observables obey the same exclusion process under both laws, so only coherent preparations and readouts distinguish them under this schedule.

Theorem 3.5 (Finite quantum memory)**.** *Under the schedule of Theorem 3.4 on the twelve-port carrier:*

(i) on the binary shell both record-discarded channels are entanglement breaking after at most 134 attempts;

(ii) on the full carrier let u, v be adjacent and $|\Psi\rangle = (|2u\rangle|0\rangle + |uv\rangle|1\rangle)/\sqrt{2}$ with a reference qubit; after t attempts the coefficient of $|2u\rangle\langle uv|$ is a_{L}^t, with $a_{\mathsf{C}} = 23/30$ and $a_{\mathsf{E}} = 7/10$, and the partial transpose on the reference has a principal 2×2 minor equal to $-a_{\mathsf{L}}^{2t}/4 < 0$; no finite power of either full channel is entanglement breaking;

(iii) every input converges to $\tau_P \otimes \rho_R$, where τ_P is the uniform binary density, with dist $\le \min\{1, \mu^{s/2} + \mu^s + \frac{D}{2}(1-\gamma)^{t-s}\}$ *for all* $0 \le s \le t$, $\mu = 5/6$ *and* $D = 66$.

Proof sketch. For the normalized Choi state J_t, $\|J_t - I/D^2\|_2^2 = D^{-2}\sum_{i>0}\lambda_i^{2t}$. The separable ball of Gurvits and Barnum [111] and the equivalence of separable Choi states with entanglement-breaking channels [113] reduce (i) to $65\, r_{\mathrm{mix}}^{2t} + 4290\, r_{\mathsf{L}}^{2t} \le 1/4355$, where $r_{\mathrm{mix}} = 1-\gamma$. The rational enclosure $559/250 < \sqrt{5} < 2237/1000$ certifies this at $t = 134$ and its failure at $t = 133$; treating all modes with r_{mix} would certify only 178. For (ii), 25 edges avoid u and four of them touch v; C keeps one coin on each of those four and E keeps neither, and the triangular structure prevents any return from the binary space. Separable states have positive partial transpose [117]; the complete argument is in [53, 94]. □

Replacing the full carrier by its attracting binary shell therefore changes an exact finite-time quantum property, although both converge to the same stationary density. The number 134 is a sufficient certificate time; the exact index of entanglement breaking on the shell is not determined.

3.6 Recovery from complete records

Preserving committed records and recovering the unknown input from them are different questions [54]. Here a seam policy may depend on committed records and classical randomness and may not inspect hidden loads or insert quantum corrections; a final decoder may act on the full load register and tape, keep the tape, use blank ancillas and return a state on the original carrier. That decoder access is stronger than the record-only access of the observer's controller.

Theorem 3.6 (Optimal recovery from complete native records)**.** *Let X be the 78 configurations or the 66-state shell and $D = |X|$. For a complete admitted history h let X_h be its compatible inputs, P_h the projector onto them, $d_h = \operatorname{rank} P_h$, and w_h the product of its seam and coin factors. Then:*

(i) the history Kraus operator is $K_h = \sqrt{w_h}\, U_h P_h$ with $U_h^\dagger U_h = P_h$ and $\sum_h w_h P_h = I$; a tape-controlled inverse outputs $\sum_h w_h |h\rangle\langle h| \otimes P_h \rho P_h$, so every initial load population is recovered;

(ii) among all final CPTP decoders the optimal entanglement fidelity for the maximally mixed input is $F_{\mathrm{opt}} = D^{-2}\sum_h w_h d_h^2 = D^{-2}\sum_{x,y} H_{xy}$, with $H_{xy} = \sum_{h:\, x,y \in X_h} w_h$;

(iii) for the two-event observer, F_{opt} equals 5543/8712 *(binary) and* 1301/2028 *(full) under C, and* 769/1452 *and* 3329/6084 *under E with the same current-sign selector applied to its decoded current;*

(iv) the maximal exactly correctable algebra on the whole input is the commutant of $\{P_h\}$, and one event with full-support random seam choice reduces it to the diagonal load algebra; nevertheless the qubit code $|0_L\rangle = (|\{0,2\}\rangle + |\{1,3\}\rangle)/\sqrt{2}$, $|1_L\rangle = (|\{0,3\}\rangle + |\{1,2\}\rangle)/\sqrt{2}$ satisfies the Knill-Laflamme condition $QP_hQ = \alpha_h Q$ for every seam, coin and record of either law and is recovered exactly after one event;

(v) under independent seam attempts $H_{xy}(t) = q_{\mathsf{L}}(x-y)^t$, with largest off-diagonal rates 5/6 *for C and* 7/10 *for E on both carriers; $F_{\mathrm{opt}}(t) \to 1/D$, and neither retained instrument is entanglement breaking at any finite horizon;*

(vi) a fixed E seam word identifies every full-carrier input exactly when at most one visited component has size one or two, and every binary input exactly when at most one component is a singleton and at most one has size two; the minimal such words have eight and seven distinct edges;

(vii) no bounded record-only adaptive policy makes C *entanglement breaking; a* C *word that doubles a minimal forest, 16 or 14 events, excludes every nontrivial exact code; the policy that repeats each forest edge until its coin first differs identifies the input almost surely, with stopping time* $T = m + \sum_{i=1}^{m} G_i$ *for independent geometric* G_i *of success probability one half,* $\Pr(T = t) = \binom{t-m-1}{m-1} 2^{-(t-m)}$ *for* $t \geq 2m$*, and* $\mathbb{E}[T] = 3m$*, namely 24 on the full carrier and 21 on the binary shell.*

Proof sketch. Every retained current determines a translation, so reversing them gives the input uniquely from the terminal configuration and h. Optimality follows from Cauchy-Schwarz on each classical branch: $\sum_a |\mathrm{Tr}(R_{ha} U_h P_h)|^2 \leq d_h^2$. For (iv), commutation is sufficient by (i), and necessity follows from equality in the Schwarz inequalities in a Stinespring dilation; the code has nonadjacent occupied pairs, equal one-port means and vanishing products $N_a N_b$ on every edge, which gives the Knill-Laflamme identity [114]. For (vi), matching E histories are those for which z is constant on every visited component. For (vii), the inputs $\{u, w\}$ and $\{v, w\}$ with u, v nonadjacent keep equal currents for at least one coin on every edge, so a history of length at most T shares a rank-two fiber with probability at least 2^{-T}; repeating an edge with the opposite coin reveals the initial parity through $d = 2j_1 + j_*$ [54, 95]. □

The minimal fixed E words are the forests $(0,1),(0,5),(2,3),(2,4),(6,8),(8,9),(7,10),(10,11)$ on the full carrier and $(1,5),(2,3),(2,4),(6,8),(8,9),(7,10),(10,11)$ on the binary shell, with component sizes $(3,3,3,3)$ and $(1,2,3,3,3)$. The full-input correctable blocks of the two-event observer have sizes $46, 10, 9, 1$ on the binary shell and $56, 10, 9, 1, 1, 1$ on the full carrier, for both laws. Existing source inverse words implement the single-seam branch reversal of (i) while keeping the classical archive [54, 95]. Native current records thus suffice to reverse classical load transport, without restoring arbitrary coherence lost across record groups. The source encoder and active recovery of Section 6.5 instead preserve the stated logical qubit between events; they do not invert the whole bare-load channel.

3.7 A coherent continuation and its selection boundary

An exact finite experiment attaches a reflection walk to the binary native branches, with coherent event access in place of a committed tape [55]. For each binary configuration x, seam e and coin c, let (y, j) be the native output and current. On the $66 \cdot 60 = 3960$ arc basis vectors define

$$S|x,e,c\rangle = |y,e,c'\rangle, \qquad c' = \begin{cases} c, & j = 0, \\ 1-c, & j \neq 0, \end{cases} \qquad A|x\rangle = \frac{1}{\sqrt{60}} \sum_{e,c} |x,e,c\rangle.$$

Every arc is checked: $S^2 = I$, with 300 exchanged pairs and 3360 fixed arcs, and the outgoing native record (y, e, c, j) is recovered bijectively from the reversed arc. Thus $A^\dagger A = I$ and $A^\dagger S A = K$, the actual symmetric native transition kernel, and $W = S(2AA^\dagger - I)$ satisfies the hypotheses of the quantum-walk spectral mapping theorem with discriminant K [146, 188]. The identity $A^\dagger W^n A = T_n(K)$, with T_n the Chebyshev polynomial, gives for the centered loads $X_{x,p} = x_p - 1/6$

$$KX = X(I - L/60), \qquad U_{n+1} - 2U_n + U_{n-1} = -U_n L/30, \qquad U_n = A^\dagger W^n A X.$$

These are compressed amplitudes, and n counts the supplied coherent continuation with no identification as duration. The binary domain is essential: on the full carrier $(2, 0, \ldots) \mapsto (1, 1, \ldots)$ has no native reverse event, so its asymmetric kernel equals $A^\dagger S A$ for no involutive unitary on the same domain. Twelve port addresses and sixty seam/coin events give a 720-arc source-compiled walk

with discriminant $K_{12} = I - L/60$; the map $F_{x,p} = (x_p - 1/6)/\sqrt{10}$ satisfies $F^*F = I - \mathbf{1}\mathbf{1}^T/12$ and $K_{66}F = FK_{12}$, so it realizes the shared mean-zero density sector. Among the sixty proper carrier rotations a unique inverse-symmetric order-five class $\mathcal{C}$ of size twelve gives $L = 6I - \frac{1}{2}\sum_{g\in\mathcal{C}} U_g$, with multiplicity-free density representation $1 \oplus 3 \oplus 5 \oplus 3$. The other order-five class is isospectral but assigns probability zero to a native seam whose actual one-attempt probability is $1/60$, so the seam records select the class.

The emitted walk word uses ten address qubits, five blank scratch qubits and 114,910 source pulses per walk with 229,820 load/export events; uniform event preparation costs a separate 36,288 pulses. The reversible circuit is checked on all 32,768 basis inputs, every clean-scratch address returns clean scratch, including 304 padding addresses, and the fifteen-qubit execution reuses raw pulse factors without target-matrix insertion or output normalization. Floating routine guards sum to $s = 829/(5 \cdot 10^9)$, giving complete output trace-norm error at most $2\delta + \delta^2$ with $\delta \le s/(1-s)$; these are numerical guards, not physical precision certificates.

A finite reference and its modular clock. Biased native event amplitudes, the compiled reversible seam shift, inverse event preparation and a herald realize $F_p = I - pL/60$ on the protected site code. From the supplied $I_{12}/12$, n successive successes give $F_p^{2n}/\operatorname{Tr} F_p^{2n}$ with success probability at least $1/12$, every failure recorded; for $p = 30t/n \le 1$ this tends to $e^{-tL}/\operatorname{Tr} e^{-tL}$ with relative-entropy and total-variation errors at most $16t^2/[n(1-4t/n)]$. The execution at $p = 1/2$, $n = 2$, $t = 1/30$ tests all twelve input columns with success $4382551/5184000$ and bound $1/105$, and a source interference readout has contrast $260343/43825510$ against zero after dephasing. This state is central on the 44-dimensional represented group algebra, where its modular flow is trivial; nontrivial modular gaps require the full protected matrix algebra [21].

Positive subordination gives square-root weights on the same register. With $F = I - hL/2$, $h = p/30$, a Poisson number of Sibuya jumps gives positive weights with

$$\sum_n w_n F^{2n} = e^{-R\sqrt{I-F^2}} = e^{-b\sqrt{L-hL^2/4}}, \qquad b = R\sqrt{h}.$$

Finite jump and count caps and dyadic rounding retain missing mass as sampler rejection. The source instance $p = R = 1/2$ uses 64 native coin records and at most four raw filters, retains each first failure and all sixteen successful interference outcomes, and has Gibbs trace-distance bound below 0.217 and spectral diagnostic 0.01344. Its finite polynomial $M = \sum_n w_n F^{2n}$ fixes $H = -\log(M/m_0)$, $m_0 = \sum_n w_n$. Raw source words execute that clock, a current-controlled intersector kick and complete spectral-pair detectors on the same register, retaining all 3,600 two-attempt native tapes, sampler and herald failures, sixteen detector addresses and twelve reference columns. The modular identity $\sigma_s = \alpha_{-s}$ holds for the ideal successful marginal $M/\operatorname{Tr} M$ on $\mathcal{B}(\mathbb{C}^{12})$; it does not hold for every retained request block.

This density supplies eleven pairs $Q_j = X_{0j}/\sqrt{2\Delta_j}$, $P_j = Y_{0j}/\sqrt{2\Delta_j}$, $\Delta_j = p_0 - p_j > 0$, with $\langle [Q_j, P_k] \rangle = i\delta_{jk}$, symmetric covariance $\delta_{jk}\coth(g_j/2)/2$, $g_j = \log(p_0/p_j)$, and frequency g_j under H. For chronological factors $U_N = e^{iA_m/\sqrt{N}} \cdots e^{iA_1/\sqrt{N}}$ built from these pair observables on independent identical copies, a parity commuting with the state and anticommuting with every A_k gives

$$\left| z_N^N - e^{-C} \right| \le \frac{D^4/24 + |C|^2/2}{N}, \qquad z_N = \operatorname{Tr}(\rho U_N),\ D = \sum_k \|A_k\|,$$

with $C = \frac{1}{2}\sum_k \langle A_k^2 \rangle + \sum_{k>j} \langle A_k A_j \rangle$. Source pointer reads execute the one-copy factors for three band representatives with all 32 work/pointer addresses retained; a growing copy register is not executed. With physical normalization $q_j = Q_j/\sqrt{\omega_j}$, $\pi_j = \sqrt{\omega_j} P_j$ and $g_j = b\omega_j$, the clock H/b

gives thermal covariance $\coth(b\omega_j/2)\cos(\omega_j t)/(2\omega_j) - i\sin(\omega_j t)/(2\omega_j)$. For m smeared factors with $R = \sum_k R_k$, $R_k^2 = \sum_j (f_{kj}^2/\omega_j + \omega_j h_{kj}^2)$, gaps $g_j, b\omega_j \geq \gamma > 0$, $\vartheta = \gamma/(1+\gamma)$ and $Z = 1/p_0$,

$$|z_N^N - e^{-C}| \leq \frac{R^4}{\vartheta^2 N}\left(\frac{(m+1)Z}{96} + \frac{1}{8}\right),$$

so $Z/N \to 0$ suffices on a fixed positive band; the family $\rho = \mathrm{diag}(2,1,\ldots,1)/(d{+}1)$ with $N = d$ has instead the limit $e^{3(\cos(1/\sqrt{2})-1)} \neq e^{-3/4}$. On the native family $D_l = 4^l L_l$ one has $\lambda_n(D_l) \geq n/81940$, hence $\mathrm{Tr}\, e^{-b\sqrt{D_l}} \leq 1 + 163880/b^2$, and with the lazy filter $p = 630h16^l \leq 1$,

$$Z \leq \frac{1 + 327760/b^2}{m_0}, \qquad P_{\mathrm{success}} \geq \frac{m_0}{240\,4^l};$$

both prepared and target gaps exceed $b/182$ when $(1-m_0)/m_0 \leq b/182$. Preparation error E with success floor $p > 2E$ gives conditional trace error at most $4E/p$, and with one-copy word error ϵ the N-copy amplification is at most $N\zeta/(1-N\zeta)$, $\zeta = 4E/p + \epsilon$. On a fixed band $0 \leq D_0 \leq \Lambda$ with $h\Lambda < 2$, retained positive weights with missing mass $\delta < a \leq e^{-b\sqrt{\Lambda}}$ give

$$\|\log(m_0)I - \log M - b\sqrt{D_0}\| \leq \frac{\delta}{a-\delta} + \frac{bh\Lambda^{3/2}}{4(1+\sqrt{1-h\Lambda/4})}.$$

A constructive growing word family sets $k = 2l + c$ with $4^c \geq b^2\max(630,\Lambda)$, $R = 2^k$, $h = b^2 4^{-k}$, $a = 4^{-k}$, jump cap 2^{6k}, count cap $6R + 2k$ and Taylor order $4k+3$; the zero-request branch keeps weight at least $f = 2^{-2R}$. For $d = 240\,4^l$, request cap K and word length P, primitive and load/export error at most $fa/(64dKP)$ gives complete stopped preparation error $E \leq fa/(32d)$ and

$$\|dS_{\mathrm{raw}} - M\| \leq 2dE, \qquad \|\log(dS_{\mathrm{raw}}) - \log M\| \leq \frac{a}{16-a}$$

for the actual successful marginal S_{raw}, without a reducing-band premise. The derived partition bound permits $N = 2^l$ copies with copy, preparation and detector errors tending to zero on a fixed positive band. Growing execution, spectral tails, accurate source services and a zero-energy anchor for the radial product remain obligations, and fixed caps with $N_c h \to 0$ force the bounded-band generator to zero; the uncapped sampler has infinite expected work because the constant sector survives every filter [21].

Selection controls. Two exact controls bound what the committed records select. For $|z| = 1$, $W_z = S[AA^\dagger + z(I - AA^\dagger)]$ has the same first event $W_z A = SA$ but $A^\dagger W_z^2 A = K^2 + z(I - K^2)$; a complete added Ramsey read gives two-step plus probability 1 for $z = 1$ and $113/150$ for $z = -1$. Multiplying each coherent record branch on seam $(0,1)$ with current $+1$ by -1 preserves every committed C and E branch on arbitrary references, yet the twisted embedding gives $\mathrm{Tr}[(A_\eta^\dagger S A_\eta)^3] - \mathrm{Tr}(K^3) = -1/900$. Both real continuations conserve load, respect all sixty proper rotations, fix the uniform coherent state and admit the time reversal $\Theta = S$ conjugation; under full event-permutation covariance, real orthogonal coins fixing the uniform vector are identity or reflection. A sign change on every moving arc preserves proper covariance and the committed instruments but gives $\mathrm{Tr}(D^3) - \mathrm{Tr}(K^3) = -1/90$ for $D = 2\,\mathrm{diag}(K) - K$. Scalar record phases cancel throughout committed-record histories. A source-executed interface test that reads the low unread record bit in the X basis gives complete X-plus probabilities $3/4$ and $1/4$ for one preparation, against $1/2$ for both after commitment, using 5380 C or 5383 E source pulses per attempt [56]. These are restrictions on selection: coherent access is a supplied interface, and the test selects no phase.

3.8 The local event clock

N4 is a reciprocal extension of the binary repair law in the construction's candidate register [46]. Each port carries a private six-level response factor, and one coherent exchange $W_e = \exp[\theta(Y \otimes \Gamma_e + X \otimes \Delta_e)]$, with $X = S_e(N_a - N_b)^2$, $Y = iS_e(N_a - N_b)$ and response generators Γ_e, Δ_e built from the endpoint response matrices, conjugates the binary repair, $K_r = W_e(L_r \otimes I)W_e^\dagger$, with supplied coupling θ. Private preparations change bare load probabilities: at fixed bare load and $\theta = 1/8$ an endpoint occupation probability moves from 0.482020673 to 0.517979327. An interval certificate proves an edge-Gram margin $\mathcal{G} \geq mI$ at $\theta = 1/16$ and $\theta = 1/8$ [46, 71, 2].

Fix an observer port v. A tick is any attempted seam incident to v, including attempts that leave its load unchanged. Under independent uniform seam selection a tick has probability $p = 1/6$ per attempt, and remote attempts continue between ticks. In independent trials the full tracial state is prepared, the binary occupation N_v is measured, the unmodified source runs until one or two ticks have occurred, and N_v is measured again, with no intermediate measurement, reset or postselection. With $f = N_v - 1/6$ and $\|f\|_\tau^2 = 5/36$, write $c_k = \mathbb{E}[(X_0 - \frac{1}{6})(X_k - \frac{1}{6})]/(5/36)$ for the endpoint correlation after k ticks [71, 72]. The numbers of Theorem 3.7(iii) are certified lower bounds; the full N4 two-tick correlation is not evaluated, and preparation, incident-event detection and bare-occupation readout are supplied operations that the detector of Section 3.1 does not implement.

Theorem 3.7 (Memory at observer-local ticks)**.** *Let $\mathcal{A}$ and $\mathcal{B}$ be the averages over the thirty seams of the remote and incident seam channels, so that the first-hit channel is $\mathcal{K} = \mathcal{B}(I - \mathcal{A})^{-1}$. Then:*

(i) $c_k = \langle f, (\mathcal{K}^*)^k f\rangle / \|f\|^2$, *and* $c_1 = 1 - 6\alpha_{\mathrm{occ}}/11$ *for the occupation compression factor* α_{occ};

(ii) the statistic

$$b_{\mathrm{tick}} = c_2 - c_1^2 - \frac{r^2(209r + 180)}{1331r^2 + 3300r + 1800}, \qquad r = 1 - c_1,$$

vanishes for the native law and for the one-term exchange controls, which have $c_1 = 5/11$ *and* $c_2 = 93/407$;

(iii) for both N4 completions, built on C *and on* E, $b_{\mathrm{tick}} \geq 3m/50$, *and the certified lower bounds are*

$$b_{\mathrm{tick}} \geq \frac{139996257}{12500000000000} \approx 0.00001119970056 \quad (\theta = 1/16),$$
$$b_{\mathrm{tick}} \geq \frac{11335290321}{50000000000000} \approx 0.00022670580642 \quad (\theta = 1/8);$$

(iv) $c_{2,\mathsf{C}} \geq c_{2,\mathsf{E}}$ *and* $b_{\mathrm{tick},\mathsf{C}} \geq b_{\mathrm{tick},\mathsf{E}}$.

Proof sketch. Each seam channel is a positive semidefinite contraction of Hilbert-Schmidt space, with spectrum in $\{0, 1/2, 1\}$ for C and $\{0, 1\}$ for E; N4 conjugation preserves this. Hence $0 \leq \mathcal{A} \leq (1-p)I$, the first-hit series converges, and the stopped maps are unital and trace preserving. Remote seams fix f, which gives (i). The closed comparator with seam mean maps $I - \frac{\alpha_{\mathrm{occ}}}{2} w_e w_e^\top$ yields the correction term in (ii) after substituting $\alpha_{\mathrm{occ}} = 11(1-c_1)/6$. With the Schur complement S of $I - \mathcal{A}$ on the centered occupations, $S \leq I$ and $b_{\mathrm{tick}} = \langle z, S^{-1}z\rangle/(p\|f\|^2) \geq \|z\|^2/(p\|f\|^2)$. The hidden defect is $z = -\frac{1}{60}\sum_e t_e S_e$ with five incident coefficients ± 1, so $\|z\|^2 \geq 5m/3600$ and $b_{\mathrm{tick}} \geq 3m/50$. On each seam $\mathsf{C} - \mathsf{E}$ is positive semidefinite, which gives (iv) [71]. □

The correction used for global attempts, $(1 - c_1)^2/10$, assigns $-16/2035$ to the native null at local ticks; the correction of Theorem 3.7(ii) is the one adapted to this stopping instrument. The

receipt and eight independent checks bind the result, including a comparison with the full 66-state native stopping kernel and a complete dimension-six quantum model of the resolvent and Schur identities [71, 97].

4 Observers, clocks and screens on one process

This section completes layer three on one native process. Its components run the native C or E instrument itself, with the uniform independent seam schedule and fair placement coin as declared scheduling inputs. Their time parameters are attempt counts and local event counts. Each result states its preparation, observer access and services; none identifies a count with laboratory duration [15].

4.1 Two overlapping observers

On the 66 binary total-two configurations, prepare the uniform state and disclose the bits at ports $(0, 1, 5)$; all eight labels stay in the output menu. Observer A accesses $(0, 1)$ and observer B accesses $(0, 5)$, so the regions overlap at port 0. Each receives its initial bits and every incident local current, including zero, and ticks when an attempted seam meets its region. The protocol requests two incidents on the union with at most two native attempts per interval; after the first it applies one native repair on $(0, 1)$ if the reconstructed n_0 is one and on $(0, 5)$ otherwise, and that repair ticks both observers.

Proposition 4.1 (Exact overlapping-observer correspondence)**.** *Normalize the native remote and incident branch counts by* 60*, write $P = A + B$ with $B = \sum_m B_m$ over incident marks m, and put $D = I - A$, $p = 2/5$ and $G = D/p$. With $K_m = B_m D^{-1}$ and $F_m = D^{-1} B_m$,*

(i) $K_m = G F_m G^{-1}$ and $\mathbf{1}^{\mathsf{T}} G = \mathbf{1}^{\mathsf{T}}$; remote projectors, the selective feedback maps and the disclosed uniform preparation commute with or are fixed by G;

(ii) for attempt cap h the marked and timeout maps are $K_{m,h} = B_m(I + A + \cdots + A^{h-1})$ and $Q_h = A^h$, their effective conjugates are $F_{m,h} = G^{-1} K_{m,h} G$ and A^h, and the terminal decoder G returns the complete unnormalized source state on every output word of every finite adaptive protocol in this commuting intervention family;

(iii) the direct endpoint execution and the effective recursion agree exactly on all 5,768 *output words,* 689 *of positive probability, including all terminal populations, and the timeout probability is* $369/625$*; and*

(iv) the unread completed transfer $F = D^{-1} B$ is self-adjoint in the metric G with $1/20 \le F \le 1$, so $H = -\log F$ is an effective Euclidean generator per completed union incident tick.

Proof sketch. Remote repairs leave the union fields fixed and every remote seam is disjoint from the feedback ports, so the conjugation by G cancels between successive reads and allowed feedback maps; induction over the protocol gives (ii). Every native seam count equals $I + \mathrm{SWAP}$, which makes $GF = B/p$ symmetric positive semidefinite, and the native triangle $(0, 1, 5)$ contributes at least $3I/60$ to B [16]. □

At every positive-weight prefix each observer's decoder, which sees only its local initial pair and incident increments, returns fields that obey native continuity, and the two port-zero values agree. The transformed finite-cap maps preserve the transported cone $G^{-1}\mathbb{R}^{66}_+$; they are not entrywise-positive stochastic maps, and the comparison is of decoded full states. The quantum extension

checks the native remote resolvent on all 4,356 matrix units with coherent preparations, untouched references and complete local clock records, and an ordinary completely positive effective instrument for local outputs is a positive mixture of native repairs; its one-particle coherent experiment retains all 81 outputs [17]. The compiled clock history replaces the ideal preparation, intervention and pointer of that experiment by words of the six-level source compiler. Exact rational propagation of the binary source arrays keeps all 512 code addresses and 1,656 clock, timeout and complement-abort outputs within 10^{-11} of the ideal history, at most 51 pulses (54 with the compiled Bell preparation), two code services per pulse and five native repair attempts [18]. The native instrument, the scheduler and the code services are specified inputs.

4.2 Joint and active event clocks

A local event clock ticks on actual occupation changes at an observer port, or on attempted repairs incident to a region, while remote repairs continue; its stopped instrument sums the unresolved wait and retains finite timeouts and conditional quantum outputs. Two such clocks on ports 0 and 72 of joined carriers read one continuing history with an observer-chosen field algebra. Exact elimination of the hidden factor gives a record-conditioned quantum memory recurrence valid for arbitrary untouched references. At completed events the experiment admits a four-mark measure-and-prepare description, while unresolved waits retain coherence that the completed instrument erases. C and E share the completed clock law and predict field-plus probabilities 35/54 and 14/27 after three native attempts. Without feedback each two-tick cycle of clock A carries B-count mean two and variance 16; instrument-valued clock conversion preserves that count law, and direct native elimination and elimination through the joint event instrument agree, including B-triggered field feedback [19].

An active event clock lets each native event incident to port 0, including a zero-current one, trigger one fixed compiled modular word $U(1) = \exp(iH)$ generated by the native three-state attempt transfer; the carrier pauses during probe services, which are charged separately. For caps $c \in \{1, 2\}$ the execution retains all 3,661 joint output labels, the first timeout has probability $(5/6)^c$ and both ticks occur with probability $[1 - (5/6)^c]^2$. Closed native histories derive a faithful thermal reference for that generator, and passive repair itself cannot act as a nontrivial modular automorphism on an invariant observer algebra [20]. On the complete movement-bit tape $B_t = |j_t|$, including zero currents, four consecutive covariances $c_n = \mathbb{E}[B_t B_{t+n+1}] - p^2$ invert to the two transfer roots $\lambda_\pm = (25 \pm \sqrt{3})/30$ with $p = 5/33$, and the same roots predict the coherent quadratures $q_X = (1 + \cos(sE_j))/2$ and $q_Y = (1 + \sin(sE_j))/2$, $E_\pm = -\log \lambda_\pm$, with no frequency fit. The one-exponential approximation matching c_0, c_1 misses c_2 by exactly $1/2673000$. The inference does not distinguish C from E, whose classical tapes agree [21].

4.3 Identification of the record law

Nine source-executed scalar measurements identify all nine real coordinates (k, g, h, m, n) of the record-law table of Theorem 2.3, and with them the coarse definite-current instrument on all 78 total-two states with arbitrary spectator and reference coherences. The execution comprises 36 compiled probes and ten complete two-event histories, five with untouched quantum references, run through the canonical observer runtime with serialized checkpoint validation and continuation. An admissible collision law agrees with E on the whole binary shell, and this experiment separates it. A reference-uniform bound carries measurement uncertainty through the finite feedback histories, so narrowing the admissible probability tubes narrows the complete-history bound [22]. The probes are synthetic; they identify an instrument within the classified family and select no law for nature.

4.4 Federations and matched screens

Federations join 20, 80 or 320 identical twelve-port carriers through the declared seam wiring, and native Kraus maps preserve signed-difference operator blocks with individual outcomes and reference correlations. The endpoint instruments fix the local Liouville action $H_{\mathrm{native}} = \sum_e (I - \Phi_e)$, with $h_e = P_1 + \frac{1}{2}P_{1/2}$ for C and $h_e = P_1$ for E on orthogonal projectors; the discrete uniform transfer is $I - H_{\mathrm{native}}/M$ for M seams. For an observed region R, maximum degree Δ and retained radius r, truncating distant ports changes the complete local record, quantum and reference output by at most $\min(1, |R|(\Delta s)^{r+1}/(r+1)!)$ at auxiliary per-seam Poisson parameter s [23]. Actual source neighborhoods are identified at levels 1/3, 2/4 and 3/5. On the 15,360- and 245,760-port screens matched radius-nine initial restrictions give complete root-observer output distance below 0.000382 at $s = 1/4$; after four local incident ticks, with no auxiliary time coordinate, the pair allowance is below 6.656×10^{-9}, and a density-1/3 preparation keeps a certified two-time field covariance above 0.15332581621. Full-screen executions retain the actual seam scheduler, current records, feedback repair costs, timeouts and serialized continuation. Conditional on a conserved total, native connected exchange derives the unique diagonal equilibrium, so the screens at totals 5,120 and 81,920 keep density 1/3, the feedback reversal of the connected four-field statistic and their finite-number correlations without product preparation [24].

A coherent observer runs on the same matched screens. A logical qubit on ports 0 and 1, possibly entangled with an untouched reference, passes through two native incident ticks and a local phase Z_a after first mark 1, while every other port is occupied with probability 1/3. A degree-two operator dual evaluates the common 222-port ball with 561 seams and all 21 capped record labels, and each full-screen completed output lies within 7.779×10^{-10} of it, uniformly over references. C and E give identical classical records in this protocol and different reference coherences [25]. The native continuum operator of the refining screens and its density, wave and response limits are stated in Section 7.

5 Field detectors on supplied actions

Layer four begins with detectors that couple native records to fields whose actions are supplied. They provide complete-output and refinement machinery that a source-derived field description must reuse; they derive no field.

5.1 The scalar detector and its refinement transfer

The scalar detector joins the native observer to the free-scalar continuum family of the theorem library [61, 62]. The scalar action, quantization, vacuum, preparation, Born readout and model clock are supplied. The field is the massive scalar on the golden tensor grids $q \in \{13, 34, 89, 233\}$, with Dirichlet boundary and mass one; after mass whitening its one-particle operator is $\mathsf{A} = I + B \otimes I \otimes I + I \otimes B \otimes I + I \otimes I \otimes B$, with $\Omega = \sqrt{\mathsf{A}}$ and $H = d\Gamma(\Omega)$. A preparation profile f and a detector profile g lie in compact left and right slabs.

One trial runs as follows. (1) Prepare one of the 66 binary configurations uniformly, repair $(0, 1)$ and retain (c_1, j). (2) Prepare the field vacuum and apply the momentum displacement of amplitude $\gamma j f$, $\gamma = 4$. (3) Evolve to model time t and read the centered binary Weyl instrument

$$K_y = \tfrac{1}{2}[e^{-iX/2} - iy\, e^{iX/2}], \qquad X = a^\dagger(v) + a(v), \qquad v = \Omega^{-1/2} g/\sqrt{2},$$

with $K_y^\dagger K_y = \frac{1}{2}[I + y \sin X]$. (4) Commit y, repair $(0, 5)$ for $y = +1$ and $(0, 1)$ for $y = -1$, and halt. Model times lie in $[19/20, 1]$, with primary setting 39/40, and carry no laboratory calibration.

Proposition 5.1 (Scalar feedback and refinement transfer)**.** *For the process above at model time t, let $r(t) = \Pr(y = +1 \mid j = 1) - \frac{1}{2}$. Then:*

(i) the branch operators $M_{(r,y,s)} = (L_s L_r) \otimes (K_y e^{-iHt} V_j)$, with V_j the controlled displacement, form one CPTP instrument on arbitrary joint source and field inputs, and each branch retains its conditional field state as a superposition of two coherent components with their relative phase;

(ii) $p(y \mid j) = \frac{1}{2} + yj\, r(t)$ with $r(t) = \frac{1}{2} e^{-\|v\|^2/2} \sin \mu(t)$ and $\mu(t) = \gamma \langle g, \Omega^{-1} \sin(\Omega t) f \rangle$;

(iii) over the 528 histories, $\mathbb{E}[j_2] = 5r/132$, $\mathbb{E}[N_{0,\mathrm{final}}] = \frac{1}{6} - 5r/132$, $\mathbb{E}[yj] = 10r/33$ and $\Pr(y = \pm 1) = \frac{1}{2}$; disconnecting the preparation or replacing y by an independent fair bit gives $\mathbb{E}[j_2] = 0$ with the same seam frequencies, and reversing the kick reverses the sign;

(iv) if $|r_q - r_\infty| \le \varepsilon_q$ on the window, then the complete labelled histories satisfy the total-variation distance $\mathrm{TV} = \frac{5}{33}|r_q - r_\infty| \le 5\varepsilon_q/33$, and $|\Delta \mathbb{E}[j_2]| \le 5\varepsilon_q/132$.

Proof sketch. The Weyl operators $e^{\pm iX/2}$ are displacements by $\pm iv/2$, which gives the branch states and (ii). Conditioned on the initial configuration and first coin, the two detector laws differ in total variation by $|j|\,|r_q - r_\infty|$, and the subsequent native update is common [62]. □

At $q = 233$ the parent certifies, uniformly for $19/20 \le t \le 1$,

$$\varepsilon \le \frac{12513633763}{250000000000}, \qquad -\frac{111322580247}{500000000000} \le r \le -\frac{34438518047}{250000000000},$$

so the finite-grid mean second current satisfies

$$-\frac{37107526749}{4400000000000} \le \mathbb{E}[j_2] \le -\frac{34438518047}{6600000000000},$$

negative throughout the window, with the continuum mean within $\frac{5}{132}\varepsilon \approx 0.001896$ [98, 63]. At $t = 39/40$ the plus probabilities for $j = +1$ are about 0.334515, 0.319421, 0.317093 and 0.316771 on $q = 13, 34, 89, 233$, and the finest-grid prediction is $\mathbb{E}[j_2] \approx -0.00694050$. A seeded execution with 200,000 fresh trials per grid tests Born outcomes against predictions computed before sampling, with reversal, donor and fixed-seam controls; at $q = 233$ all 12,487,168 field modes contribute. An independent diagonalization gives largest relative frequency errors 2.137%, 0.2875% and 0.04116% at $q = 13$, 34 and 89 against the supplied dispersion $\omega^2 = 1 + \pi^2(n_x^2 + n_y^2 + n_z^2)$ [90]. A local kick-drift-kick extension through the six-neighbour stencil runs at $q = 13$ and 34 and transfers the combined spatial and temporal bound through the same factors 5/33 and 5/132; at $q = 233$ the temporal term is at most 0.000008071512 and the combined probability error at most 0.050062606564 [61].

5.2 The retained two-read continuation

The continued observer keeps the disturbed field after its first read [64]. The field is read at model times 19/20 and 1; the first bit y_1 selects the second native seam, the conditional field evolves to time 1, the second bit y_2 selects the third seam, and the observer halts with five records readable. This continuation uses exact continuous spectral evolution and carries its own error bounds.

Proposition 5.2 (Two reads of one field)**.** *In the continued protocol:*

(i) all 2,112 labelled histories are retained; the record-prefix counts at the six phases are $1, 4, 8, 24, 48, 124$, so the 209 reached prefixes have an eight-bit encoding and the 124 terminal words need seven bits;

(ii) with the Gram kernel G_{ab} of the two detector vectors, $V = G_{11} = G_{22}$, $C = \operatorname{Re} G_{12}$, $c = 2 \operatorname{Im} G_{12}$ (so that $[X_1, X_2] = ic$), $S_a(j) = e^{-V/2} \sin(j\mu_a)$, $D = \cos(c/2)$ and $R_j = \frac{1}{2}[e^{-(V-C)} \cos(j(\mu_1 - \mu_2)) - e^{-(V+C)} \cos(j(\mu_1 + \mu_2))]$, the joint law is

$$p_j(y_1, y_2) = \tfrac{1}{4}[1 + y_1 S_1(j) + y_2 D\, S_2(j) + y_1 y_2 R_j];$$

(iii) on the positive-current preparation, performing the first read changes the second plus probability by $(-r(t_2))[1 - \cos(c/2)]$, certified greater than 0.000022281162 at $q = 233$ and greater than 0.0000252366 in the declared compact continuum; a commuting Gaussian control with the same means and real covariance has $D = 1$ and no such disturbance; the zero-current vacuum correlation exceeds 0.045619388905 at $q = 233$;

(iv) the total variation distance of the complete labelled history from the declared continuum is at most 1, 0.451806, 0.052189606637 and 0.025709913643 on $q = 13, 34, 89, 233$, including the zero-current histories.

Proof sketch. A state is stored as $\sum_n c_n D(i \sum_a h_{na} v_a)|\alpha\rangle$ with exact half-integer shifts, its Gram matrix is explicit in G, and both terms of each Kraus operator are applied before the branch norm is taken. For four atoms with Walsh-moment differences a, b, d, $\mathrm{TV} = \frac{1}{2} \max(|a|, |b|, |d|, \frac{1}{2}(|a| + |b| + |d|))$, bounded in directed rational arithmetic [64]. □

At $q = 233$ the execution gives $\mathbb{E}[j_3] \approx -0.00130126$ and $\mathbb{E}[y_1 y_2 \mid j = 0] \approx 0.0514838$; deleting the first field record from the forecast raises its Brier risk by about 0.00064677. The 50,000-trial runs per grid do not resolve the small disturbance statistically.

5.3 The source-compiled two-site apparatus

The two-site apparatus executes preparation, evolution and pointer as emitted twelve-port control words [65, 66]. Its action lies outside the golden refinement family, so its occupation limit carries no spatial refinement. The Hamiltonian is

$$H = \tfrac{1}{2}(P_0^2 + P_1^2 + Q_0^2 + Q_1^2) + \tfrac{\kappa}{2}(Q_0 - Q_1)^2, \qquad \kappa = \tfrac{9}{32},$$

with normal-mode frequencies 1 and 5/4 and $d \in \{4, 8, 16\}$ number states per site. The executed Hamiltonian uses the compressed squares $Q_d^2 + \frac{d}{2}|d-1\rangle\langle d-1|$ and $P_d^2 + \frac{d}{2}|d-1\rangle\langle d-1|$, so that $H_d = \Pi_d H \Pi_d$. The prepared state is the normalized projection of the coupled vacuum onto total occupation at most $d-2$; at $d = 16$ its binary amplitude tree costs 5,379 native pulses. The current impulse is $e^{ijQ_{0,d}}$, 64 or 128 symmetric split steps reach $t = 1$, and the pointer coupled by $e^{-iZQ_{1,d}/2}$ and read in the Y basis gives $\Pr(y = +1 \mid j) = \frac{1}{2} + j r_d$ and $\mathbb{E}[j_2] = 5 r_d / 132$.

Theorem 5.3 (Complete ideal error budget of the two-site instrument)**.** *For the declared two-site action, model time $0 \le t \le 1$ and native current $j \in \{-1, 0, 1\}$:*

(i) with A the onsite and B the bond part of H_d and $S(h) = e^{-ihA/2} e^{-ihB} e^{-ihA/2}$,

$$\|S(t/n)^n - e^{-it(A+B)}\| \le \frac{t^3}{n^2}\Big(\frac{\|[A,[A,B]]\|}{24} + \frac{\|[B,[A,B]]\|}{12}\Big),$$

uniformly over inputs and references;

(ii) the prepared state differs from the infinite coupled vacuum in trace distance by less than 0.007576, 0.000079899 and 1.0324×10^{-8} at $d = 4, 8, 16$;

(iii) the trace distance between the complete ideal finite instrument, including its classical outcome and unnormalized terminal field branches, and the infinite two-oscillator instrument is below 0.799447, 0.077171 *and* 0.000196 *at* $d = 4, 8, 16$, *and the bound tends to zero as* d *grows;*

(iv) at $d = 16$ *and* $n = 128$ *the combined occupation and splitting error is below* 0.001464436, *and the ideal source-word instrument satisfies* $0.514957066204 \leq \Pr(y = +1 \mid j = 1) \leq 0.517885937293$, *an interval that excludes* $1/2$.

Proof sketch. Item (i) is a defect bound for the symmetric product formula [120] with nested commutators bounded by exact rational Frobenius norms. The infinite state is a displaced squeezed vacuum throughout; the Gaussian generating function bounds the number tails along impulse, evolution and both pointer arms, and projected Duhamel comparison with the orthogonal leakage gives (iii) without dividing by a Born probability. Item (iv) adds a rational enclosure of the infinite reference probability [66, 99]. □

The executed plus probabilities for a positive unit current, with their rational splitting bounds, are 0.516680958 (0.000103957) and 0.516679563 (0.000025990) at $d = 4$ with 64 and 128 steps, 0.516424174 (0.000693013) and 0.516422803 (0.000173254) at $d = 8$, and 0.516423330 (0.005076875) and 0.516421959 (0.001269219) at $d = 16$. The infinite reference has detector variance $9/20$ and mean $\mu(t) = \frac{1}{2}(\sin t - \frac{4}{5}\sin(5t/4))$, so $\Pr(y = +1 \mid j) = \frac{1}{2}(1 + e^{-9/40}\sin(j\mu(t)))$, equal to 0.516421501748 at $t = 1$ and $j = 1$ [65]. The $d = 16$, 128-step protocol uses nine work qubits, 18,612,125 native pulses and 37,224,250 proper-code load and export events, and retained 200,000-trial runs record the reversal, disconnected and donor controls.

6 The common interacting source

6.1 Construction

The common source places quantum electric flux, a charged fermion pair and a neutral scalar on one evolving work register. The sharp protocol measures it twice and feeds its records into the register and the declared native integer primitive [73, 74]; its four-read smooth extension joins compiled native quantum channels and encoded recovery (Section 6.5). The action, preparation, coupling and model-coordinate schedule are construction inputs.

Order four fermion modes as L_0, R_0, L_1, R_1. The two channels carry charges $q_0 = +1$ and $q_1 = -1$ and one fermion each; $b_j = 0$ means left and $b_j = 1$ right occupation of channel j. The electric link has basis $|\ell\rangle$, $\ell \in \mathbb{Z}$, with $\mathcal{E}|\ell\rangle = \ell|\ell\rangle$ and $U|\ell\rangle = |\ell + 1\rangle$. The isometry

$$W|b_0, b_1\rangle = |\text{modes } b_0 \text{ and } 2 + b_1 \text{ occupied}\rangle \otimes |-b_0 + b_1\rangle$$

maps onto four orthonormal occupation and flux vectors.

Proposition 6.1 (Flux-dressed matter code)**.** *On the image of* W*:*

(i) both operator Gauss constraints hold, $G_L W = G_R W = 0$ *with* $G_L = \mathcal{E} - Q_L$ *and* $G_R = -\mathcal{E} - Q_R$, *the supported flux set is* $\{-1, 0, 1\}$ *inside the infinite link, and fermion parity is even;*

(ii) the dressed hopping terms $h_j = c_{Lj}^\dagger U^{q_j} c_{Rj} + c_{Rj}^\dagger U^{-q_j} c_{Lj}$ *preserve both ambient Gauss values and the code, with* $W^*(h_0 + h_1)W = X_0 + X_1 = H_{\text{hop}}$ *and* $W^*\mathcal{E}^2 W = \frac{1}{2}(I - Z_0 Z_1)$, *and* $\|[H_{\text{hop}}, \mathcal{E}^2]\| = 2$*;*

(iii) with $\theta = \operatorname{atan2}(4,3)$, $u = (3I - 4iX)/5$, $U_\theta = u \otimes u = e^{-i\theta H_{\mathrm{hop}}}$ *and* $V_\theta = e^{-i\theta\mathcal{E}^2} = \operatorname{diag}(1, z, z, 1)$, $z = (3 - 4i)/5$, *the echo* $U_\theta^* V_\theta U_\theta$ *applied to* $|00\rangle$ *gives right occupation of channel* 0 *with probability exactly* 576/3125, *and without* V_θ *the probability is zero.*

Proof sketch. On every image column $Q_L = -b_0 + b_1 = \mathcal{E}$ and $Q_R = -\mathcal{E}$. The four-mode algebra of canonical anticommutation relations (CAR) is checked on all 16 occupation states with independent Jordan-Wigner signs, and all entries of the echo unitaries are Gaussian rationals [74]. □

The echo runs on the source controls with 22 native pulses and 44 coherent service events. Two neutral scalar sites join the same register, and with onsite quadratics compressed from the infinite oscillator the declared dimensionless action is

$$\begin{aligned} H &= H_{\mathrm{hop}} + \mathcal{E}^2 + H_{\mathrm{scalar}} + H_{\mathrm{int}}, \\ H_{\mathrm{scalar}} &= \sum_{j=0,1} \tfrac{1}{2}[P_j^2 + (1+\kappa)Q_j^2] - \kappa Q_0 Q_1, \qquad H_{\mathrm{int}} = g\sqrt{2}\, Q_0(n_{L0} - n_{R0}), \end{aligned} \tag{3}$$

with $\kappa = 9/32$ and $g = 1/4$. The interaction commutes with both Gauss generators and with fermion parity and couples the sectors, $[H_{\mathrm{int}}, H_{\mathrm{hop}}] = 2ig\sqrt{2}\, Q_0 Y_0$. The experiment proceeds in four steps. (1) Native repair on $(0,1)$ records a current j, which controls the gauge pulse $V_j = e^{-ij\theta X_0}$. (2) The register evolves for interval 1 (U_1), and a source pointer commits the right occupation a of the first charged species (P_a). (3) Only a selects the hopping pulse $F_1 = (3I - 4iX_0)/5$, with $F_0 = I$, and the register evolves for interval 1/2 (U_2). (4) A second pointer commits the sign b of Q_0 (R_b), and only b selects the second native seam, $(0,5)$ for $b = 1$ and $(0,1)$ otherwise.

Proposition 6.2 (One joint instrument with retained histories)**.** *The operators* $K_{(a,b)} = R_b U_2 F_a P_a U_1$ *satisfy* $\sum_{a,b} K_{(a,b)}^\dagger K_{(a,b)} = I$, *and* $M_{(r,a,b,s)} = (L_s^{e(b)} L_r^{(0,1)}) \otimes (K_{(a,b)} V_j)$ *form one CPTP instrument on arbitrary joint native and quantum inputs. Under the uniform binary preparation and the declared quantum vector* ψ, *the* $66 \cdot 2 \cdot 4 \cdot 2 = 1{,}056$ *histories have probabilities* $p(a,b \mid j)/264$ *with* $p(a,b \mid j) = \|K_{(a,b)} V_j \psi\|^2$, *each bound to its terminal density* $|v_{j,a,b}\rangle\langle v_{j,a,b}|/264$, *and the first native record keeps the masses* 14/33, 14/33, 5/66, 5/66. *For a complete record-controlled instrument* $B_y = \sum_r |r\rangle\langle r| \otimes K_{(r,y)}$ *with channel* $\Lambda(X) = \sum_y B_y X B_y^\dagger$, *every old record projector* Q_r *satisfies* $\Lambda^*(Q_r) = Q_r$, *and the block readback* $E_{\mathrm{block}}(X) = \sum_r Q_r X Q_r$ *and the public readback* $E_{\mathrm{public}}(X) = \sum_r \operatorname{Tr}(Q_r X Q_r) Q_r / \dim W$ *are distinct conditional expectations.*

Proof sketch. Summing $M^\dagger M$ over s, then (a,b), then r gives the identity. Each B_y commutes with Q_r, so $\Lambda^*(Q_r) = Q_r \sum_y B_y^\dagger B_y = Q_r$; every reported probability is checked against the norm of the retained unnormalized vector [74, 100]. □

At 512 steps per interval the emitted words use 4 working qubits, 176,182 native pulses and 352,364 coherent load and export events at $d = 2$, and 6 qubits, 1,879,628 pulses and 3,759,256 events at $d = 4$. Disconnected, feedback-disabled, fixed-seam and independent-bit controls isolate the causal effect of the records [73].

6.2 The feedback contrast at finite and infinite occupation

Theorem 6.3 (Finite feedback contrast)**.** *For the common source with intervals* 1 *and* 1/2*:*

(i) every palindromic product $F(h)$ *of the component exponentials whose generators sum to* H *satisfies* $\|F(t/n)^n - e^{-itH}\| \le S^3 |t|^3/(3n^2)$ *with* $S \le 3 + (1+2\kappa)(2d-1) + |g|\sqrt{4d-2}$, *on all inputs and references; at* $n = 512$ *and* $t = 1$ *the half-diamond bounds are below* 0.000728 *for* $d = 2$ *and* 0.004184 *for* $d = 4$*;*

(ii) at $d = 2$ and zero native preparation current, exact rational Taylor arithmetic of degree 64 with $\|H\| \le 109/16$ encloses the unconditional probability of a positive scalar coordinate and the feedback contrast for the continuous finite Hamiltonian:

$$\begin{aligned} &\textit{feedback enabled:} && [0.370697726608684,\ 0.370697726608685],\\ &\textit{feedback disabled:} && [0.402359206973491,\ 0.402359206973492],\\ &\textit{contrast:} && [-0.031661480364808,\ -0.031661480364807]; \end{aligned}$$

(iii) for the ideal source word at 512 steps per interval the contrast lies in

$$[-0.033297300667839,\ -0.030025660061775],$$

which excludes zero.

Proof sketch. The multinomial Leibniz expansion bounds the third derivative of a product of unitaries by S^3, palindromic symmetry gives second-order agreement, and unitary telescoping gives (i). For (ii) the integral Taylor remainder gives $\|T_m(t) - e^{-itH}\| \le (St)^{m+1}/(m+1)!$, and at $m = 64$ the probability radius is below 7.091×10^{-37}; measurements and feedback contract trace distance, which gives (iii) [74]. □

With $g = 0$ the gauge-only feedback cannot change the scalar marginal, and a coarse splitting or inadequate Taylor degree yields an interval containing zero, which the evaluator reports as no resolved signal. The finite pointer measures the sign projector of the truncated coordinate, which differs from the infinite half-line effect; at $d = 2$ the two differ in norm by $\frac{1}{2} - 1/\sqrt{2\pi} \approx 0.10105772$ [78], so the infinite-occupation statement needs the infinite detector.

Theorem 6.4 (Infinite-occupation feedback contrast)**.** *Take the action (3) on two infinite oscillator Fock spaces, the infinite free coupled vacuum, proportional to* $\exp[-(a_0^\dagger - a_1^\dagger)^2/36]|0,0\rangle$ *with squared normalization $4\sqrt{5}/9$, the blank gauge code, zero native preparation current, the matter record and feedback of Section 6.1, and the half-line readout of Q_0. Then:*

(i) the unconditional positive-coordinate probability with feedback minus that without lies in $[-0.024045670143,\ -0.024043450955]$, and without the interaction the contrast is exactly zero;

(ii) both matter records and both half-line outcomes retain their quantum states, including the post-read infinite occupation tail;

(iii) along $d = 2^k$, with exact palindromic factors and $n = d^2$ steps per interval, the complete finite sign instruments converge in trace distance to this infinite instrument, terminal states included, and the finite-family contrast is strictly negative for all sufficiently large d.

Proof sketch. The linear force is relatively bounded by the coupled oscillator Hamiltonian, so H is self-adjoint and finite Hermite polynomials lie in every $\mathcal{D}(H^k)$. Each of 24 steps of length $1/16$ applies a degree-16 Taylor polynomial of H, charges its remainder exactly, truncates above total occupation 16 and rounds to a 2^{-112} grid, charging both differences. Half-line probabilities are $r + s/\sqrt{2\pi}$ with exact rationals, and π is enclosed by Machin's formula. For (iii), $P_d Q_0 P_d \to Q_0$ in the strong resolvent sense and the spectral measure of Q_0 has no atom at zero [75, 121, 118]. □

The $d = 2$ finite experiment has contrast about -0.03166148; replacing only its detector by the half-line effect gives about -0.02526221, and removing the cutoff gives the interval of Theorem 6.4. The signal survives both limits, which change its size; the interval is specific to zero native preparation current and the declared two-site action [75].

6.3 Occupation refinement and complete-output comparisons

Let P_d project both oscillators onto occupations below d and let $N = n_0 + n_1$ be the total oscillator occupation, a different object from the closure input N.

Theorem 6.5 (Uniform pre-readout occupation bound)**.** *Let the matter start in an arbitrary state entangled with an untouched reference, let the scalar start in the fixed coupled vacuum, and let the finite preparation retain total occupations* $0, 2, \dots, d-2$*. Then:*

(i) $M(s,t) = \langle e^{sN} \rangle$ *satisfies* $\partial_t M \le \frac{11}{16} \sinh(s)\, \partial_s M + \frac{25}{32} \sinh(s)\, M$ *through the matter measurement and feedback;*

(ii) at total model time $T = 3/2$ *the resulting moment bound holds with base* $\varrho = 113/64 \ge 7/4$ *and* $M < 13$*, where* $\varrho \le e^{s(T)}$*;*

(iii) $\|P_d H (1 - P_d)\| \le C_d = \frac{\kappa}{2}\sqrt{d(d+1)} + \frac{3\kappa}{2} d + \frac{1}{4}\sqrt{d}$*, and the retained-history trace distance up to the final scalar read is at most* $\left[(\sqrt{2 v_d} + T C_d \sqrt{\tau_d})^2 + \tau_d\right]^{1/2}$ *with* $\tau_d = M/\varrho^d$ *and* v_d *the omitted preparation probability, an* $O(d\, \varrho^{-d/2})$ *rate;*

(iv) this bound is below 0.011408*,* 0.000002498 *and* 6.175×10^{-14} *at* $d = 32, 64, 128$*.*

Proof sketch. In common and difference modes only the squeezing term and the force change the number; weighted commutator, Cauchy and Young inequalities that hold for matter-entangled vectors give (i), and integration along $\tanh(s(t)/2) = e^{-11t/16} \tanh(s_0/2)$ from the initial value $\sqrt{80/17}$ at $\varrho_0 = 8$ gives (ii). The record dilation and feedback commute with N, and projected Duhamel comparison gives (iii) [76]. □

Theorem 6.6 (Fourth moment and strong convergence of the sharp instrument)**.** *Let* $w(N) = (N+2)^4$ *and let* $M(t) = \langle w(N) \rangle$ *be summed over the unnormalized record branches. Then:*

(i) $|\dot{M}| \le C M$ *with* $C = 8\kappa + 4|g| = 13/4$*, and every finite preparation has* $M(0) \le M_0 = 143748821/8192000$*;*

(ii) the record-dilation vectors immediately before the final read satisfy

$$\|\Psi_d - \Psi\| \le \left[(e_0 + I_T L_d)^2 + \frac{M_0 e^{CT}}{(d+2)^4}\right]^{1/2} = O(1/d),$$

with $I_T = \sqrt{M_0}\,(e^{CT/2} - 1)/(C/2)$*,* $L_d = \kappa/(d+2) + |g|/(d+2)^{3/2}$ *and* e_0 *the preparation error;*

(iii) the sign projectors of $Q_{0,d}$ *converge strongly to those of* Q_0*, and the complete unnormalized classical and quantum history, together with its native continuation, converges in trace norm for all three native currents and both feedback settings.*

Proof sketch. On occupation at most m, $\|H\psi\| \le 4(m+2)\|\psi\|$ and H raises occupation by at most two, so finite-particle vectors are analytic and core convergence gives strong resolvent convergence [119, 121]. The moment coefficients are bounded by 8κ and, through a positive polynomial certificate, $4|g|$; the gauge measurement and rotations preserve $w(N)$ summed over branches [77]. □

Strong convergence supplies no finite-d rate for the sharp readout. The complete-output certificate supplies fixed-state enclosures at $d \in \{2, 32, 64, 128\}$ for the zero-current preparation and the named rectangular occupation codes [78, 101].

Theorem 6.7 (Complete outputs of the sharp detector)**.** *For zero native preparation current, with both matter records and both sign outcomes retained:*

(i) for two complete post-read outputs with rank-one branches, the half trace norm of their difference is $\frac{1}{2}\sum_b \sqrt{(p_b+q_b)^2-4|z_b|^2}$*, summed over earlier records, where* p_b*,* q_b *are the branch weights and* z_b *their overlap;*

(ii) the complete continuous-output distances between the finite and the infinite sharp histories lie in the intervals of Table 6; at $d=2$ *the ideal 512-step source budget widens them to* $[0.454336, 0.455973]$ *with feedback and* $[0.456824, 0.458461]$ *without;*

(iii) every output state supported in the rectangular occupation code of $d=256$ *lies at distance above* 0.122 *from the true output of either prepared history;*

(iv) for the vacuum control, whose two sign probabilities are both $1/2$*, the best approximation by a state supported in the code with* $d=2m$ *has distance between* $\sqrt{3/(8\pi)}\,m^{-1/4}$ *and* $\sqrt{4/(3\pi\sqrt{3})}\,m^{-1/4}$*, hence of order* $d^{-1/4}$*, and distance at most* ε *requires* $d \geq 9/(32\pi^2\varepsilon^4)$*.*

Proof sketch. Item (i) is the trace norm of a difference of rank-one operators. For (iii), the operator exchanging the normalized inside and outside directions of each branch is a norm-one witness, giving distance at least $\sum_{a,b}\sqrt{a_{ab}t_{ab}}-\epsilon_{\mathrm{poly}}$; half-line Hermite matrix elements are rational up to $1/\sqrt{2\pi}$ [110]. For (iv) the omitted vacuum mass per sign branch obeys $1/(4\pi\sqrt{m}) \leq T_{2m} \leq 2/(3\pi\sqrt{3m})$ [78]. □

Table 6: Certified complete continuous-output distances between the finite sharp history and the infinite sharp history, zero native preparation current, rounded outward [73, 101].

Onsite dimension d	Feedback enabled	Feedback disabled
2	[0.455154, 0.455155]	[0.457642, 0.457643]
32	[0.203713, 0.226530]	[0.203935, 0.226752]
64	[0.180917, 0.180924]	[0.181090, 0.181097]
128	[0.152201, 0.152204]	[0.152341, 0.152344]

At $d=64$ the pre-readout error is below 0.000002498, so the sharp detector dominates the final error: accurate sign probabilities do not certify an accurate post-measurement state.

6.4 The smooth Ramsey instrument

A detector variant keeps the preparation, evolution, first record and feedback and replaces the final sharp read by a source Ramsey pointer with work-space Kraus maps

$$M_0 = e^{-i\pi/4}\cos\left(\tfrac{Q_0}{2}-\tfrac{\pi}{4}\right), \qquad M_1 = e^{i\pi/4}\cos\left(\tfrac{Q_0}{2}+\tfrac{\pi}{4}\right),$$

and effects $\frac{1}{2}(I \pm \sin Q_0)$ [79]. Bit zero denotes the plus effect, a detector meaning different from the sharp sign record, so the sharp-sign certificates do not carry over.

Theorem 6.8 (Smooth instrument)**.** *For the declared initial vacuum family, uniformly over the three native current kicks and both feedback settings, with* $T=3/2$ *and the constants of Theorem 6.6:*

(i) the complete recorded instrument, including the pointer interaction and both unnormalized outcomes, lies within trace distance $\min(1, E_d)$ *of its infinite counterpart, where* $A_T = \sqrt{M_0}e^{CT/2}$ *and*

$$E_d = \Big[\Big(e_0 + I_T L_d + \frac{A_T(e-1)}{2(d+2)^{3/2}}\Big)^2 + \frac{M_0 e^{CT+2}}{(d+2)^4}\Big]^{1/2} = O(1/d);$$

(ii) with $n(d) = d^2$ *symmetric steps per interval the combined ideal temporal and occupation error is* $O(1/d)$*;*

(iii) on normalized inputs in the form domain of the infinite Hamiltonian the unconditional increment of the full energy is exactly $1/8$ *in the declared dimensionless convention.*

Proof sketch. Along the pointer parameter the generator $\frac{1}{2}Z_{\mathrm{anc}}Q_0$ obeys $dM/ds \le 2M$, which gives the pointer term. For (iii), $f_0(q) = \cos(q/2 - \pi/4)$ and $f_1(q) = \cos(q/2 + \pi/4)$ satisfy $\sum_b f_b^2 = 1$, $\sum_b f_b f_b' = 0$ and $\sum_b (f_b')^2 = \frac{1}{4}$, so the kinetic term gains exactly $\frac{1}{8}\|\psi\|^2$ [79]. □

The bound is about 0.2063 at $d = 64$, 0.04096 at $d = 256$ and 0.008840 at $d = 1024$, and vacuous at the executed $d = 2$ and $d = 4$ [73]; the energy identity belongs to the infinite oscillator and fails at a finite cutoff [79, 90].

6.5 Continued instruments, compiled native repair and encoded memory

One execution engine carries the sharp and smooth protocols and the extension

$$U(1),\ P \to r_0,\ F(r_0),\ U(1/2),\ M \to r_1,\ U(1/2),\ M \to r_2,\ F(r_2),\ U(1/2),\ P \to r_3,$$

with P the complete gauge instrument, M the smooth instrument and F the gauge feedback, model time 5/2 and all sixteen unnormalized work/reference branches retained, with no intermediate work preparation [80]. Complete Kraus families preserve every old-prefix effect, and imported checkpoints are validated by prefix reexecution; this supplies neither physical quantum serialization nor cloning. Carrying a moment bound A and an error e from stage to stage, with $L_H = 9/[32(d+2)] + 1/[4(d+2)^{3/2}]$ and $L_S = 1/[2(d+2)^{3/2}]$, each unitary interval of length t adds $A(e^{at} - 1)L_H/a$ $(a = 13/8)$, each smooth read $A(\mathrm{e} - 1)L_S$, and the final tail gives $E_d = [e^2 + A^2/(d+2)^4]^{1/2}$; with $n(d) = d^2$ the complete output distance is $O(1/d)$ for the scalar vacuum family and vacuous at the executed $d = 2, 4$; the classical native attachment enumerates 4,224 joint outcomes at each executed cutoff. A separate state-specific certificate retains all sixteen outputs for the blank-gauge and coupled vacuum preparation and each $j \in \{-1, 0, 1\}$ [81]; the rank-one trace-norm identity, complete-channel contractivity and directed rational evaluation at 512 splits per interval give

$$\max_{j\in\{-1,0,1\}} D_{d,j}^{\text{ideal product},\infty} < \begin{cases} 0.264092, & d = 2, \\ 0.053956, & d = 4. \end{cases}$$

This bound transfers through the native and protected-memory attachments for arbitrary admitted native and reference inputs with this independently prepared work state; it is not an arbitrary work-input bound and excludes floating execution and physical service errors.

Coherent native attachment. Let $L_h^{A,e}$ be the compiled native Kraus operator for law $A \in \{\mathsf{C}, \mathsf{E}\}$, including its fair coin, and $K_r^{j(h)}$ the work branch including the first record's current-controlled kick. Source routing implements seam $(0, 5)$ by exchanging port labels 1 and 5. With

$e(0) = (0, 5)$ and $e(1) = (0, 1)$,

$$M_{h,r,s} = (L_s^{A,e(r_2)} L_h^{A,(0,1)}) \otimes K_r^{j(h)}, \qquad \sum_{h,r,s} M_{h,r,s}^\dagger M_{h,r,s} = I_{78} \otimes I_W,$$

complete on arbitrary entangled native, work and reference inputs [82]. The execution keeps all 512 native amplitudes, padding and abort outcomes, with 295,106 terminal basis outcomes for C and 820,166 for E. For input $(|\{2,3\}\rangle + e^{i\theta}|\{1,2\}\rangle)/\sqrt{2}$ and independent work, the native interference effect gives $p_C(\theta) = B + \frac{1+q}{8}\cos\theta$ and $p_E(\theta) = B$, with $q = \Pr(r_2 = 0 \mid j = 0)$; the C phase contrast is about 0.349153278 at $d = 2$ and 0.351351028 at $d = 4$, against zero for E. Cyclic Gray ordering of the shared nine-qubit controlled-flip macro reduces its source pulse count from 11,782 to 2,044 while preserving the full 512-dimensional operator, including global phase, inverse and padding [196, 82, 83].

Active encoded memory. The code $V|0\rangle = (|\{0,2\}\rangle + |\{1,3\}\rangle)/\sqrt{2}$, $V|1\rangle = (|\{0,3\}\rangle + |\{1,2\}\rangle)/\sqrt{2}$ has the flagged Knill–Laflamme identity $V^\dagger L_h^\dagger L_h V = p_h I_2$, $p_h = k_h/4$, $k_h \in \{0,1,2\}$. A source encoder and record-controlled full-register unitary R_h realize $R_h L_h V = \sqrt{p_h} V$ [83]. Correcting after each native event, with final seam $(2,3)$ for $r_2 = 0$ and $(0,1)$ otherwise,

$$M_{h,r,s}^{\text{corrected}}(V \otimes I_W) = \sqrt{p_h p_s^{e(r_2)}}(V \otimes K_r^{j(h)}),$$

which preserves the logical and reference marginal after summing all work and record outputs, including initially entangled inputs. The passive control without recovery has complete plus/minus output distance $1 - \bar{q}$, with $q_j = \Pr(r_2 = 0 \mid j)$ and $\bar{q} = (q_{-1} + 2q_0 + q_1)/4$, so every later decoder errs by at least $\bar{q}/2$ on one input, while the corrected distance is one; for fixed seams $(0,1), (2,3)$ passive Bell fidelity is $1/2$ and corrected fidelity one. Encoding, blank registers, source precision, storage and record services are supplied. This is active protection of the stated qubit before readout, not arbitrary-carrier recovery, fault tolerance or an autonomous observer.

6.6 Observer readouts and the fixed-reference clock

The readouts below extract observer representations, modular flow and regional structure from the same executed history [84]. The quantum algebra, state pairing and access choices are premises; the Born rule is not derived from native repair.

Proposition 6.9 (Observer representations of the executed history)**.** *Let v_{ab} be the four unnormalized terminal vectors of the history at $d \in \{2,4\}$ and $\rho_{\text{hist}} = \bigoplus_{a,b} |v_{ab}\rangle\langle v_{ab}|$ on $\bigoplus_{a,b} M_{4d^2}$.*

(i) For a positive functional ω the Gelfand-Naimark-Segal (GNS) construction $\langle [A], [B] \rangle = \omega(A^*B)$, $\pi(X)[A] = [XA]$, $\Omega = [I]$ has dimension dr on a block M_d whose density has rank r. The record and work algebra has GNS dimension $4(4d^2)$, that is 64 and 256; the four committed record projectors give a four-dimensional classical GNS space; the site-0 observer that forgets both records has GNS dimension d^2, that is 4 and 16; and the regional maps $J_R[A] = [\text{embed}_R(A)]$ are isometric and intertwine the regional actions.

(ii) For faithful ρ_0 the modular group $\sigma_s(A) = \rho_0^{is} A \rho_0^{-is}$ moves number observables; the commutative record algebra has identity flow; states conditioned on the sharp terminal bit are nonfaithful on the full scalar algebra, with rank at most $d/2$.

(iii) *At $d = 2$, $\rho_0 = \frac{1}{2}(I - \delta X)$ with $\delta = P(b = 0) - P(b = 1)$ and modular frequency $\nu = \log(P(b = 0)/P(b = 1))$. A source-compiled continuation of every terminal branch by $Z(\pi/4)$, $X(s\nu/2)$ and a fresh number pointer preserves the old two-bit record marginal and has readout law $P(1 \mid s) = \frac{1}{2}(m + \delta \sin(\nu s))$, where m is the total branch mass, equal to one for the normalized history.*

(iv) *Scalar regional restrictions are completely positive conditional expectations, normalized partial trace followed by identity embedding; they are unital and idempotent and in general not multiplicative. The centered components of the actual Hamiltonian give an interaction graph that is a path through scalar site 1, scalar site 0 and the shared charged-matter and link sector.*

Proof sketch. For $\rho = V \operatorname{diag}(\lambda) V^*$ of rank r the map $[A] \mapsto AV\sqrt{\operatorname{diag}(\lambda)}$ is the GNS isometry [104]; the modular identities, including the finite Kubo-Martin-Schwinger (KMS) condition, are checked with this sign convention [105], and at $d = 2$ the probe implements the Schrödinger dual of σ_s [84]. □

In the executed history $\nu \approx 0.5292248981159262$ and $\delta \approx 0.2586046272446501$ [102]; ensemble frequencies read a modular interval on a declared monotone window, and full-cycle aliases are indistinguishable. The post-measurement site-0 state taken as its own modular reference fails to reproduce the interacting dynamics: its modular flow preserves it while its density changes under H with derivative norms about 0.2086 at $d = 2$ and 0.6603 at $d = 4$ [90]. A fixed reference on the full work algebra removes this obstruction [85].

Proposition 6.10 (Fixed-reference modular relation)**.** *Let H be the finite interacting Hamiltonian on the work space of dimension $4d^2$, $\beta > 0$ and $\sigma_\beta = e^{-\beta H} / \operatorname{Tr} e^{-\beta H}$. Then*

$$\sigma_\beta^{is} A \, \sigma_\beta^{-is} = e^{-i\beta s H} A e^{i\beta s H} = \alpha_{-\beta s}(A), \qquad \alpha_t(A) = e^{itH} A e^{-itH},$$

so continuous Hamiltonian segments correspond to modular flow at $t = -\beta s$. For the full matrix algebra this identification of the two automorphism groups forces the Gibbs form of the reference. With record blocks $\Sigma_\beta = \bigoplus_r q_r \sigma_\beta$, $q_r > 0$, the weights cancel and every record projector is fixed. The actual state $\omega(t) = e^{-itH} \omega e^{itH}$ can change while σ_β is stationary, and $\operatorname{Tr}(\omega(t)E) = \operatorname{Tr}(\omega \, \sigma_{-t/\beta}(E))$ for every effect E.

Proof sketch. $\log \sigma_\beta = -\beta H - \log Z_\beta \, I$, and equality of the generators requires $[\log \sigma + \beta H, A] = 0$ for every matrix A, whose commutant is scalar [85, 109]. □

The reference, β and full-work access are declared, and measurements and feedback act outside this one-parameter group. The receipt uses $\beta = 1$ and reference weights $(1/10, 2/10, 3/10, 4/10)$, with source continuations at model times 0, 1/2 and 1 appending a gauge-number pointer to every post-sharp branch and retaining all eight outcomes [85].

6.7 History repair of the completed record

Theorem 6.11 (Local repair of the completed history)**.** *Fix $d \in \{2, 4\}$, the native current j and the feedback setting. Let twelve event nodes carry unnormalized work vectors subject to the compatibility equations $x_{\text{root}} = \text{prepared}(j)$, $x_1 = U_1 x_{\text{root}}$, $x_{(a)} = P_a x_1$, $y_{(a)} = F_a x_{(a)}$, $z_{(a)} = U_2 y_{(a)}$ and $w_{(a,b)} = B_b z_{(a)}$, of depths 0 to 5. Then:*

(i) *exactly one assignment satisfies all equations;*

(ii) *the common fixed set of the local updates $x_v \leftarrow A_v x_{\text{parent}(v)}$ equals that compatibility set;*

(iii) *every node-fair schedule reaches the compatible assignment from arbitrary initial vectors after finitely many updates and stays there; five covering rounds suffice, a forward topological pass needs eleven updates, and a reverse order may need 55;*

(iv) *with subtree weights w_v, equal to* 11, 5, 4, 3 *and* 1 *at depths one to five, the potential $\Phi = \sum_{v \neq \mathrm{root}} w_v \|x_v - A_v x_{\mathrm{parent}(v)}\|$ decreases by at least the repaired residual at every active ideal repair;*

(v) *every deterministic extraction from the completed assignment, including branch probabilities, terminal densities, GNS representations and modular parameters, is independent of the schedule.*

Proof sketch. Agreement at depth zero forces agreement at each further depth. A node whose ancestors are correct becomes correct permanently at its next selection, and after the kth covering round every node of depth at most k is correct. Updating v zeroes its residual and changes each child's residual by at most $\|r_v\|$, since all maps are contractions [86]. □

The solved object is the complete branching instrument law for given apparatus inputs; its uniqueness selects no realized record pair, and the auxiliary solver is a host computation distinct from a native hardware process.

7 One repair history: geometry, continuum, clocks and fields

This section derives what one native repair history supplies on the carrier and on the refining spherical source. Its statements are dimensionless carrier displacements, attempt counts and event counts; physical position, length scale and propagation speed need their own operational identification.

7.1 Native geometry, causal cones and field relations

Proposition 7.1 (Native geometric readout)**.** *(i) Exact enumeration of all 4,680 state, seam and coin branches gives the one-attempt mean $T = I - L/60$ under uniform seam selection.*

(ii) With Q removing the constant mode and $C_n = QT^{2n}Q$, the normalized response $12C_n/\operatorname{Tr} C_n$ tends to $G = 4P_{\mathrm{slow}}$, where P_{slow} projects onto the rank-three band $5 - \sqrt{5}$; all eleven nonconstant modes are present at every finite attempt count.

(iii) For the readout $x(n) = \sum_p n_p u_p$ with frame directions u_p, a committed current j on the seam (a, b) gives the relative displacement $x(n') - x(n) = j(u_b - u_a)$, computable from the record without hidden loads.

(iv) The two-event controller spans two displacement directions. A declared continuation that sends the final scalar bit to $(0, 5)$ or $(0, 7)$ after the first event re-executes all 1,056 second events, and its three displacement directions $(2, 0, 0)$, $(1, 1-\varphi, \varphi)$, $(1, 1-\varphi, -\varphi)$, with $\varphi = (1+\sqrt{5})/2$, have determinant 4.

Proposition 7.2 (Native causal cones)**.** *For a fixed seam schedule and shared coins, the load at a port v after the schedule is a function of the initial loads and coins in its backward dependency set. An initial difference supported on D influences v only along a time-ordered path, so $\operatorname{dist}(D, v) \leq$ edge crossings on that path $\leq$ elapsed seam events, and at most one edge is crossed per causal layer of disjoint simultaneous events.*

Exhaustive checks cover all 78 states for 27 schedule and coin pairs, and two executed examples show influence at graph distance two arriving only after two properly ordered seams [88]. One edge per causal layer is a native signal speed of at most one seam per ordered repair event; remote repairs can carry a difference to a receiver before its first local tick [15]. Units are observer conventions [90]: if a common operational propagation law fixed a dimensionless speed c_{model}, choosing the SI speed of light would impose only $L/T = c_{\mathrm{SI}}/c_{\mathrm{model}}$ on the model's length and time units, an operational action constant would add $ET = \hbar_{\mathrm{SI}}/\hbar_{\mathrm{model}}$, and one overall scale would stay free. The construction supplies no c_{model} relating its native distance readout to its modular clock.

Proposition 7.3 (Native field relations)**.** *(i) Every event satisfies the discrete continuity law* $x' - x = -b_e j$ *with* $b_e = \mathbf{e}_a - \mathbf{e}_b$.

(ii) Conditional on the past and the selected seam, $\mathbb{E}[j] = d/2$ *and* $\mathrm{Var}(j) = (|d| \bmod 2)/4$.

(iii) Under the uniform calibration the mean load obeys $\mu_{k+1} - \mu_k = -L\mu_k/60$*, exactly, by* $\mathcal{K}X = XT$ *for the 78-state kernel* $\mathcal{K}$ *and load matrix* X*; the deviation* $V = \frac{1}{2}\|\mu - \frac{1}{6}\mathbf{1}\|^2$ *decreases strictly unless the mean is constant, while every binary trajectory keeps* $\|x - \frac{1}{6}\mathbf{1}\|^2 = 5/3$.

(iv) The slow band has mean multiplier $q = (55+\sqrt{5})/60 \approx 0.953934466291663$*, every nonconstant multiplier lies in* $(0, 1)$*, and the second-difference coefficient* $q + 1/q - 2 > 0$ *has the opposite sign to the coefficient* $-4\sin^2(\omega\Delta t/2) \le 0$ *of an undamped oscillator; no positive rescaling of the attempt count turns this closed mean law into a wave law.*

Proof sketch of Propositions 7.1 to 7.3. The fair coin averages a seam to its endpoint mean, which gives the mean operator and the conditional current moments; the other two positive response eigenvalues have strictly smaller magnitude than $1 - (5 - \sqrt{5})/60$, which gives the limit in Proposition 7.1(ii). Conservation gives the displacement identity. For the causal cones, scanning the schedule backwards and adding both endpoints of every seam that touches the dependency set proves the dependency claim by induction [87, 88, 89]. □

The recovered frame also supplies an unbounded three-component *axial circulation*. With oriented incidence matrix B, seam current J_n and columns $W_{(a,b)} = \hat{u}_a \times \hat{u}_b$ of the unit frame vectors, exact source identities give $WB^T = 0$, $WW^T = 8I_3$, $\mathbb{E}[J_n \mid x_n] = -B^T x_n/60$ and $\mathbb{E}[J_n J_n^T] = I_{30}/198$ in the uniform binary equilibrium. Hence $Z_N = \sum_{n<N} W J_n$ is a record-readable martingale with $\mathrm{Cov}(Z_N) = (4N/99)I_3$, and $N^{-1/2} Z_{\lfloor Nt \rfloor}$ converges to three-dimensional Brownian motion with covariance $(4/99)\min(t, s)I_3$ and Cameron–Martin form $(99/8)\int |\dot{z}(t)|^2\, dt$ in a rescaled attempt count. A three-event face loop restores the carrier state while adding a nonzero counter increment. The counter's memory grows with the observation horizon; it supplies neither a radial field nor a physical trajectory [12].

7.2 The common continuum operator

On the declared spherical source family, the native cell Green operators converge in norm along one common subsequence to a positive compact injective operator G on the mean-zero observation space. Its inverse $A = G^{-1}$ is densely defined and self-adjoint, and restoring the constant zero mode gives a conservative strongly local regular Dirichlet form with domain H^1 on the supplied flat-polyhedron comparison surface, energy bounded above and below by positive multiples of the gradient energy, and heat-trace spectral dimension two. A boundary-action test excludes a boundedly invertible L^2 energy identification of A with the static boundary action of a smooth compact three-dimensional Dirichlet bulk with smooth boundary, whose Steklov eigenvalues grow like $\sqrt{n}$; bounded-energy

internal rank-three modes vanish at this scaling [12]. An exact terminal reduction gives local dual-edge conductances in $[5/8, 15/19]$ with Green-operator error at most $182\,2^{-l}$ relative to native cell means, so this weighted graph has the same continuum subsequences.

A positive native lag readout realizes the Poisson extension. For $K_l = I - L_l/(2M_l)$, Sibuya jump probabilities $p_n = \binom{2n}{n}/[4^n(2n-1)]$ and a Poisson number of jumps with mean R give $\mathbb{E}K_l^N = e^{-R\sqrt{I-K_l}}$. Restricting jumps to b and their count to j, with overflow retained, has maximum native lag bj and operator error at most $\min(1, R\binom{2b}{b}4^{-b} + R^{j+1}/(j+1)!)$; at $R = 1/2$, $b = 64$, $j = 8$ the full 66-state source gives 0.034581 within the bound 0.036. Taking $R_l = r\sqrt{2M_l 4^l}$ yields the compressed limit $e^{-r\sqrt{A}}$; here r labels a positive mixture of native lags, not a physical radius.

Full native density correlations converge to $e^{-\tau A}$, and the admitted coherent continuation converges strongly to $\cos(t\sqrt{A})$. The full-space extension $\bar{A} = A \oplus 0$ satisfies $\|\mathbf{1}_F e^{-\tau\bar{A}}\mathbf{1}_E\| \le e^{-d(E,F)^2/(2\tau)}$, and its cosine propagator vanishes between these supports when $|t| < \sqrt{2}\,d(E,F)$ in the declared flat comparison distance. Two consecutive long observation blocks of one history supply a continuous classical Gaussian wave readout with covariance $G\cos((t-u)\sqrt{A})$ in $C([0,T], H^{-2}(S^2))$, with observation count taken to infinity before spatial refinement [12]. A declared static filter $A_l^{1/4}$ changes it to $A_l^{-1/2}\cos((t-u)\sqrt{A_l})$, twice the real part of the free-field vacuum covariance; the stationary increments obey $2A_l^{-1/2}(I - \cos(\Delta t\sqrt{A_l})) \le (\Delta t)^2\sqrt{A_l}$ and, with $\sum_m \langle h_{jm,l}, \sqrt{A_l} h_{jm,l}\rangle \le 12j^2$ for centered cell averages of spherical harmonics, the geometric H^{-s} increment moment is at most $12[1 + 1/(2s-3)](\Delta t)^2$ for $s > 3/2$, giving convergence in $C([0,T], H^{-s}(S^2))$. The wave readout is declared classical processing, distinct from passive density waiting and from a quantum vacuum.

7.3 Native density histories and their limits

At level l, let $V_l = 240\,4^l$, $q_l = V_l/3$ and $\chi_l = q_l(V_l - q_l)/[V_l(V_l-1)]$. In the uniform canonical equilibrium, with independent uniform seams and fair repair, the actual snapshot readout of a mean-zero test function repeated on its twelve native ports,

$$Y_l(\tau; f) = \frac{\sum_p f_l(p)[n_{\lfloor 1260\,16^l\tau\rfloor}(p) - 1/3]}{\sqrt{V_l\chi_l}},$$

converges jointly at finitely many times to a stationary Gaussian family with covariance $\langle f, e^{-|\tau-u|A} g\rangle$, refining space and native attempt count together. Representing each unread repair as a fair identity-or-swap, the conditional profile overlaps concentrate with variance at most $4\|g\|_\infty^2\|f\|_{\ell^2}^2/V_l^2$ uniformly in attempt count, and a conditional finite-population central limit theorem applies. The declared filter $G_l^{1/2}$ gives covariance $Ge^{-|\tau-u|A}$. The law is reflection positive with one-particle semigroup $e^{-\tau A}$, generator A and not $\sqrt{A}$. The interpolated density converges in $C([0,T], H^{-3}(S^2))$ and its Green-filtered readout in $C([0,T], H^{-2}(S^2))$, by a forward–backward martingale decomposition with $g = 2(I+P_l)^{-1}f$ and harmonic tail bounds; the step paths have the same Skorohod limits [12]. For $\phi = \Phi/\sqrt{2}$ a mode of eigenvalue λ has covariance $e^{-\lambda|\tau-u|}/(2\lambda)$, so the Gaussian precision is the Euclidean form

$$S_E(\phi) = \frac{1}{2}\int_{\mathbb{R}} (\|\partial_\tau\phi\|^2 + \|A\phi\|^2)\,d\tau,$$

with reconstructed Hamiltonian $d\Gamma(A)$ and spatial precision A^2. The action follows from the native readout law and has diffusive dispersion.

Two restrictions bar identifying this sector with a relativistic field. On the shorter attempt scale $b_l = 1260\,8^l$ both the density and Green-filtered paths freeze to their initial field, with expected squared supremum increments at most $50(T2^{-l} + a_l^{-1})$, $a_l = 1260\,16^l$. For a complete linear readout with bounded injective dense-range composite map $B = TC^{1/2}$, $C = I$ or G, the left polar decomposition $B = D_0^{1/2}U$ gives

$$D_0^{-1/2} D_\tau D_0^{-1/2} = U e^{-\tau A} U^*,$$

so normalization preserves the ordered decay rates $\lambda_n(A) \asymp n$. No common positive clock factor turns these into the $n^{1/3}$ Euclidean decay rates of a complete stationary free scalar on a smooth compact three-dimensional manifold without boundary, and the same argument excludes every spatial dimension $d \geq 2$, whose rates grow like $n^{1/d}$; a continuous nonconstant time change preserving the semigroup is a scalar factor. The excluded object is the complete linear Gaussian identification; restricted boundary observations, nonlinear dictionaries and interacting fields need their own analysis, and a causal microscopic theory may have this diffusive sector as an effective description [12].

7.4 Incident clocks and active waves

Let $a_l = 1260\,16^l$, $M_l = 630\,4^l$ and let a fixed event set contain $d_l > 0$ seams. Its incident counter $N_l(n)$ includes zero-current repairs, with $p_l = d_l/M_l$; write $T_l(k)$ for the attempt of its kth tick. For $n_* = \lceil a_l(T + \delta) \rceil$,

$$\Pr\left[\max_{n \leq n_*} |N_l(n) - np_l| > a_l p_l \delta\right] \leq \frac{n_*(1 - p_l)}{a_l^2 p_l \delta^2},$$

and on the complementary event the normalized count $C_l(t) = N_l(\lfloor a_l t \rfloor)/(a_l p_l)$ and the inverse clock $U_l(t) = T_l(\lfloor a_l p_l t \rfloor)/a_l$ differ from t by at most $\delta + 1/a_l$ and $\delta + 1/(a_l p_l)$ on $[0, T]$. A primitive port has $d_l = 5$ or 6 and a whole carrier $d_l = 33$. The density path theorem transfers to these local count parametrizations without independence between clock marks and fields [14]. Passive port access chosen before each seam draw from earlier records also works: integer credits $R_n = \sum_{j \leq n} (30/d_j) I_j$ have conditional mean $30/M_l$ per attempt, deviation probability at most $n_*(M_l - 5)/(5a_l^2\delta^2)$, and inverse clock error at most $\delta + M_l/(5a_l)$.

For fixed incidence counts d_i and shared-seam counts c_{ij}, centered clock counts divided by $\sqrt{a_l/M_l}$ converge to Gaussian increments with covariance $\min(t, u)c_{ij}$, asymptotically independent of the density snapshots. Adjacent degree-five port clocks have limiting correlation $1/5$, the endpoints of an external seam $1/6$, and nonadjacent clocks zero. These are native model predictions under the stated schedule and preparation, without an atomic-clock or proper-time dictionary [14].

The admitted coherent walk can use the same tick. A protected wave register receives one fixed step W_l per incident event at a degree-d port, including zero current, while the timing source pauses during coherent services. With $Q_l = 630\,16^l$, $D_l = 4^l L_l$, $\beta = 2d/\sqrt{630}$, $\eta = \delta + 1/a_l$ and $e_l = \Lambda/[6(4Q_l - \Lambda)]$, on the band $0 \leq D_l \leq \Lambda < 4Q_l$ and the count good event,

$$\sup_{\tau \leq T} \left\| \cos(N_l(\lfloor a_l \tau \rfloor) \arccos K_l) - \cos(\beta\tau\sqrt{D_l}) \right\| \leq \beta\sqrt{\Lambda}\,[\eta + (T + \eta)e_l],$$

so the compressed wave converges to $\cos(\beta\tau\sqrt{A})$. Step error ϵ_l accumulates at most $N_{\max}\epsilon_l$ with $N_{\max} = O(\sqrt{Q_l})$, so $\sqrt{Q_l}\epsilon_l \to 0$ suffices. An exact control retains 237,600 two-attempt clock histories; the continuing Ramsey-plus probability is $1297/1320$ against $7901/7920$ when zero-current triggers are dropped. The wave step is compiled apparatus; passive repair does not supply it [14].

7.5 An encoded quantum field and its modular clock

The classical covariance does not determine commutators. N independent proper-code registers in $|0\rangle$ give $Q_N = \sum_j X_j/\sqrt{2N}$ and $P_N = \sum_j Y_j/\sqrt{2N}$, with commutator $i\sum_j Z_j/N$, $\mathbb{E}Q_N P_N = i/2$ and $\mathbb{E}Q_N^4 = 3/4 - 1/(2N)$. For Weyl factors $W_N(v) = e^{i(sQ_N+tP_N)}$ the quantum central limit theorem [1] gives

$$\lim_{N\to\infty} \mathbb{E}\prod_j W_N(v_j) = \exp\left[-\frac{1}{4}\left|\sum_j v_j\right|^2 - \frac{i}{2}\sum_{j<k}\sigma(v_j,v_k)\right], \qquad \sigma((s,t),(s',t')) = st' - ts',$$

the CCR vacuum with its ordered-product phase. Source pulses execute four codes and one Ramsey pointer with all 32 outcomes retained; the pointer quadratures give $0.7452257133 - 0.3013609949i$ for the declared three-factor word, reversal conjugates the phase and the tracial preparation removes it. The native density bands fix the mode rotation $e^{i\theta_l Z/2}$ per incident trigger, $\theta_l = \arccos(1-\lambda/(2Q_l))$, with encoded frequency $\Omega_l = \sqrt{Q_l}\theta_l \to \sqrt{\lambda}$, ordered covariance $e^{-i\Omega_l(t-u)}/(2\Omega_l)$ and kick response $\sin(\Omega_l(t-u))/\Omega_l$, executed for both kick signs at three native bands [12]. Polarized preparation, mode encoding, storage, pulses and readout are specified services. The sine kernel $A^{-1/2}\sin(t\sqrt{A})$ vanishes outside the comparison cone for mean-zero tests; unrestricted smearings add tP_{const}, and restoring the compact massless zero mode gives a free particle with no normalizable ground state, so the mean-zero field is no complete local net.

For $\rho_r = (I + rZ)/2$, $0 < r < 1$, and $H = \Omega(I - Z)/2$, the finite Gibbs and Kubo–Martin–Schwinger identity gives $\sigma_s = \alpha_{-bs}$ for every real s exactly when $\log\frac{1+r}{1-r} = b\Omega$, that is $r = \tanh(b\Omega/2)$; equal polarization at unequal frequencies fails a common b, the pure endpoint is not faithful and the tracial endpoint has trivial flow. With $Q_N = \sum X_j/\sqrt{2Nr}$ and $P_N = \sum Y_j/\sqrt{2Nr}$ the ordered copy limit has thermal covariance $C(\delta) = [\cos(\Omega\delta) - ir\sin(\Omega\delta)]/(2r\Omega)$ with $C(t - ib) = C(-t)$, executed on two codes with retained purifying references. The native bound $\lambda_n(A) \geq n/49920$ gives $\operatorname{Tr} e^{-b\sqrt{A}} \leq 99840/b^2$, so the mean-zero Fock space has the faithful normal state $Z_b^{-1}e^{-b\,d\Gamma(\sqrt{A})}$ with the same modular flow, and under $t = \kappa\tau$, $\kappa = 2d/\sqrt{630}$, its modular parameter is $s = -\kappa\tau/b$. The common b is a measurable preparation condition, not a repair-selected temperature.

7.6 The product field, its clock and its Lorentzian density

Independent long blocks of one native history supply radial innovations. Their mean-zero covariance is $B_l = G_l - P/(2520\,16^l)$, and a second block adds a normalized constant-direction readout, giving $B_l + P_{\text{const}}$. On a declared interval $(0, L)$ with Dirichlet ends, a mode with $h = \sqrt{\lambda}$ has covariance

$$C_\lambda(r,s) = \frac{\sinh(h\min(r,s))\sinh(h(L-\max(r,s)))}{h\sinh(hL)}, \qquad C_0(r,s) = \frac{\min(r,s)(L-\max(r,s))}{L},$$

realized by applying $Q_{rs}^{1/2}(B_l + P_{\text{const}})^{-1/2}$ to successive blocks, with Q_{rs} the conditional innovation. The whitening means the radial direction carries no native information beyond the spectrum of A. Complete 96-event current tapes check the finite decoder. The spatial subsequence gives the product operator $\mathcal{B} = -\partial_r^2 + \bar{A}$ with action $\frac{1}{2}\int_0^L(\|\partial_r\phi\|^2 + \mathcal{E}(\phi))\,dr$; for $L = \pi$ its count satisfies $\Lambda^{3/2}/(12\sqrt{2}) \leq N_{\mathcal{B}}(\Lambda) \leq 49921\Lambda^{3/2}$ for $\Lambda \geq 3$, and its tensor form gives comparison wave speed one in $d^2_{\text{product}} = |r-s|^2 + 2d^2_{\text{screen}}$. On a radial grid the sampled bridge has an explicit sine-mode covariance, and a spectral filter realizes the thermal covariance $\coth(b\sqrt{\mathcal{B}}/2)/(2\sqrt{\mathcal{B}})$ from the same records; the full-port generator and the cell Schur generator have compressed resolvents

within $1/(8\,4^{l})$ at each positive shift, so both give the same product target. Radial labels, filters, boundaries and block access are specified operations.

For a full-port angular mode set $a_l = \sqrt{Q_l}\arccos(1-\lambda/(2Q_l))$ and $c = \sqrt{\rho_{k,h}}$. Each incident trigger applies the source Z pulse of angle $a_l/(2\sqrt{Q_l})$ and the radial X pulse of angle $c/(2\sqrt{Q_l})$; anticommutation gives $\omega_{l,k} = \sqrt{a_l^2 + c^2}$ with no fitted product frequency, and the split word differs from the joint Hamiltonian by at most $na_lc/(4Q_l)$ after n triggers. The faithful reference is polarized along $(cX + a_lZ)/\omega_{l,k}$ with $r = \tanh(b\omega_{l,k}/2)$; ordered two-time measurements retain all 32 outputs. A deterministic packet $v \propto (12I - L)^{1024}(\mathbf{1}_{\mathrm{cell}\,0} - \mathbf{1}_{\mathrm{cell}\,1})$ on the 240-port carrier has Rayleigh quotient $\lambda \simeq 0.044442251696$, residual $\|(L-\lambda)v\| < 0.000304$ and squared cell compression above 0.996458773594; Householder encoders and a Bell reference read it after incident-triggered phase, with 25-trigger Chebyshev discrepancy about 0.000149026 at $M = 630$, at a cost of 1,747,068 source pulses. Copy and spectral limits followed by radial refinement give

$$C_b(t) = \frac{\coth(b\sqrt{\mathcal{B}}/2)\cos(t\sqrt{\mathcal{B}}) - i\sin(t\sqrt{\mathcal{B}})}{2\sqrt{\mathcal{B}}}, \qquad D(t) = \frac{\sin(t\sqrt{\mathcal{B}})}{\sqrt{\mathcal{B}}}, \qquad \sigma_s = \alpha_{-bs},$$

and with $C_L = L(49920 + L^2/9)/3$, $\operatorname{Tr} e^{-b\sqrt{\mathcal{B}}} \le 6C_L/b^3$, so the encoded Fock algebra has the faithful normal state $Z_b^{-1}e^{-b\,d\Gamma(\sqrt{\mathcal{B}})}$ with $s = -\kappa\tau/b$. Vanishing count tolerances require $4^l\eta_l^2 \to \infty$ and complete-step errors $\sqrt{Q_l}\epsilon_l \to 0$. Radial design, common b, preparation, mode access and source precision are declared services [12].

Lorentzian density. The regular strongly local angular form with its H^1 domain admits a measurable uniformly elliptic tensor in each chart [132, Theorem 7.2 and Corollary 7.3], recoverable from complete native records: with the disclosed total-one kernel $K_l = I - L_l/(2M_l)$, including every zero-current hold, the ordered limits

$$\lim_{t\downarrow 0}\lim_{l}\int \phi\,\frac{\bar{P}_l(t)(fg) - f\bar{P}_l(t)g - g\bar{P}_l(t)f + fg}{2t}\,d\mu = \int \phi\,a^{ij}\partial_i f\partial_j g\,d\mu$$

recover weighted tensor averages. In barycentric face coordinates the one-attempt increment tensor obeys $\frac{3}{7}a_{\mathrm{bare}} \le a_{\mathrm{screen}} \le \frac{15}{19}a_{\mathrm{bare}}$ with $a_{\mathrm{bare}} = \begin{pmatrix} 1/36 & -1/72 \\ -1/72 & 1/36 \end{pmatrix}$, so the one-attempt-first limit reads a different tensor. The radial product adds $a^{rr} = 1$. With coordinate measure $\mu(x)\,d^3x$ the chosen wave action has coefficient density $C = \mu\,\mathrm{diag}(-1, a)$, and

$$g = \sqrt{-\det C}\,C^{-1} = \mu\sqrt{\det a}\;\mathrm{diag}(-1, a^{-1}), \qquad \sqrt{-\det g}\,g^{-1} = C$$

has signature $(-,+,+,+)$ almost everywhere, with weak wave equation $\partial_\alpha(C^{\alpha\beta}\partial_\beta\phi) = 0$ [193, Equations (23)–(26)]. The signature is that of the chosen action; the derived content is the measurable tensor a. The native measure is the radial pushforward of flat face area, with density $\rho_F(u) = c/(n_F\cdot u)^3$, $c = \pi r^2/(5a_0)$, on a face of unit normal n_F and plane distance r. Adjacent faces agree on their common edge but their spherical gradients differ by $-3c(n_F - n_G)/(n_F\cdot u)^4 \neq 0$, so the density is not C^1, and since $C^{tt} = -\mu$ the exact dictionary in this atlas fails the smooth-density premise. An alternative smooth dictionary would need a separate argument.

The density is globally Lipschitz relative to smooth sphere volume. Smoothing only the angular variables and the tensor, with the unitaries $U_\varepsilon f = \sqrt{\rho_\varepsilon/\rho}\,f$ between the changing L^2 spaces, gives transported Mosco convergence and then norm-resolvent convergence with gap $(\pi/L)^2$. Thermal covariance and sine response converge in operator norm, and ordered Weyl laws, finite adaptive Ramsey histories and finite detector outputs converge. Hence for any fixed finite admissible menu

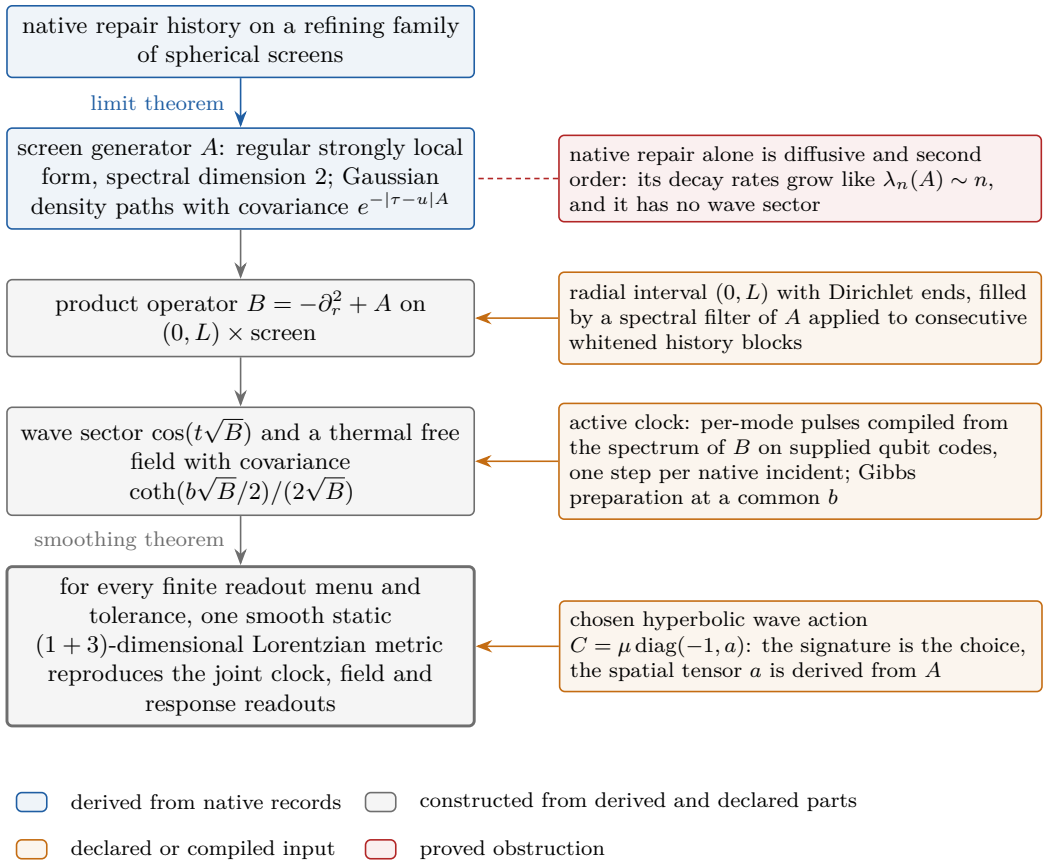


Figure 3: What the effective $(1+3)$ scalar description derives from native records and what it declares. Blue: derived from one native repair history by a limit theorem. Grey: constructed from derived and declared parts. Orange: declared or compiled inputs. Red: the proved obstruction that rules out a wave sector from passive native repair. The smoothing theorem is an existence result for fixed finite readout menus.

on bounded wave times and any positive tolerance one smooth static $1+3$ metric approximates the joint readouts. Both sides are free fields on the same supplied codes; the theorem removes the roughness of the native coefficients, not the declared radial product, clock or state, and gives no computable rate. In the smooth comparison $g = N^2\,\mathrm{diag}(-1, a^{-1})$, $N^2 = \mu\sqrt{\det a}$, the canonical momentum is $\mu\partial_t\phi$ and the thermal covariance and response have the standard static CCR/KMS normalization with a reflecting radial boundary.

Angular observations do not select the radial law. For the 240-port L, three 720-port products $B_R = I_3 \otimes L + R \otimes I$ with the unit path, fourfold path and unit triangle Laplacians satisfy $J^*F(B_R)J = F(L)$ for the radial average J and every spectral function, while a radial-contrast probe has cosine second derivatives $-5/2, -10, -3$.

For positive finite-measure regions E, F of the product, let $d(E,F) = \sup_u(\operatorname{ess\,inf}_F u - \operatorname{ess\,sup}_E u)$ over bounded local-form functions with $\Gamma_{\mathcal{B}}(u) \le 1$. Davies–Gaffney estimates and the short-time theorem of Ariyoshi and Hino [125, Theorem 2.7], $\lim_{\tau\downarrow 0} -4\tau \log\langle \mathbf{1}_F, e^{-\tau\mathcal{B}}\mathbf{1}_E\rangle = d(E,F)^2$, combined with heat-cosine transmutation give

$$\inf\{t>0 : \|\mathbf{1}_F \cos(t\sqrt{\mathcal{B}})\mathbf{1}_E\| > 0\} = \inf\{t>0 : \|\mathbf{1}_F \mathcal{B}^{-1/2}\sin(t\sqrt{\mathcal{B}})\mathbf{1}_E\| > 0\} = d(E,F).$$

This is the sharp finite-propagation front of the chosen wave equation, fixed by its intrinsic metric; it supplies no independent propagation speed, and same-field modular time has onset $|s| = d/b$.

7.7 Interacting readouts on the product

Retain the angular constant and radial profiles $\sqrt{2}\sin r$ and $\sqrt{2}\sin(2r)$ on $(0,\pi)$. The projected local quartic polynomial is

$$W = \tfrac{3}{2}(q_1^4 + q_2^4) + 6q_1^2q_2^2, \qquad H = \tfrac{1}{2}(p_1^2 + q_1^2 + p_2^2 + 4q_2^2) + \frac{g_{\mathrm{eff}}}{24}W,$$

with $g_{\mathrm{eff}} = (\lambda/\pi)\int \mu_{\mathrm{screen}}^2 \sqrt{\det a_{\mathrm{screen}}}\, dx$ for a constant covariant coupling; $g_{\mathrm{eff}} = 1$ is an input. At occupation cutoff 4×4, native source pulses prepare the purification of $e^{-\beta H}/\operatorname{Tr} e^{-\beta H}$ and execute two reads with feedback, retaining all work, reference and record outputs. For $\beta = 1/4$, intervals $1/16$ and 64 split steps, the complete-output distance from an independent exponential is below 1.97×10^{-8}; deleting the interaction or feedback changes the record law by about 7.31×10^{-4} or 1.95×10^{-3}. The worst branch uses 217,110 source pulses, 434,220 load/export events and two native repairs. A separate protocol advances this Hamiltonian on every incident event of two native timing attempts, retaining all 3,600 tapes and 4,900 field-prefix outputs, with timeout probability $35/36$ and complete error below 3.44×10^{-9}. On the refined clock family, incident rate $R_l = 2d4^l$ gives a controlled inverse-count comparison; each metric readout uses its own clock pullback, and no universal physical conversion is inferred.

For $g \geq 0$ the positive quartic form has a Friedrichs Hamiltonian $H \geq H_{\mathrm{harm}}$, Hermite compressions form a core, and

$$Z_d \longrightarrow Z, \qquad 0 < Z \leq \frac{e^{-3\beta/2}}{(1-e^{-\beta})(1-e^{-2\beta})}, \qquad \rho_d \longrightarrow \rho \quad \text{in trace norm},$$

so every fixed finite complete adaptive history converges and the limiting Gibbs state has $\sigma_s = \alpha_{-\beta s}$. This proves occupation existence for the two-mode theory, without a small-code certificate: the complete $d = 4$ versus $d = 8$ output distance is about 0.56045, and $d = 8$ versus $d = 12$ about 0.24501. An inverse control recovers $(g_{\mathrm{eff}}, \beta, c) = (1, 1/4, 1)$ within 10^{-4} from six noiseless source probabilities and predicts a withheld interval within 10^{-8}, with local Jacobian singular values about $0.264713, 0.001489, 0.000110$. For fixed covariant λ the smoothed metrics give convergent quartic projections. The quartic excites omitted modes; no invariant two-mode band, interacting ultraviolet limit or finite-cutoff microcausality follows [43].

The reference-state boundary. Increasing native-return caps match modular phase magnitudes to the cosine limit, and a fixed cap freezes the low-band phase. On the full configuration algebra the uncapped reference approaches normalized trace with half-trace error at most $3/(8\,2^l)$, its density-wave sector has vanishing weight, and fixed-parameter modular evolution becomes trivial in normalized-trace L^2. If the relevant fluctuation moments converge, any limiting scalar canonical commutator in this bounded-overlap regime is zero. Refinement-rescaled time, other states and other fluctuation regimes are not excluded [12].

8 A common matter metric and its gravitational response

This section is the interface of the executed construction with the gravitational layer of the tower. The conditional scalar description supplies a concrete dictionary for comparing matter and geometry, and the same dictionary yields the static-product control that any responsive geometry must

overcome. The counts, ranks and identities quoted here are retained in the receipts of the matter and common-field contracts of the public repository. Write its four-dimensional coefficient density and metric as

$$C = \mu \operatorname{diag}(-1, 1, a), \qquad g = \sqrt{-\det C}\, C^{-1} = N^2 \operatorname{diag}(-1, 1, a^{-1}), \qquad N^2 = \mu\sqrt{\det a},$$

where a is a positive two-dimensional screen tensor and $\mu > 0$. These identities use the complete action density, including its measure. For the measurable native limit, smooth representatives are understood with the stated finite-menu approximation theorem and smearing transport.

8.1 Common principal symbols and finite field operators

Choose an orthonormal coframe e with $g = e^t \eta e$. Transporting Clifford matrices through its inverse gives

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} I, \qquad (\gamma^\mu p_\mu)^2 = (g^{\mu\nu} p_\mu p_\nu) I.$$

The scalar wave symbol, both Weyl principal determinants and the Maxwell symbol therefore have the same characteristic cone. The Maxwell gauge direction is retained, leaving two physical polarizations on a nonzero null covector. A change of clock coordinate rescales all field generators and the reference inverse temperature together, preserving βH. This pointwise dictionary does not select the fields' dynamics or chirality.

For a finite incidence factor C_0 on the declared angular–radial product,

$$D = \begin{pmatrix} 0 & C_0^\dagger \\ C_0 & 0 \end{pmatrix}, \qquad D^2 = \operatorname{diag}(C_0^\dagger C_0, C_0 C_0^\dagger).$$

The node block is the scalar spatial operator $B = C_0^\dagger C_0$. CAR second quantization gives a conditional fermion construction with node propagation $\cos(t\sqrt{B})$ and node-to-edge propagation $-iC_0 \sin(t\sqrt{B})/\sqrt{B}$. The full finite construction retains its extra edge modes. A factor matching the finite angular frequency is also tested; its ideal joint generator is not the exact Floquet generator of the finite angular/radial split word. Their difference retains the specified approximation budget.

Completing the actual tetrahedral cone through faces and tetrahedra gives a separate full-cochain control. Its dimensions are $(13, 42, 50, 20)$, its differential ranks are $(12, 30, 20)$, and its Betti numbers are $(1, 0, 0, 0)$. The complete metric $d + d^\dagger$ construction removes the truncated cone's thirty edge-cycle zero modes, preserves its scalar action block, and retains the scalar constant. This cone is a different global domain from the angular–radial cylinder. Differential-form parity does not establish physical Weyl chirality.

The smooth product admits a spin structure, but scalar radial Dirichlet conditions do not by themselves define a self-adjoint first-order spinor operator. A tested reflecting extension squares to a Neumann component and a Dirichlet component. Spin-connection and curvature terms also enter: a curved spinor square is not a componentwise scalar Laplacian. These domain and connection hypotheses are explicit in the common-field tests.

8.2 Area, volume and the entropy variation

The same product metric gives exact induced measures

$$dA = \mu\, d^2x, \qquad dV = \mu N\, dr\, d^2x.$$

Thus the scalar action measure supplies a whole-screen area dictionary. Fixed counting measure alone does not fix physical three-volume. Radial slices in this product have equal area and are not concentric small geodesic spheres.

Native constrained maximum-entropy completion and its joint counting first law hold on their finite algebraic variation domains. They do not identify fixed cap marginal or fixed total dimension with fixed proper volume. In the exact selected-reference test, forty-five four-port caps have identical counted edge entropy and crossing-seam counts ranging from ten to eighteen. No nonzero universal constant equates that entropy to this boundary proxy. The statement concerns a finite two-dimensional boundary proxy, rather than the whole-screen area above.

The family

$$\mu(z) = \mu_0 e^{-2z/3}, \qquad a(z) = e^{2z} a_0$$

keeps proper volume fixed on the declared radial coordinate interval and decreases its boundary area. At fixed model-coordinate inverse temperature, the cavity thermal entropy also decreases. Hence the trial functional $S_{\text{thermal}} + A/(4G_N)$ is not stationary for positive G_N on these variations. This is a specified thermal-cavity comparison; it is not an identification with local vacuum entanglement or a proof that repair generates those variations.

The distinction between observable algebras matters as well. For an executed symmetric mixture of $\pm\sqrt{\epsilon}$ displacements, the unread work density has positive first-order entropy and energy derivatives at fixed reference Hamiltonian. The retained sign-plus-work density has zero entropy derivative. Both outcomes and the purification are retained. A gravitational entropy identification must specify which region algebra and which records enter its bulk term.

8.3 A fixed-product obstruction and a necessary dynamic response

Proposition 8.1 (Radial null response of the static product)**.** *For a smooth metric*

$$g = f(x,y)\,\mathrm{diag}(-1, 1, h(x,y)), \qquad f > 0,$$

with positive screen metric h, both radial null contractions of its Einstein tensor vanish. With the Gaussian reference and other sources fixed, the same geometry cannot satisfy the Einstein balances for that reference and every coherent scalar excitation at positive gravitational coupling.

Proof. The (t, r) factor has Lorentz symmetry and all metric coefficients are independent of t, r. Its Ricci components obey $R_{tt} + R_{rr} = 0$ and $R_{tr} = 0$, also verified by direct differentiation for arbitrary smooth f, h. For $k = \partial_t \pm \partial_r$, $g(k,k) = 0$, so trace and cosmological metric terms vanish and $G(k,k) = 0$. The restored angular-constant scalar sector admits

$$\phi = A \sin(\pi r/L)\cos(\pi t/L).$$

It obeys the product wave equation and the declared radial Dirichlet condition. Its action-derived coherent relative stress satisfies

$$\Delta T(k,k) = (k\phi)^2 = \frac{A^2\pi^2}{2L^2} > 0 \quad \text{at } t = 0,\ r = L/4.$$

Subtracting the reference and excited Einstein equations contradicts $\Delta G(k,k) = 0$. This uses relative stress and supplies no absolute vacuum-energy prescription. □

A coherent amplitude changes stress quadratically; this argument is not a claimed failure of the linear entropy first law. A symmetric mixture of opposite displacements supplies a first-order state

direction, with its observable algebra specified separately. Independently changing compensating sources and geometries outside the static product class are outside the proposition.

For the smooth spherical extension

$$ds^2 = -2du\,dv + R(u,v)^2 d\Omega^2,$$

direct curvature calculation gives $G_{uu} = -2R_{uu}/R$. The null Einstein balance consequently requires

$$R_{uu} = -4\pi G_N R T_{uu}, \qquad \theta_u = -\frac{\theta^2}{2} - 8\pi G_N T_{uu}, \qquad \theta = \partial_u \log(4\pi R^2).$$

Here u is affine along the declared null curves. These are necessary area-response equations in this symmetry class, not a selected repair law or a solution of the full Einstein system. The alternative finite nine-null theorem also requires its independent balances; a pointwise tetrad map alone does not supply them.

The exact block, null-stress and focusing algebra is formalized in the Lean library of the public repository, and the source matrices, curvature identities, complete finite records and approximation controls are executed with the observer-particle model. Their conditional realization establishes neither observed Standard Model masses nor generalized-entropy stationarity on a physical refinement family.

9 Prediction protocols

A response without a fitted coefficient. In the stationary binary number sector, disclose local occupations and apply one extra fair native repair on seam (a,b) exactly when $n_a = 1$. Detailed balance gives, for any fixed future record event F whose continuation does not consult the earlier pulse mark,

$$\Pr_{\text{feedback}}(F) - \Pr_{\text{passive}}(F) = \tfrac{1}{2}\mathbb{E}_{\text{passive}}[(n_b - n_a)\mathbf{1}_F].$$

Complete capped clock and field forecasts and the native spatial Green response satisfy this identity exactly, with every trigger, coin and current label retained [12]. No matched public intervention dataset exists for these forecasts.

A controlled continuum response. For a disclosed port profile h, a uniform seam (a,b) executes its unchanged fair repair with probability $[1 + \epsilon(h_b - h_a)(n_a - n_b)]/2$, and the clock counts executed repairs, including zero currents. The exact linear occupation response is $2\chi K(I - K^r)h$, and on the refining source

$$\partial_\epsilon \mathbb{E}\eta_l(g)|_0 \longrightarrow 2\langle g, (I - e^{-\tau A})f\rangle.$$

At $\tau = 1$ with four selection slots per target repair, stopped likelihood bounds control the nonlinear mean error by $36\epsilon^2\|g\|$, and for a unit-axis flat coordinate at $\epsilon = 10^{-6}$, $\liminf_l \mathbb{E}_\epsilon \eta_l(f) \geq \epsilon\, 9/32792 - 36\epsilon^2 > 0$ [12]. In the selection-slot clock the derivative converges to $\langle g, R_\tau f\rangle$ with $R_\tau = 2(I - e^{-\tau\bar{A}})$, and positivity improvement of the heat semigroup gives, for nonzero nonnegative Lipschitz profiles at distance $r > 0$,

$$0 < -\langle g, R_\tau f\rangle \leq 2e^{-r^2/(2\tau)}\|f\|\|g\|, \qquad \tau > 0.$$

This derivative has no exact finite support cone. Reading its control, support and slot parameter as physical local interventions and time would contradict an exact relativistic cone; that identification is not derived, and a causal microscopic theory with diffusive effective observables is consistent with it.

9.1 The phase experiment

The classical repair law fixes the diagonal of every admissible record law and none of its coherences (Theorem 2.3). A fixed interference experiment reads the multiplier g that separates the members, and its count protocol, comparing the executed laws C and E, is fixed before any data [69, 70]. All members are ordinary quantum instruments, so the test separates record laws and makes no claim to separate OPH from quantum mechanics.

Write $x = \{2, 3\}$, $z = \{1, 2\}$ and $w = \{0, 2\}$. Prepare $|\phi_\theta\rangle = (|x\rangle + e^{i\theta}|z\rangle)/\sqrt{2}$ for $\theta \in \{0, \pi, \pi/2\}$, or the incoherent mixture $\frac{1}{2}(|x\rangle\langle x| + |z\rangle\langle z|)$, apply one repair on the seam $(0, 1)$, retain its record, and measure $P_+ = |\phi_0\rangle\langle\phi_0|$ and $P_- = I - P_+$ inside each committed record block.

Proposition 9.1 (Phase law of the two record instruments)**.** *For the preparations above:*

(i) the nonzero recorded source densities of C *are* $\frac{1}{2}|\phi_\theta\rangle\langle\phi_\theta|$ *for* $(c, j) = (0, 0)$, $\frac{1}{4}|x\rangle\langle x|$ *for* $(1, 0)$ *and* $\frac{1}{4}|w\rangle\langle w|$ *for* $(1, -1)$*; those of* E *are* $\frac{1}{4}|x\rangle\langle x|$ *for* $(c, d) = (0, 0)$ *and* $(1, 0)$, $\frac{1}{4}|z\rangle\langle z|$ *for* $(0, -1)$ *and* $\frac{1}{4}|w\rangle\langle w|$ *for* $(1, -1)$*;*

(ii) the plus probabilities are $p_{\mathsf{C}}(\theta) = \frac{3}{8} + \frac{1}{4}\cos\theta$ *and* $p_{\mathsf{E}}(\theta) = \frac{3}{8}$*, and the incoherent preparation gives* $3/8$ *for both; the phase-reversal contrast is* $1/2$ *for* C *and zero for* E*;*

(iii) all four preparations have identical classical load probabilities, the readout preserves the probability of every old record, and no record is erased and no outcome postselected.

Proof sketch. On the seam $(0, 1)$ the configuration x has contrast zero for both coins, and z has contrast -1, current 0 for coin zero and current -1 for coin one, which moves it to w. The coherent sum survives only in the C branch $(0, 0)$, where both currents vanish; E separates x from z by their contrasts [70]. □

Table 7: Plus probabilities of the four fixed settings [69, 44].

Preparation	C_0	C	E	T
Phase 0	$\frac{3}{8} + \frac{\sqrt{2}}{4}$	5/8	3/8	3/8
Phase π	$\frac{3}{8} - \frac{\sqrt{2}}{4}$	1/8	3/8	3/8
Phase $\pi/2$	3/8	3/8	3/8	3/8
Incoherent mixture of x and z	3/8	3/8	3/8	3/8

For every admissible record law the plus probability is $p_\Phi(\theta) = \frac{3}{8} + \frac{1}{2}\,\mathrm{Re}(g\,e^{-i\theta})$, since the empty seam of x stays in place and z waits with probability $\frac{1}{2}$ while keeping coherence g with x. The settings 0, π and $\pi/2$ therefore estimate $\mathrm{Re}\,g$ and $\mathrm{Im}\,g$ for whichever law is realized; laws with equal g, such as E and T, are separated by the same construction with the full seam $\{0, 1\}$ in place of z, which gives $\frac{1}{2} + \frac{1}{2}\,\mathrm{Re}(k\,e^{-i\theta})$ [44].

The count protocol fixes four settings with 8,192 fresh independent attempted trials each and no optional stopping. Each attempt falls in one of twelve common bins of coin, current and outcome, or aborts. At each setting it scores the twelve bin indicators, the plus indicator and the abort indicator, $m = 4(12 + 2) = 56$ statistics, with sampling half-width $b = 1/32$, independently established apparatus probability budget $\eta = 1/64$ and familywise sampling requirement $\delta = 0.001$.

Proposition 9.2 (Sampling certificate)**.** *If the attempted trials of each setting are independent, then with* $n = 8192$ *and* $b = 1/32$ *the probability that any of the 56 observation intervals* $[\hat{p}-b, \hat{p}+b]$ *misses its actual probability is at most* $112e^{-16} < 112/S_{32} < 0.000012606 < \delta$*, where* $S_{32} = \sum_{k=0}^{32} 16^k/k!$*.*

Proof. Hoeffding's inequality [112] gives $2e^{-2nb^2} = 2e^{-16}$ for each statistic, a union bound over 56 statistics gives $112e^{-16}$, and $S_{32} < e^{16}$; the final comparison is exact rational arithmetic [70]. □

A candidate is rejected when some observation interval is disjoint from its ideal probability widened by η. Under the apparatus contract, rejection occurs only when an observation interval fails, and both candidates use the same intervals; if the calibration guarantee fails with probability at most β, the combined bound is $112/S_{32} + \beta$. An abort has ideal probability zero, and a resolved excess returns the verdict that the apparatus budget is exceeded. Without supplied counts the verdict is "no data", and software demonstrations are labelled synthetic [69]. One survivor discriminates the conditional instruments, two survivors do not separate them, and two rejections mean neither specified apparatus model explains the counts; in every case the same counts estimate g.

9.2 The source-executed implementation

Both complete recorded instruments, the four preparations and the readout also run as emitted twelve-port control words [67, 68]. A configuration is written as the seam pair and the ten spectators: the six endpoint pairs use three bits and the spectators six, placing the 78 configurations injectively in a nine-qubit register with 434 unused addresses. A local permutation with blank record bits realizes each repair from phase-exact transpositions, and the fair coin is prepared by a source pulse and recorded. The encoding is a mathematical isometry, not a physical interface from occupation states.

Proposition 9.3 (Exact source realization of the two instruments)**.** *In the nine-qubit encoding J:*

(i) the encoded recorded channels intertwine the native ones,

$$\mathcal{E}_{\mathrm{code}}(J\rho J^*) = (I_{\mathrm{record}} \otimes J)\, \mathcal{E}_{\mathrm{native}}(\rho)\, (I_{\mathrm{record}} \otimes J^*),$$

on all 6,084 matrix units of the 78-state carrier and therefore with every untouched reference; the complete Gram identity has 156 nonzero Kraus entries;

(ii) E *is* C *followed by a three-pulse source controlled-NOT (CNOT) that copies the seam-total parity p, with contrast decoded as $d = 2j + (2c-1)p$; relative to the retained coin and current labels one additional binary record is necessary and sufficient, and forgetting it leaves parity dephasing without restoring the coherence of* C*;*

(iii) the readout is the Lüders instrument [116] of the rank-one plus projector and its complement, realized by coupling the plus projector to a blank pointer and measuring only the pointer; the minus outcome is one rank-77 projector whose internal coherences survive, and reading the full computational basis would implement a different instrument.

Proof sketch. For fixed current the endpoint map is an injective translation, and for fixed contrast the spectators and endpoint sum reconstruct the input, so every input column of the joint coin, record and source isometry has two entries $2^{-1/2}$ with disjoint supports; an independent enumeration checks all 6,084 matrix units. For (ii), $\{2,3\}$ and $\{1,2\}$ share the C record $(0,0)$ and have contrasts 0 and -1 [68]. □

The complete attempted trial costs 32,006 to 32,010 native pulse requests, with one proper-code load and one export per pulse; a readout costs 26,632 pulses, a local C repair 5,370 and a local E repair 5,373, with peak live width 11 qubits for C and 12 for E [67]. Executed terminal states

reproduce Table 7, and on synthetic counts drawn from executed branch norms the evaluator keeps the generating candidate and rejects the other. The compositional error contract

$$N_p u + N_l\, l + N_e\, e + p + z + c + a + r + s + m + g + n \le \frac{1}{64},$$

with pulse, load and export counts and their channel errors and whole-trial allowances for preparation, initialization, coin, randomizer, records, storage, measurement, coefficient compilation and numerical execution, gives sufficient conditions for $\eta = 1/64$; an equal allocation at the largest schedule is $1/6146496$ per term. The same source alphabet, code services and classical transitions admit two quantum instruments whose phase-zero observable differs by $1/4$, so these ingredients alone do not select the record law.

9.3 Retrospective audit of the closure coupling output

The detector of Section 3.4 evaluates the structured-running branch of the closure map; the flagship synthesis reports a second, self-consistent closure map with a much smaller residual [3], and this construction consumes the bare source map only. Its output, approximately 136.994835177413, falls short of the 2022 CODATA value $\alpha^{-1} = 137.035999177(21)$ [108] by about 300.388218 parts per million, and exact rational arithmetic shows its narrow numerical interval disjoint from the band of five reported standard uncertainties [69, 70]. The band is a declared comparison allowance, not a confidence interval. The audit rejects the additional hypothesis that this output is the complete physical endpoint with zero missing correction; no fit is performed and P is unchanged.

9.4 A sequential design on one carrier

The phase test above assumes fresh preparations. A sequential design runs it on one continuing carrier in its binary total-two sector without resets or independent trials. Each trial applies 379 unread native attempts to the state the carrier holds, performs the fixed-seam native current herald, whose useful yield is $5/33$, and on either useful branch executes the source mixer, three unread native attempts, a second mixer and the native coin and current pointer; a failed herald counts as a trial. After exactly 3,927 trials the two locked likelihoods are compared on all six counts, and an impossible record returns a verdict outside both models. Conditional-law bounds on the dependent transcript give a wrong-candidate probability of at most $3/400$ under the exact-source model; a uniform service allowance of $1/1570800$ per complete trial would suffice for a one-percent bound, and that allowance is not authenticated [26]. Two consecutive trials per candidate and their reference controls are executed.

9.5 Native records against public data

Published records test precise attachments of native records without a laboratory realization of the carrier; each test is retrospective and states its readout hypotheses.

The binary endpoint map is an identity or a local exchange, so diagonal preparations stay diagonal under recorded operations and local classical feedback. Under independent settings and local factorized responses the diagonal branch obeys $P(\text{win}) \le 3/4$ in the CHSH game, that is $S \le 2$; a control that lets one side read both settings wins every trial, and the native example on the 66-state carrier, with all 1,056 histories, gives $S = 26/33$. Replaying the deposited loophole-free records of Hensen et al. [29] reproduces 245 eligible trials, 196 wins and $S = 16036094/6619647 \approx 2.42250$, with a published fixed-count tail bound near 0.03917 [27]. This rejects the local classical

reading of the diagonal branch under that experimental contract; it does not test the quantum C/E instruments.

Two native ports, or two regions of one carrier, counting incident repairs under the uniform scheduler have independent interval increments with exact count laws. Attaching the published 300-second blocks of the Ar^{13+}/Yb^{+} frequency ratios of King et al. [30] to the normalized count ratio, through a gain of at least one, a declared bracket on the Yb^{+} frequency, an absolute error of 10^{-12} per block and direct phase-count readback, gives conditional probability bounds below 10^{-104} for two ports and below 10^{-22} over all ordered pairs of the 3,433 regional event sets, after charging every choice of subwindow, gain and retained blocks [28]. This rejects the direct attachment with its readout assumptions; the native event clocks are unaffected as clocks of the model, and a clock calibration needs an explicit model of the measuring apparatus.

Published trapped-ion records test a supplied experimental model executed through the same source interface: two calibrated preparation parameters predict 100 reserved complex readouts with per-spin-mean root-mean-square error 0.06647, and a changed-potential transfer predicts 125 derived observations with root-mean-square error 0.2876427868 in two-point position-estimator units [11]. They test the supplied oscillator model, whose action and apparatus do not follow from native repair.

10 The premise map

The premise map relates the construction to the conditional theorems of the library [47, 48] and is the starting ledger of the joins of Section 11. It applies one rule: a result "H implies C" discharges an input I only when C establishes I on the same domain with the required errors and H has independent support; a change of regulator, state or code, or a calibration relabelled as a derivation, discharges nothing. An antecedent is either discharged for the finite objects of the construction by execution or proof, or supplied, and discharge on the finite domain differs from discharge of the physical premise on a physical refinement tower. Six of the seven claim records of Table 3 consume no canonical physical premise; the detector record consumes the accepted closure inputs P and N and the selected coupling map, and lists the same-quantity bridge for P and the physical Thomson readback contract as unsupplied [2]. The crosswalk represents all 84 canonical premise identifiers in 27 groups and inventories 343 registered claims and 661 Lean modules, without individually instantiating every listed claim [47]. Table 8 summarizes seven theorem routes.

Finite entropy and the gravitational premise. For positive sector counts d_i, normalized probabilities p_i and fixed $M = \sum_i d_i$, the counting entropy obeys

$$S_{\mathrm{count}} = H(p) + \sum_i p_i \log d_i = \log M - D(p\|d/M) \le \log M.$$

The maximizer is $p_i = d_i/M$. In the positive-real relaxation, joint fixed-M variations have zero first derivative there and second derivative $-\sum_i (p_i' - d_i'/M)^2/p_i$. This is no operation varying integer dimensions, and with bulk factors full saturation also requires maximally mixed bulk blocks. Uniform native preparation realizes counting weights. Fixed count is not identified with fixed physical volume, nor the logarithmic sector term with a geometric area term of universal coefficient [13]. For every cap of the wave reference's return-length law, an absolute bound holds on the full $D = \binom{V}{V/3}$ configuration space:

$$\Delta = \log D - S(\sigma) \le \frac{2025}{64V}, \qquad I_\sigma(R : R^c) \le \log J + \Delta,$$

where J counts the allowed regional particle numbers. A complete base-face patch has $V/20$ ports and $3\,2^l$ crossing seams. Its stationary reduced entropy is extensive in port count, while the return reference's mutual information divided by crossing-seam count vanishes. Neither observable therefore has a nonzero universal coefficient times that boundary count. This does not exclude interpreting a whole spherical screen as a bulk boundary; that physical region map is a premise [13].

Table 8: Theorem routes, antecedents discharged on the declared construction, and supplied antecedents [47, 43].

Route	**Derived from native records (N) or constructed on declared inputs (D)**	**Supplied**
Consensus and repair [6]	Committed-record preservation; the exact gap of the record-discarded channels; schedule-independent repair of the completed common-source history; the exact law of two overlapping observers; federation output bounds below 0.000382 at $s = 1/4$ and 6.656×10^{-9} after four incident ticks.	Repair completeness, local diamonds and transactional descent for an unrestricted observer menu; conflicting updates; global history repair by native hardware; all-level federation routing.
Quantum records [7]	(D) All-input CPTP maps with coherent write isometries and complete binary instruments; the exact correctable algebra, optimal recovery and active encoded memory; the compiled joint native/work channel; GNS representations of the actual observation state; the quantum correspondence on 4,356 matrix units and the coherent screen observer within 7.779×10^{-10}.	Hilbert representation, subsystem split, Born readout, coherent preparation and readout services (not available from diagonal native resources, Section 2.5), a physical encoding of loads in the code.
Record capacity [7]	M_0 for each declared record interface with a carrier projector certificate; $\ln M_0 \le N$ on a declared carrier.	Complete-fiber closure, a physical selector and the horizon identification behind $\Lambda \ell_\star^2 = 3\pi/N$.
Gauge structure [8]	Branch charge covariance of $Q = g(P)N$; a four-state CAR and link code with operator Gauss constraints and even parity.	Physical current and action for the complete response algebra, coupling attachment, nonabelian and chiral matter.
Scalar continuum [7, 3]	Refinement transfer and occupation limits with complete-output bounds; the infinite-occupation interacting signal. (N) The native H^1 form of spectral dimension two, Gaussian density paths, incident-clock parametrizations, the precision $-\partial_\tau^2 + A^2$ and active incident-clock waves [12]. (D) A thermal CCR field on the declared radial product with a shared active clock; a quartic projection with interacting Gibbs readouts and a fixed-band occupation limit.	The physical scalar action, quantization and vacuum; inertia; a calibrated clock; a spatially refining interacting family.
Modular clocks [9]	Finite modular flow of actual observation states; $t = -\beta s$ for a fixed reference; a source-prepared finite reference with its positive-log clock; an active native-transfer clock whose quadratures four movement-bit covariances predict [20, 21]. (D) Faithful encoded product-field states sharing the active clock.	A physical reference and its complete modular dynamics; Bisognano-Wichmann identification; common null translations; clock calibration; the relation to causal depth.

Table 8 continued

Route	Derived from native records (N) or constructed on declared inputs (D)	Supplied
Gravity [9]	A rank-three response and axial circulation from committed records; native causal cones; the common spherical continuum with density and incident-clock limits [12]; exclusion of a bounded energy identification with a smooth compact three-dimensional Dirichlet bulk. (D) A measurable Lorentzian density of the product wave with its recovered angular tensor; the exact measure fails C^1; radial-average observations do not select the radial architecture.	Physical region, metric and area attachment; geometric meaning of counting-entropy stationarity; small-ball geometry; conserved stress; universal coupling and normalization; a responsive metric.

Four retained audit controls fix exact boundaries of the construction [43]. Each rejects a named identification or an overly broad inference; none rejects the theorem premises.

Complete-carrier center. The centers of M_{66} and M_{78} consist of scalars, so the unextended working algebra cannot supply twelve nonzero primitive central ports. Binary occupations are noncentral, overlap and sum to $2I$, and total-two occupations also admit the eigenvalue two. The native matrix algebra is kept as a sector; a separate twelve-block central-address extension supplies a central algebra with port-controlled operations, and occupation measurement is a separate contract.

Marked triplets. The slow Gram matrix satisfies $G^2 = 4G$, $\operatorname{Tr} G = 12$ and $(5I - A)G = (5-\sqrt{5})G$. One fixed order-five symmetry has trace $(1-\sqrt{5})/2$ on the slow frame and $(1+\sqrt{5})/2$ on the lower response triplet, which excludes an invertible intertwiner for the same action, so spatial and internal response domains are kept distinct. The comparison separates these two specified triplets only; it neither shows every internal triplet inequivalent to the frame nor identifies carrier rechartings with physical rotations. The local rank-three carrier band and the refining global low band are distinct. The conditional Spin–matter construction has independent commuting factors, and its locked diagonal action fails to give a uniform fermionic central sign. A physical separation requires the spatial and internal actions to be identified on the same clocked interacting response and preserved by its dynamics.

Feedback class. With the two endpoint loads in the uniform mixture of their one-particle configurations, a rational number-preserving exchange gives endpoint probabilities 769/1250 and 481/1250 for the two private preparations, a difference of 144/625. Four Kraus operators certify that it is CPTP, and its private commutant has dimension one, which excludes every single-product exponential representation, alternate logarithm branches included. The no-feedback theorems for reduced loads and single products hold within their stated classes, and this channel marks the class boundary.

Configuration equilibrium. For $A = I + L_{\text{path3}}$ the controls $\varepsilon = \pm 1/16$ keep the normalized configuration Gibbs marginal and change the full reference and the vacuum. For a stiffness response $Y = \gamma(A + \varepsilon A^2)$ one has $Y_{13} = \gamma\varepsilon$ and a third-mode fit residual $6\gamma\varepsilon$, while the actual acceleration is $-Yq$. The theorem identifying inertia with the same capacity keeps that premise, and equal equilibrium configurations do not select the kinetic law.

11 Ordered joins, acceptance conditions and stopping tests

The completion target is one native repair and memory process whose observers' compatible records and interventions admit the observed 3+1-dimensional quantum-field description, with its matter and interactions, and a common classical-gravity limit, each as a controlled effective description in its regime of validity [31, 43]. Observer clocks, the defect and maintenance-energy mechanism, the interacting field and particle description and the realized Einstein premises must come from that same process. Component benchmarks keep their own contracts and are not pooled into that realization. Hardware, protocol and implementation choices are design variables, and no uniqueness theorem for a chosen rule is a prerequisite; closure-derived P and N are accepted inputs that discharge no other interface.

Every comparison of observable laws uses one complete-output criterion. For every protocol p of a declared menu,

$$\sup_{p} \tfrac{1}{2} \left\| D_{\mathrm{native}}(\rho_{\mathrm{native}}(p)) - D_{\mathrm{eff}}(\rho_{\mathrm{eff}}(p)) \right\|_1 \leq \epsilon,$$

with failures, unresolved waits and admitted references retained in both outputs and decoders that keep the claimed observations. An acceptance names the protocol menu, decoders, refinement family and error budget before the comparison [43]. Exact equality of limiting observable laws is the mathematical target; finite simulations and experiments are assessed with declared approximation and measurement errors, and agreement must be constrained by measurement, since a consistent world model can describe physics different from that of nature.

Finite constructive joins. A separately versioned campaign applies existing finite theorems to four executed component joins [33]. Its clock inventory covers all 66 transported two-port cores at the declared modular parameter, with supplied coherent services and complete failure accounting. A joint instrument advances the transported modular detector and a protected field walk on the same native incident history; a control preserving both separate marginals changes their joint law. This operational comparison does not require equal generators. The positive native-return wave reference remains available even though the tested three-charge reference fails direct substitution for it.

On the binary two-load configuration complex, 300 movement edges and 495 declared triangle/square cells give integral first homology $\mathbb{Z}/2$. Actual C/E records realize an odd exchange loop, but its native branch amplitude is positive: a history class alone supplies no fermionic phase or protected particle. Complete native currents also give the positive form $D(v) = v^T L v/396$. In the existing declared product-field symbol, rest gap and independent radial curvature agree in the ideal limit; the executed split words have rest/inertia ratio $\chi \cot \chi$ with a controlled subdivision error. This does not assign mass to the history class. Reference-like localization, the physical momentum and clock maps, and one defect action supplying rest energy, inertia and stress remain open. These finite joins add no physical comparison or integrated acceptance.

A compiled Ramsey source service. On the finite density representation of the existing native coherent walk, one additional Ramsey arm gives a 16-qubit controlled source program. Its walk uses 123,646 source pulses; the full-register reversible shift is checked on all 65,536 basis addresses. The 720-arc mean-zero density sector has an exact isometry into the original 3,960-arc binary-source walk, but the full binary-arc unitary is not compiled. Source arrays execute the Ramsey preparation, controlled walk and terminal analysis, and complete two-attempt C/E joint laws retain clock-export aborts, both field outcomes and untouched references. One selected four-attempt continuation keeps the wave register and three FIFO exports without fresh field preparation. It is a selected

unnormalized history, not an exhaustive four-attempt distribution. Numerical routine guards and counted proper-code loads/exports are separate from physical apparatus accuracy and calibrated seconds [36].

A bounded repair/reference alternative. A separately frozen finite family replaces the fair placement coin on a seam (a, b) by probability $\theta_a/(\theta_a + \theta_b)$ of placing the one endpoint load at a, for declared positive port weights θ and a fixed source-independent seam selector. Complete C and E record instruments then fix the faithful binary-shell product-pair reference $\pi(x) \propto \prod_i \theta_i^{x_i}$ and covary with its modular action. This is a changed local law, with the inherited native branch support; its full branch algebra has no nontrivial reducing source charge. The earlier transported three-charge reference cannot be made reversible by this endpoint-only rule: its per-configuration weights obey $w_1^2/(w_0 w_2) = 125/64$, whereas this product family requires one [34]. Neither its weights nor a physical clock are selected by these finite statements.

Reference balance and relational charge. For the unchanged uniform-seam C/E law, the complete binary-shell population kernel is symmetric on 300 movement edges and has the unique uniform reversible measure. Holding the transported clock reference fixed produces a one-attempt total-variation drift of 17/750 or 17/600 for the two tested cores. A recorded two-port core that follows every nonzero current instead gives a reversible joint source/core measure on 66^2 states. Its overlap charge $q = |x \cap c|$ is preserved by every complete record block; the three sectors are connected separately and have stationary weights 2/5, 5/9, 2/45. The charge's native gradient form is zero. These are source-plus-controller sectors, not a localized intrinsic particle or an energy splitting. The reduced diagonal distribution converges to the joint stationary measure from the stated fixed-core preparation under independent uniform seams; adaptive feedback and the full archived record are outside that stationarity result [33].

Sharp sectors and active memory. An exact test of the same binary source/core charge finds that every state in a sharp $q = 0, 1, 2$ sector differs from the declared transported reference on the full ten-port exterior by trace distance at least 3/5, 4/9, 43/45, respectively; conditioned references attain these bounds. For an uninterrupted complete native C/E history, every positive closed odd exchange flag has a rank-one source input domain. The result prevents that flag from carrying an arbitrary unknown coherent qubit through a final decoder, but intermediate active recovery is a different operation grammar. The existing source-executed $q = 1$ code attains the all-seam exact-correction dimension bound two. Its recovery leaves a moved core with $q = 1$ probability 7/8 unless an explicit classical core reset is appended. The code has native movement probability 1/6 per uniform-seam attempt, an excess of 1/66 over the uniform-source baseline, while coherent recovery work is outside that statistic. Neither the code nor this activity identifies a localized reference-preserving particle or its rest energy [35].

A joint readout of the incident count. The compiled modular detector and density-wave Ramsey service also support an operational comparison when the native tape is hidden from the readout. Conditional on a successful preparation and terminal read in the declared two-attempt policy, let $N \in \{1, 2\}$ count port-0 incidents. Rational enclosures give a reduction in optimal squared-error risk from the joint detector/field output, relative to the detector alone, exceeding 0.0027325 for C and 0.0027329 for E. Independent resampling of the field bit with both separate marginals preserved removes the gain. The finite preparation and complete source-abort allowances retain a gain above 0.00272 under the declared numerical routine guards. This is inference of one

dimensionless shared count. The stochastic readout does not instantiate a strictly monotone reading, a two-event affine calibration or its distinct third-event check from the finite clock interfaces, and supplies no calibration in seconds [37].

Optimal small-cap screening for the corrected qubit. For every exterior cap size $1 \le k \le 10$, hypergeometric averaging gives the minimum over sharp-q states of the worst trace distance over all k-port exterior caps to the transported reference. The conditional reference attains the bound. At $q = 1$, the unrestricted one-port value is $8/225$. Exact all-seam correction imposes a stronger restriction: the finite flagged-code theorem forces nonedge source support and scalar port-occupation compressions. An exterior port adjacent to either core endpoint can then carry no code occupation. Edge cores have four safe exterior ports, distance-two cores have two, and opposite cores have none. A crossed four-port qubit attains worst one-port distance $61/450$, optimal among exact all-seam sharp-$q = 1$ qubits on an edge core and improved from the earlier code's $82/225$. All complete raw C/E branches satisfy its correction conditions; source pulse recovery for this new code remains uncompiled. Its full exterior differs by at least $4/9$. Separately, a sharp-$q = 2$ pair starting on any native edge exits that edge set with hazard $1/15$ per uniform-seam attempt, excluding an invariant edge bond in this finite candidate [38].

Same-code activity and unselected sector action. The improved code and its predecessor both have native moving-current probability $1/6$ per uniform-seam attempt. The stationary co-moving $q = 1$ ensemble has rate $5/33$, but averages over core labels, so their difference $1/66$ is a specified diagnostic comparison. On the joint source/core graph, each connected q sector has a uniform stationary measure. Any positive reweighting of those three sectors remains stationary for the same reduced diagonal source/core population law under uniform seams. Doubling the $q = 1$ weight leaves its conditional record law unchanged while shifting a separately declared Gibbs modular contrast by $-\log 2$. Thus native stationarity and the current form do not select the rest gap. The earlier code's mean repair plus recovery costs, 439409 and 863483/2 source pulses for C/E, belong to the distinct fixed-seam $(0, 5)$ protocol; they are not a measured uniform-seam energy. The finite action-measure uniqueness theorem applies within each connected sector, and the proper-time/inertia and Ward theorems require the same candidate's action, kinetic and stress maps [39].

11.1 The joins in dependency order

The joins below are ordered by acceptance dependency [32]. Each states what must be shown on the same process, the results of this paper it starts from, and the test that would stop it. A theorem premise assigned to a join must be proved on the actual model and retained domain, with its state, algebra, operation, clock and refinement maps bound in the premise map of Section 10, before the theorem's conclusion is used.

The admissible class. A versioned class of carrier and refinement, repair and memory laws, record instruments, resources, preparations, clocks and interpretation maps is fixed before validation, together with the observable domains, Standard Model, neutrino and gravity targets, allowed calibrations, input ancestry, uncertainties, budgets and decision thresholds. It distinguishes the fixed C/E law, the permitted quantum record implementations of Theorem 2.3, and any bounded alternative-law family. *Stopping test:* if the declared class is empty or jointly inconsistent, that exclusion is proved. An unbounded unspecified escape class leaves a whole-program exclusion unarmed, and an unfinished freeze is neither confirmation nor falsification.

One continuing native observer. One replayable native repair and memory process prepares, acts, reads, retains records and continues on overlapping accessible regions, with coherent resources, ancillas, initialization, read and write, retention, feedback, timeout and recovery services each constructed or counted, and complete classical and quantum outputs including untouched references, failures and continuation. It starts from the finite observers of Sections 3 and 6, the compiled native attachment and active encoded memory of Section 6.5, and the overlapping observers and matched screens of Section 4; several of these run on separate schedules with separate error budgets, and their joins need their own implementation and error analysis. In parallel the theorem library is mapped onto the model: every theorem the chain needs receives a native interface and an owning join, and results that no join needs leave the active chain with their provenance. *Stopping test:* a source-access, coherence, retention or resource impossibility rejects the tested implementation class. The coherence floor of Section 2.5 is such a result for diagonal native resources: it makes coherent services a necessary resource of that class.

Continuing regime, faithful reference and local modular clocks. The same process prepares a reference with support and faithfulness on the accessible observer algebras, keeps a continuing regime, executes the modular actions and clock readouts, and derives compatible local clock comparisons and the common normalization used by energy and propagation. Passive repair, dimensionless modular flow, event counts and physical duration stay distinct, and tracial and compressed-algebra controls are retained. It starts from the fixed-reference relation and modular representations of Section 6.6, the finite reference and positive-log clock of Section 3.7, the event clocks of Sections 3.8 and 4.2, and the incident clocks and faithful encoded-field states of Sections 7.4 to 7.6. Two retained limits bound it: passive repair cannot act as a nontrivial modular automorphism of an invariant observer algebra, and the uncapped native-return reference tends to the trace. *Stopping test:* a proof that the frozen preparation and access grammar forces trivial modular action, incompatible regional flows or no admissible physical clock map; a failed chosen reference alone is a candidate obstruction.

A localized native defect and its protecting invariant. An actual native state or observable sector has a localized core, a reference-like exterior and a boundary charge invariant under the full admitted repair law, defined independently of a target mass, with autonomous protection or a controlled lifetime, every seam and branch retained, and quantum and record sectors checked separately. *Stopping test:* an exhaustive invariant and localization no-go over every sector and allowed refinement of the frozen defect class stops this mechanism. The classical and collision exclusions of Section 11.2 do not exhaust arbitrary quantum or history sectors.

Maintenance energy, kinetics and conserved stress. One intrinsic action and energy, and one kinetic response, are derived for the same defect, with reference subtraction, clock normalization and an independently justified stress and Ward identity. Action, energy, entropy production, event counts and host work stay distinct; moving and rest responses share one signal speed with no species-dependent conversion, and interaction energy and reference or cutoff dependence are accounted. *Stopping test:* a proved inability to define a common positive energy, kinetic and stress dictionary in the frozen class, or a mandatory incompatibility between maintenance and propagation. A scalar code cost or a failed ansatz has narrower scope.

Transport, particle identity and currents. The same excitations carry charges, spin and statistics, family structure, gauge-current couplings and allowed creation, fusion, annihilation and

decay, with transport and charge conservation shown on the native process, leakage and recovery bounded, extra modes and chirality retained, and stable defects, metastable excitations and massless collective modes distinguished. It starts from the charged realization of Section 3.4 and the flux-dressed code of Section 6.1. *Stopping test:* incompatible required charge, spin, statistics, current or transport structures exclude the covered defect realization.

An interacting chiral 3+1 field description. A refining same-process observation law has a nontrivial interacting chiral 3+1-dimensional effective description, compared through the complete-output criterion with locality, quantum correlations, positivity and controlled errors. Every taste, chirality, dark mode and residue is retained until a derived decoupling map removes it, kinetic, Higgs and current responses are obtained jointly, and the photon-containing theory stays gapless, with massive thresholds, confinement observables and any pure Yang–Mills gap stated separately. It starts from the occupation limits of Section 6, the continuum operator and its density, clock and wave limits of Section 7, the declared radial product with its smoothing theorem and the fixed-band quartic projection. *Stopping test:* a covered chirality, locality, positivity, continuum or interaction no-go stops the frozen realization class. A supplied scalar product, finite free symbol or pole list does not pass this join.

The physical dictionary and the matched spectrum. A source-selected spectrum, mixing and interaction packet in compatible physical conventions uses the same defects, energy, kinetics and clocks. The charged benchmark of nine running Yukawa couplings and four Cabibbo–Kobayashi–Maskawa parameters is derived together, with source and decoder revisions, empirical ancestry, matching scheme and scale, inputs, covariance and theory uncertainty bound before the comparison; neutrino, unstable and confined sectors keep their own obligations. *Stopping test:* a wrong unavoidable matched spectrum with an independently justified error budget rejects the fixed candidate, and the class only if every permitted nuisance and source choice is excluded. Calibration on known data is not independent validation.

Responsive geometry and causal clocks. A source-derived event, causal and geometry dictionary has an effective Lorentzian domain, compatible observer directions, local clock comparisons and count, volume and length maps with controlled errors, and responds to admitted matter changes. It starts from the rank-three response and causal cones of Section 7.1 and the common matter metric of Section 8, and it keeps the finite rank-three carrier geometry distinct from physical 3+1 event geometry. *Stopping test:* a mandatory incompatibility between source geometry, clocks and nonzero matter stress excludes the covered common realization. The static radial-product failure is a scoped control, not a universal gravity no-go.

Realized Einstein premises. A premise-by-premise same-tower witness supplies geometrically normalized local modular flow whose directional charges define the matter stress used in propagation, its Ward identity, the edge-center entropy split and area normalization, the modular first law, generalized-entropy stationarity at fixed proper volume with controlled small-ball remainders and enough null directions, universal coupling and one positive Newton constant [9]. A stationary Markov law, a Gibbs reference or a stationary conditional record entropy is not this variation; the counting-entropy and boundary-proxy controls of Section 10 and the cavity comparison of Section 8 are retained. *Stopping test:* a proof that a necessary entropy sign, modular or stress normalization, or stationarity variation is incompatible across every allowed same-process realization. An unmet premise is an open join, not a theorem failure.

The composed gravitational response. The conditional Einstein theorem is applied with a checked map for every premise, giving the Einstein equation up to its cosmological term with one positive Newton constant and universal coupling of the same defect energy and stress, and finite and refinement errors are propagated into gravitational observations. A quantum spin-two graviton lies outside the classical theorem. *Stopping test:* a counterexample satisfying every theorem premise refutes the implication; a necessary failure of joint gravitational response excludes the frozen program only with coverage of the whole admissible class.

Joint independent tests. One frozen physical dictionary is compared with independent field, particle and gravity observations through complete joint and conditional laws and intervention responses, with predeclared rejection rules, eligible data, nuisance ranges, covariance and theory errors. It starts from the prospectively fixed count protocol, the sequential design and the retrospective replays of Section 9; the phase test needs a realized preparation and observable and an independently established service budget, and it measures where the realized record law lies in the classified family. *Stopping test:* the fixed formulation is rejected when an unavoidable prediction fails a valid frozen comparison; a whole-class empirical exclusion requires every admitted realization and nuisance choice to fail.

Across all joins an obstruction review records, for every exclusion, the target version, mandatory premises, quantified class, evidence, independently checked coverage, surviving escape routes and consequence. A mathematical error, a candidate failure, a class no-go, an empirical rejection, an inconclusive search and a budget stop are kept distinct. Whole-program falsification requires a verified contradiction in indispensable premises, a proof excluding every frozen realization of a required join, or an unavoidable independently excluded prediction; a finite unsuccessful search, an unconstructed map or an exhausted budget proves no impossibility.

11.2 Negative controls on the tested law

Four results reject specified mechanisms of the tested law. Each is a fact about that law and its stated class.

The native double-load defect is transient. On the twelve-port carrier, the only configuration functions invariant under every admitted seam and coin are constants, and the binary shell is the unique closed sector. A collision state carries one double load above that shell. Under the uniform seam probe, the complete C and E instruments both satisfy $\sum_h K_h^\dagger P_{\mathrm{coll}} K_h = \frac{5}{6} P_{\mathrm{coll}}$ for the projector onto the twelve collisions, so after n attempts the collision probability is $(5/6)^n \operatorname{Tr}[(P_{\mathrm{coll}} \otimes I)\rho]$ for arbitrary quantum inputs and untouched references. The mean number of attempts to split is six, and the double load never hops before splitting. Load differences telescope around all twenty faces for all 78 states, so this gradient map carries no holonomy. Holding one port fixed creates sectors only by disabling its five repairs, which changes the law. The record costs of the corrected memory code have expectations independent of the logical state, so they cannot split logical-state masses [40]. No persistent quantum subspace supported on the collisions exists under this evolution; other quantum and record-encoded sectors are untested.

The static radial product fails null balance. For the smooth static product metrics of the conditional scalar description, both radial null Einstein contractions vanish, while a coherent scalar excitation in the restored angular-constant Dirichlet sector has positive relative null stress. That geometry cannot satisfy the Einstein balances for the reference and the excitation at positive coupling with other sources fixed (Proposition 8.1). A responsive geometry is required.

Passive repair has no wave sector. The record-discarded native channels are Hilbert–Schmidt self-adjoint with spectrum in $[0, 1]$ and scalar fixed algebra (Theorem 3.4). The executed quantum channel strictly attenuates a coherent mode, which rules out unitary conjugation on this carrier, and the half-step map of its principal logarithm is not positive, so it generates no completely positive semigroup [11]. The closed mean law is damped (Proposition 7.3(iv)), ballistic rescaling freezes the density paths, and the native decay rates grow like Laplacian eigenvalues, so no common clock factor gives a wave dispersion (Section 7.3). Every wave in this paper is declared record processing or a compiled apparatus step on supplied codes; other dilations or generators are not excluded.

The tested mass laws miss the charged hierarchy. The declared linear Weyl–Higgs candidate gives charged-lepton coefficient ratios $1 : 1.1903 : 1.4982$ against the observed $1 : 206.77 : 3477.23$, and the whole complex affine class bounds the largest-to-smallest mass ratio by 543.55 against an observed lower bound of 3470 [41, 42]. These are conditional diagnostics of unmatched coefficients under a direct pole-ratio identification; no maintenance-energy map underlies them.

12 Reproducibility

Every executable result has a replayable receipt. Package manifests bind source, document and receipt artifacts by SHA-256. The repository's verification entry point recomputes these hashes, replays retained receipts and compares canonical bytes. Exact certificates must match exactly, and floating diagnostics carry declared replay tolerances. Independent test suites check the executable statements against separately written oracles, including adversarial mutations. Analytic limit proofs state their domains, normalizations and order of limits; a finite receipt does not replace those proofs. Model sources and the premise crosswalk are available at the pinned revisions cited with their results [12, 47]; the flagship synthesis and basis document are cited on the main branch [3, 4].

Statements and declarations

Funding. The authors declare that no external grants or dedicated third-party research funding were received for the preparation of this manuscript.
Competing interests. The authors participate in OPH-related research, software, simulation, or prototype-development programs and may receive professional or reputational benefit from this work. These relationships are disclosed. The authors declare no other competing interests.
Data availability. The model documents, executable sources, retained receipts and claim records supporting this study are available in the public OPH repository at the baseline revision [2] and the revisions cited with individual results.
Corresponding author. Correspondence should be addressed to Bernhard Mueller, bernhard@floatingpragma.ai.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Aharonov_Bohm_loop_phase [103].
- analytic_vector_self_adjointness_and_core_strong_resolvent_convergence [168, 177].
- binary_Kraus_Stinespring_instrument [115, 184, 172].

- Bose_Hubbard_lattice_action [150].
- canonical_lattice_scalar_quantum_simulation [152].
- Caputo_Liggett_Richthammer_exclusion_spectral_gap [106].
- classical_CHSH_local_response_bound_and_published_Bell_null_contract [133, 29].
- classical_controlled_quantum_channels [194].
- classical_Weyl_eigenvalue_counting_law [178].
- closed_quadratic_form_Galerkin_and_Gibbs_trace_convergence [183].
- coherent_state_evolution_under_positive_quadratic_action [140, 134].
- coisometry_involution_quantum_walk_spectral_mapping [188, 146].
- conditional_Hoeffding_Azuma_concentration_for_randomized_Gram_witnesses [112, 126].
- conditional_law_sequential_coupling [160].
- conditional_probability_and_Brier_decomposition [128, 167].
- convex_Schrodinger_spectral_gap_comparison [123].
- Davies_Gaffney_finite_wave_propagation_equivalence [135].
- densitized_scalar_wave_metric_identity [193].
- Doob_L2_martingale_maximal_inequality [122].
- elliptic_coefficient_Mosco_convergence [187].
- exact_rational_arithmetic_with_directed_root_and_pi_enclosures [166, 179].
- finite_acyclic_dependency_induction_and_contractive_residual_inequality [153, 127].
- finite_CAR_representation_and_gauge_invariant_sector_isometry [151, 105, 157].
- finite_Gibbs_reference_modular_Hamiltonian_identity [145, 105].
- finite_Kraus_Stinespring_construction [184, 158, 131].
- finite_Lindblad_master_equation [142, 163].
- finite_matrix_block_elimination_and_stopped_instrument_resolvents [198, 170, 137].
- finite_matrix_conditional_expectations_and_dual_fixed_observables [192, 124].
- finite_matrix_GNS_representation_and_modular_automorphisms [104, 190].
- finite_state_martingale_functional_central_limit_theorem [197, 159].
- finite_stationary_endpoint_correlations [160].
- first_hit_channel_geometric_series [156].
- Gauss_Hermite_quadrature_remainder [189, 171].
- Gaussian_Green_covariance_and_generalized_free_field_construction [182, 141].
- Gaussian_Weyl_characteristic_function [147, 195].
- Hajek_finite_population_central_limit_theorem [161].
- Hermite_half_line_projector_matrix_elements [189, 171].
- Hoeffding_finite_sample_concentration_and_union_bound [112].
- integrated_short_time_local_Dirichlet_heat_asymptotics [125].
- irreducible_positive_self_adjoint_semigroup [155].
- number_moment_Gronwall_and_projected_Duhamel_comparison [144, 174].
- Poisson_factorial_moments_and_finite_dependency_cones [138, 175].
- positive_semigroup_fractional_extension [185].
- Primary mathematical and physical sources [143, 169].
- published_trapped_ion_anharmonic_dynamics [165].
- published_trapped_ion_preparation_and_characteristic_readout [165].
- quantum_IID_fluctuation_Weyl_limit [1].
- quantum_relative_entropy_data_processing [162].
- Ramsey_binary_quantum_instrument [176].
- Richardson_extrapolation [186].
- round_sphere_finite_perimeter_isoperimetric_inequality [129].
- Schur_complement_positive_resolvent_bound [198, 148].
- spectral_projector_strong_convergence_at_nonatomic_endpoints [118, 154].

- spectral_theorem_for_self_adjoint_operators [118, 154].
- spherical_harmonic_orthogonality_and_Laplacian_spectrum [171].
- Steklov_boundary_Weyl_law [139].
- strong_resolvent_and_spectral_projection_convergence_at_continuity_points [118, 154].
- strongly_local_Sobolev_form_tensor_representation [132].
- symmetric_product_formula_integral_Taylor_remainder [149, 130].
- tensor_product_CP_conditional_expectations [191, 173].
- total_variation_contraction_under_finite_classical_continuation [136, 160].
- trace_distance_contraction_and_unitary_channel_composition [180, 169].
- Walsh_Gray_diagonal_unitary_synthesis [196].
- zero_error_independence_number [181, 164].

References

[1] Jakšić, V., Pautrat, Y., and Pillet, C.-A. (2009). *A quantum central limit theorem for sums of IID random variables*. Author manuscript, Sections 3–4. https://jaksic.xyz/papers_pdf/qcltiid.pdf

[2] FloatingPragma (2026). *Observer Patch Holography*: papers, Lean library, executable model, receipts and claim registries. Public repository; model documents, receipts, papers and the basis document are cited at the release tag of this paper, and frozen custody records at their registration revisions. https://github.com/FloatingPragma/observer-patch-holography/tree/r2053

[3] Mueller, B., Poneder, M., Osika, A., Xue, K., Cassie, B., Nguyen, P., Kim, J., Matscheko, D., Friedman, D. A., Hill, J., Glynn, W. T., Visser, M. A., Anirudha, K. A., and de La Fournière, B. (2026). *Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/flagship/from_observer_consensus_to_standard_physics.pdf

[4] FloatingPragma (2026). *The OPH basis*. Basis document: the closure hypothesis, the no-outside rule, the agreement condition and the executable specification. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/BASIS.md

[5] FloatingPragma (2026). *OPH premise reference*. Formal statements of the theorem-library premises A1, A2 and A3. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/AXIOM_REFERENCE.md

[6] Mueller, B., Xue, K., Kim, J., Anirudha, K. A., Matscheko, D., and Hill, J. (2026). *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/reality_as_consensus_protocol.pdf

[7] Mueller, B., Poneder, M., Osika, A., Xue, K., Cassie, B., Nguyen, P., and Friedman, D. A. (2026). *Observers Are All You Need*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/observers_are_all_you_need.pdf

[8] Mueller, B., Poneder, M., Osika, A., Xue, K., Nguyen, P., Matscheko, D., and Friedman, D. A. (2026). *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[9] Mueller, B., Poneder, M., Osika, A., Xue, K., Nguyen, P., Visser, M. A., Matscheko, D., and de La Fournière, B. (2026). *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[10] Mueller, B., Osika, A., Xue, K., Cassie, B., de La Fournière, B., and Friedman, D. A. (2026). *Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer-Patch Holography*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/screen_microphysics_and_observer_synchronization.pdf

[11] FloatingPragma (2026). *The runnable OPH model*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/README.md

[12] FloatingPragma (2026). *A common native continuum operator, wave observations, modular phases and repair response*. Executable model, analytic bounds and retained source receipts. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/README.md

[13] FloatingPragma (2026). *Joint counting stationarity and its geometric boundary*. Exact entropy identity and finite source controls. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/code/geometry/ENTROPY_FIRST_LAW_AUDIT.md

[14] FloatingPragma (2026). *Native incident clocks, predictable access and active waves*. Analytic count and inverse-clock bounds, correlated clock fluctuations and exact native controls. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/relative_observer_geometry/README.md

[15] FloatingPragma (2026). *Observer-relative time, fields and shared records*. Recovery assessment of the native process results. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/OBSERVER_RELATIVE_RECOVERY.md

[16] FloatingPragma (2026). *One process, overlapping observer records and an effective transfer*. Executable model and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer_correspondence/README.md

[17] FloatingPragma (2026). *Coherent quantum histories on overlapping native clocks*. Executable model and receipts. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/quantum_observer_correspondence/README.md

[18] FloatingPragma (2026). *Source-compiled apparatus in a complete observer-clock history*. Executable model and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/compiled_clock_history/README.md

[19] FloatingPragma (2026). *Joint local clocks and quantum fields with explicit memory*. Executable model and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/joint_clock_process/README.md

[20] FloatingPragma (2026). *An executable modular clock for the native record-transfer description*. Executable model and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/transfer_modular_probe/README.md

[21] FloatingPragma (2026). *Fixed observer predictions and complete-count comparison*. Modular clock recovery and native repair response. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/predictions/README.md

[22] FloatingPragma (2026). *Identifying the complete current-resolving repair law*. Executable model and receipts. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/repair_identification/README.md

[23] FloatingPragma (2026). *A local action for native quantum observations*. Executable model, locality bounds and screen comparisons. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/native_action/README.md

[24] FloatingPragma (2026). *Joint time and field laws on matched source screens*. Executable model and receipts. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/observer_time_fields/REFINEMENT.md

[25] FloatingPragma (2026). *Coherent observer outputs on matched source screens*. Executable model and receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/coherent_screen_observer/README.md

[26] FloatingPragma (2026). *A finite sequential native-law prediction protocol*. Executable model and receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/closed_prediction_protocol/README.md

[27] FloatingPragma (2026). *Classical record responses and published Bell measurements*. Replay of the deposited records. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/predictions/BELL_RECORDS.md

[28] FloatingPragma (2026). *OPH predictions and public-data comparisons*. Retrospective audit, including the direct optical-clock attachments. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/docs/research/PUBLIC_DATA_PREDICTION_AUDIT.md

[29] Hensen, B. et al. (2015). Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. *Nature* 526, 682–686. https://doi.org/10.1038/nature15759

[30] King, S. A. et al. (2022). An optical atomic clock based on a highly charged ion. *Nature* 611, 43–47. https://doi.org/10.1038/s41586-022-05245-4

[31] FloatingPragma (2026). *OPH and the executable-model hypothesis*. Statement of the hypothesis, its five conjunctive obligations and their acceptance conditions. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/OPH_EXECUTABLE_HYPOTHESIS.md

[32] FloatingPragma (2026). *Executable OPH completion program*. Ordered joins, retained foundations, acceptance conditions and stopping rules. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/COMPLETION_PROGRAM.md

[33] FloatingPragma (2026). *Finite clock, defect and energy constructions*. Versioned source contracts, complete clock services and joint readouts, native exchange homology, and independent mode-inertia checks. https://github.com/FloatingPragma/observer-patch-holography/tree/main/model/clock_defect_energy

[34] FloatingPragma (2026). *Bounded local repair family and faithful reference*. Exact source laws, retained records, theorem applications and finite class boundary. https://github.com/FloatingPragma/observer-patch-holography/tree/main/model/local_repair_family

[35] FloatingPragma (2026). *Exchange history, sharp sectors and active memory*. Exact full-exterior bounds, complete native record fibers and source-executed recovery/readout controls. https://github.com/FloatingPragma/observer-patch-holography/tree/main/model/exchange_sector

[36] FloatingPragma (2026). *Source-pulse Ramsey control of the native density wave*. Complete C/E clock/field outputs, retained source services and finite continuation controls. https://github.com/FloatingPragma/observer-patch-holography/blob/main/model/clock_defect_energy/RAMSEY_SERVICE.md

[37] FloatingPragma (2026). *A shared-event count read from the compiled clock and field*. Complete source records, rational risk bounds and finite theorem interfaces. https://github.com/FloatingPragma/observer-patch-holography/blob/main/model/clock_defect_energy/CLOCK_COUNT_INFERENCE.md

[38] FloatingPragma (2026). *Local screening and native-edge binding tests*. Exact cap minimax, all-seam code geometry and native first-exit bounds. https://github.com/FloatingPragma/observer-patch-holography/tree/main/model/particle_screening

[39] FloatingPragma (2026). *Same-code current and sector-action ambiguity*. Common candidate definition, counted correction work and sectorwise action limits. https://github.com/FloatingPragma/observer-patch-holography/tree/main/model/active_code_energy

[40] FloatingPragma (2026). *Native defect decision*. Exact sector, collision-survival, load-gradient and memory-cost controls on the twelve-port carrier. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/native_defect_decision_results.json

[41] FloatingPragma (2026). *Physical interpretation of the executable model*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/PHYSICAL_AUDIT.md

[42] FloatingPragma (2026). *Particle construction: results, assumptions and research tasks*. Mass-response guide with the hierarchy diagnosis. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/research/OPH_PARTICLE_MASS_RESPONSE.md

[43] FloatingPragma (2026). *OPH model construction from local repair*. Construction guide with completion target, acceptance criteria and audit controls. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/CONSTRUCTION.md

[44] FloatingPragma (2026). *Admissible record laws for native repair*. Model document and exact receipt. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/record_laws/README.md

[45] FloatingPragma (2026). *The coherence table and the recorded instrument*. Exact control on the actual carrier. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/record_laws/INSTRUMENT_CONTRACT.md

[46] FloatingPragma (2026). *Concrete repair-law candidates*. Candidate register. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/CANDIDATES.md

[47] FloatingPragma (2026). *OPH reverse-engineering table: model and theorem premises*. Generated premise map and concrete theorem bindings. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/PREMISES.md

[48] FloatingPragma (2026). *Machine-readable premise crosswalk*. Source of the generated premise map. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/tracking/premise_register.json

[49] FloatingPragma (2026). *OPH interpretation and working terms*. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/INTERPRETATION_AND_TERMS.md

[50] FloatingPragma (2026). *Runnable finite observers*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/README.md

[51] FloatingPragma (2026). *Finite current-record controller: mathematical account*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/DERIVATION.md

[52] FloatingPragma (2026). *Native observer records reuse the existing capacity construction*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/CAPACITY.md

[53] FloatingPragma (2026). *Quantum mixing of the native C/E repairs*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/QUANTUM_MIXING.md

[54] FloatingPragma (2026). *Recovery with complete native records*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/RECORD_RECOVERY.md

[55] FloatingPragma (2026). *Coherent continuation: native attachment and nonselection*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/observer/COHERENT_CONTINUATION.md

[56] FloatingPragma (2026). *Selecting one source, interface and repair law*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/SOURCE_SELECTION.md

[57] FloatingPragma (2026). *P-dependent electromagnetic observation and native feedback*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/p_response/README.md

[58] FloatingPragma (2026). *Charged realization, binary instrument and feedback*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/p_response/DERIVATION.md

[59] FloatingPragma (2026). *Detector access and the conditional capacity budget*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/p_response/ACCESS_CAPACITY.md

[60] FloatingPragma (2026). *Validation of the native-record electromagnetic adapter*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/p_response/VALIDATION.md

[61] FloatingPragma (2026). *A scalar detector inside the native observer protocol*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/scalar_response/README.md

[62] FloatingPragma (2026). *A native record coupled to a scalar field instrument*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/scalar_response/DERIVATION.md

[63] FloatingPragma (2026). *Conditional detector prediction*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/scalar_response/PREDICTION.md

[64] FloatingPragma (2026). *Two reads of the retained field*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/scalar_response/CONTINUATION.md

[65] FloatingPragma (2026). *Scalar preparation, evolution and detection through source controls*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/source_scalar/README.md

[66] FloatingPragma (2026). *A finite source implementation of the two-site scalar instrument*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/source_scalar/DERIVATION.md

[67] FloatingPragma (2026). *Execute the native C/E discrimination experiment from source controls*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/source_discrimination/README.md

[68] FloatingPragma (2026). *Source implementation of the complete C/E discriminator*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/source_discrimination/DERIVATION.md

[69] FloatingPragma (2026). *Fixed observer predictions and complete-count comparison*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/d85db928b240224c1306ada12961f18d9df9d199/model/predictions/README.md

[70] FloatingPragma (2026). *Exact record-instrument discrimination*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/predictions/DERIVATION.md

[71] FloatingPragma (2026). *N4 memory survives the observer's local tick clock*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/local_tick/README.md

[72] FloatingPragma (2026). *N4 occupation memory at observer-local ticks*. Protocol specification. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/local_tick/SPEC.md

[73] FloatingPragma (2026). *A common interacting observer process*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/README.md

[74] FloatingPragma (2026). *One finite interacting source: derivation and theorem attachments*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/DERIVATION.md

[75] FloatingPragma (2026). *The interacting feedback signal survives infinite occupation*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/INFINITE_HISTORY.md

[76] FloatingPragma (2026). *Occupation refinement of the interacting retained history*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/OCCUPATION.md

[77] FloatingPragma (2026). *Occupation limit of the same interacting observer history*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/OCCUPATION_REFINEMENT.md

[78] FloatingPragma (2026). *Infinite coordinate-sign readout*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/READOUT.md

[79] FloatingPragma (2026). *A smooth binary observer with a complete occupation error bound*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/SMOOTH_REFINEMENT.md

[80] FloatingPragma (2026). The finite executable observer specification. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/COMPOSITION.md

[81] FloatingPragma (2026). Complete-output certificate for the continued smooth history. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/SMOOTH_HISTORY.md

[82] FloatingPragma (2026). Coherent native repairs around one continuing quantum history. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/QUANTUM_NATIVE.md

[83] FloatingPragma (2026). Active encoded memory in the continuing observer. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/ENCODED_MEMORY.md

[84] FloatingPragma (2026). *Hilbert space, modular clocks and operational regions*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/OBSERVER_READOUTS.md

[85] FloatingPragma (2026). *A fixed thermal reference for the changing common history*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/THERMAL_CLOCK.md

[86] FloatingPragma (2026). *One complete history from local constraint repair*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/HISTORY_REPAIR.md

[87] FloatingPragma (2026). *The existing three-dimensional response carrier on actual observer records*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/SLOW_GEOMETRY.md

[88] FloatingPragma (2026). *Exact native causal cones*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/CAUSALITY.md

[89] FloatingPragma (2026). *Field relations of the existing native repair*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/NATIVE_FIELDS.md

[90] FloatingPragma (2026). *Testing the readouts before assigning physical units*. Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/CALIBRATION.md

[91] FloatingPragma (2026). Retained receipt of the exact native observer histories and calculation checks. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/results/native.json

[92] FloatingPragma (2026). Retained receipt of the 20,000-trial sequential native execution. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/results/run.json

[93] FloatingPragma (2026). Retained receipt of the record capacity and compact controller. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/results/capacity.json

[94] FloatingPragma (2026). Retained receipt of the quantum mixing and memory certificates. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/results/quantum_mixing.json

[95] FloatingPragma (2026). Retained receipt of complete-record recovery and identifying schedules. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/results/record_recovery.json

[96] FloatingPragma (2026). Retained receipt of the P-dependent detector. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/p_response/results.json

[97] FloatingPragma (2026). Retained receipt of the local-tick memory certificate. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/local_tick/results.json

[98] FloatingPragma (2026). Retained receipt of the scalar detector on four grids. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/scalar_response/results.json

[99] FloatingPragma (2026). Retained receipt of the two-site source apparatus. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/observer/source_scalar/results.json

[100] FloatingPragma (2026). Retained receipt of the common interacting source. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/results.json

[101] FloatingPragma (2026). Retained receipt of the complete sharp-output comparisons. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/sharp_output_results.json

[102] FloatingPragma (2026). Retained receipt of the observer readouts and the modular clock. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/common_source/bridge_results.json

[103] Aharonov, Y., and Bohm, D. (1959). "Significance of electromagnetic potentials in the quantum theory." *Physical Review* **115**(3), 485–491. https://doi.org/10.1103/PhysRev.115.485

[104] Bratteli, O., and Robinson, D. W. (1987). *Operator Algebras and Quantum Statistical Mechanics 1: C^*- and W^*-Algebras, Symmetry Groups, Decomposition of States*, 2nd ed. Springer, Berlin.

[105] Bratteli, O., and Robinson, D. W. (1997). *Operator Algebras and Quantum Statistical Mechanics 2: Equilibrium States, Models in Quantum Statistical Mechanics*, 2nd ed. Springer, Berlin.

[106] Caputo, P., Liggett, T. M., and Richthammer, T. (2010). "Proof of Aldous' spectral gap conjecture." *Journal of the American Mathematical Society* **23**(3), 831–851. https://arxiv.org/abs/0906.1238

[107] Choi, M.-D. (1974). "A Schwarz inequality for positive linear maps on C^*-algebras." *Illinois Journal of Mathematics* **18**(4), 565–574.

[108] Mohr, P. J., Newell, D. B., Taylor, B. N., and Tiesinga, E. (2025). "CODATA recommended values of the fundamental physical constants: 2022." *Reviews of Modern Physics* **97**, 025002. https://doi.org/10.1103/RevModPhys.97.025002

[109] Connes, A., and Rovelli, C. (1994). "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories." *Classical and Quantum Gravity* **11**, 2899–2917. https://doi.org/10.1088/0264-9381/11/12/007

[110] Olver, F. W. J., et al., eds. *NIST Digital Library of Mathematical Functions*, Chapter 18, Orthogonal Polynomials. https://dlmf.nist.gov/18

[111] Gurvits, L., and Barnum, H. (2002). "Largest separable balls around the maximally mixed bipartite quantum state." *Physical Review A* **66**, 062311. https://doi.org/10.1103/PhysRevA.66.062311

[112] Hoeffding, W. (1963). "Probability inequalities for sums of bounded random variables." *Journal of the American Statistical Association* **58**(301), 13–30. https://doi.org/10.1080/01621459.1963.10500830

[113] Horodecki, M., Shor, P. W., and Ruskai, M. B. (2003). "Entanglement breaking channels." *Reviews in Mathematical Physics* **15**(6), 629–641. https://arxiv.org/abs/quant-ph/0302031

[114] Knill, E., and Laflamme, R. (1997). "Theory of quantum error-correcting codes." *Physical Review A* **55**, 900–911. https://doi.org/10.1103/PhysRevA.55.900

[115] Kraus, K. (1983). *States, Effects, and Operations: Fundamental Notions of Quantum Theory*. Lecture Notes in Physics 190, Springer, Berlin.

[116] Lüders, G. (1950). "Über die Zustandsänderung durch den Meßprozeß." *Annalen der Physik* **443**, 322–328. https://doi.org/10.1002/andp.19504430510

[117] Peres, A. (1996). "Separability criterion for density matrices." *Physical Review Letters* **77**, 1413–1415. https://doi.org/10.1103/PhysRevLett.77.1413

[118] Reed, M., and Simon, B. (1980). *Methods of Modern Mathematical Physics I: Functional Analysis*, revised and enlarged ed. Academic Press, New York.

[119] Reed, M., and Simon, B. (1975). *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness*. Academic Press, New York.

[120] Strang, G. (1968). "On the construction and comparison of difference schemes." *SIAM Journal on Numerical Analysis* **5**(3), 506–517. https://doi.org/10.1137/0705041

[121] Teschl, G. (2014). *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*, 2nd ed. Graduate Studies in Mathematics 157, American Mathematical Society, Providence.

[122] Acciaio, B., Beiglböck, M., Penkner, F., Schachermayer, W., and Temme, J. (2013). "A trajectorial interpretation of Doob's martingale inequalities." *The Annals of Applied Probability* **23**, 1494–1505. https://doi.org/10.1214/12-AAP878

[123] Andrews, B., and Clutterbuck, J. (2010). "Proof of the fundamental gap conjecture." arXiv:1006.1686. https://arxiv.org/abs/1006.1686

[124] Arias, A., Gheondea, A., and Gudder, S. (2002). "Fixed points of quantum operations." *Journal of Mathematical Physics* **43**, 5872–5881. https://doi.org/10.1063/1.1519669

[125] Ariyoshi, T., and Hino, M. (2005). "Small-time asymptotic estimates in local Dirichlet spaces." *Electronic Journal of Probability* **10**, 1236–1259. https://doi.org/10.1214/EJP.v10-286

[126] Azuma, K. (1967). "Weighted sums of certain dependent random variables." *Tohoku Mathematical Journal* **19**, 357–367. https://doi.org/10.2748/tmj/1178243286

[127] Banach, S. (1922). "Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales." *Fundamenta Mathematicae* **3**, 133–181. https://doi.org/10.4064/fm-3-1-133-181

[128] Brier, G. W. (1950). "Verification of forecasts expressed in terms of probability." *Monthly Weather Review* **78**, 1–3. https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

[129] Cavalletti, F., and Mondino, A. (2018). "Isoperimetric inequalities for finite perimeter sets under lower Ricci curvature bounds." *Rendiconti Lincei - Matematica e Applicazioni* **29**, 413–430. https://doi.org/10.4171/RLM/814

[130] Childs, A. M., Su, Y., Tran, M. C., Wiebe, N., and Zhu, S. (2021). "Theory of Trotter Error with Commutator Scaling." *Physical Review X* **11**, 011020. https://doi.org/10.1103/PhysRevX.11.011020

[131] Choi, M. D. (1975). "Completely positive linear maps on complex matrices." *Linear Algebra and its Applications* **10**, 285–290. https://doi.org/10.1016/0024-3795(75)90075-0

[132] Cipriani, F. (2006). "Dirichlet forms as Banach algebras and applications." *Pacific Journal of Mathematics* **223**, 229–249. https://msp.org/pjm/2006/223-2/pjm-v223-n2-p03-p.pdf

[133] Clauser, J. F., Horne, M. A., Shimony, A., and Holt, R. A. (1969). "Proposed experiment to test local hidden-variable theories." *Physical Review Letters* **23**, 880–884. https://doi.org/10.1103/PhysRevLett.23.880

[134] Combescure, M., and Robert, D. (2012). *Coherent States and Applications in Mathematical Physics*. Springer. https://doi.org/10.1007/978-94-007-0196-0

[135] Coulhon, T., and Sikora, A. (2006). "Gaussian heat kernel upper bounds via Phragmén-Lindelöf theorem." *arXiv preprint arXiv:math/0609429*. https://doi.org/10.1112/plms/pdm050

[136] Dobrushin, R. L. (1956). "Central limit theorem for nonstationary Markov chains. I." *Theory of Probability & Its Applications* **1**, 65–80. https://doi.org/10.1137/1101006

[137] Dörfler, F., and Bullo, F. (2011). "Kron Reduction of Graphs with Applications to Electrical Networks." https://doi.org/10.48550/arXiv.1102.2950

[138] Feller, W. (1968). *An Introduction to Probability Theory and Its Applications, Volume 1*. Wiley. ISBN 0-471-25708-7.

[139] Girouard, A., and Polterovich, I. (2014). "Spectral geometry of the Steklov problem." *arXiv preprint arXiv:1411.6567*. https://doi.org/10.48550/arXiv.1411.6567

[140] Glauber, R. J. (1963). "Coherent and incoherent states of the radiation field." *Physical Review* **131**, 2766–2788. https://doi.org/10.1103/PhysRev.131.2766

[141] Glimm, J., and Jaffe, A. (1987). *Quantum Physics: A Functional Integral Point of View*. Springer, New York. https://doi.org/10.1007/978-1-4612-4728-9

[142] Gorini, V., Kossakowski, A., and Sudarshan, E. C. G. (1976). "Completely positive dynamical semigroups of N-level systems." *Journal of Mathematical Physics* **17**, 821–825. https://doi.org/10.1063/1.522979

[143] Grady, L. J., and Polimeni, J. R. (2010). *Discrete Calculus*. Springer. https://doi.org/10.1007/978-1-84996-290-2

[144] Gronwall, T. H. (1919). "Note on the derivatives with respect to a parameter of the solutions of a system of differential equations." *Annals of Mathematics* **20**, 292. https://doi.org/10.2307/1967124

[145] Haag, R., Hugenholtz, N. M., and Winnink, M. (1967). "On the equilibrium states in quantum statistical mechanics." *Communications in Mathematical Physics* **5**, 215–236. https://doi.org/10.1007/BF01646342

[146] Higuchi, Y., Segawa, E., and Suzuki, A. (2016). "Spectral mapping theorem of an abstract quantum walk." arXiv preprint, version 2. arXiv:1506.06457. https://arxiv.org/abs/1506.06457

[147] Holevo, A. (2011). *Probabilistic and Statistical Aspects of Quantum Theory*. Edizioni della Normale, Pisa. https://doi.org/10.1007/978-88-7642-378-9

[148] Horn, R. A., and Johnson, C. R. (2012). *Matrix Analysis*. Cambridge University Press. https://doi.org/10.1017/CBO9781139020411

[149] Jahnke, T., and Lubich, C. (2000). "Error Bounds for Exponential Operator Splittings." *BIT Numerical Mathematics* **40**, 735–744. https://doi.org/10.1023/A:1022396519656

[150] Jaksch, D., Bruder, C., Cirac, J. I., Gardiner, C. W., and Zoller, P. (1998). "Cold bosonic atoms in optical lattices." *Physical Review Letters* **81**, 3108. https://doi.org/10.1103/PhysRevLett.81.3108

[151] Jordan, P., and Wigner, E. (1928). "Über das Paulische Äquivalenzverbot." *Zeitschrift für Physik* **47**, 631–651. https://doi.org/10.1007/BF01331938

[152] Jordan, S. P., Lee, K. S. M., and Preskill, J. (2014). "Quantum Computation of Scattering in Scalar Quantum Field Theories." *Quantum Information and Computation* **14**, 1014–1080. arXiv:1112.4833. https://arxiv.org/abs/1112.4833

[153] Kahn, A. B. (1962). "Topological sorting of large networks." *Communications of the ACM* **5**, 558–562. https://doi.org/10.1145/368996.369025

[154] Kato, T. (1995). *Perturbation Theory for Linear Operators*. Springer, Berlin (Classics in Mathematics). https://doi.org/10.1007/978-3-642-66282-9

[155] Keller, M., Lenz, D., Vogt, H., and Wojciechowski, R. (2011). "Note on basic features of large time behaviour of heat kernels." arXiv:1101.0373. https://arxiv.org/abs/1101.0373

[156] Kemeny, J. G., and Snell, J. L. (1976). *Finite Markov Chains*. Springer-Verlag, New York (Undergraduate Texts in Mathematics). ISBN 0-387-90192-2.

[157] Kogut, J., and Susskind, L. (1975). "Hamiltonian formulation of Wilson's lattice gauge theories." *Physical Review D* **11**, 395–408. https://doi.org/10.1103/PhysRevD.11.395

[158] Kraus, K. (1971). "General state changes in quantum theory." *Annals of Physics* **64**, 311–335. https://doi.org/10.1016/0003-4916(71)90108-4

[159] Landim, C., Milanés, A., and Olla, S. (2006). "Stationary and Nonequilibrium Fluctuations in Boundary Driven Exclusion Processes." https://doi.org/10.48550/arXiv.math/0608165

[160] Levin, D. A., and Peres, Y. (2017). *Markov Chains and Mixing Times*. American Mathematical Society. https://doi.org/10.1090/mbk/107

[161] Li, X., and Ding, P. (2016). "General forms of finite population central limit theorems with applications to causal inference." *arXiv preprint arXiv:1610.04821*. https://doi.org/10.48550/arXiv.1610.04821

[162] Lindblad, G. (1975). "Completely positive maps and entropy inequalities." *Communications in Mathematical Physics* **40**, 147–151. https://doi.org/10.1007/BF01609396

[163] Lindblad, G. (1976). "On the generators of quantum dynamical semigroups." *Communications in Mathematical Physics* **48**, 119–130. https://doi.org/10.1007/BF01608499

[164] Lovász, L. (1979). "On the Shannon capacity of a graph." *IEEE Transactions on Information Theory* **25**, 1–7. https://doi.org/10.1109/TIT.1979.1055985

[165] McGarry, C., Chalermpusitarak, T., Schwennicke, K., Scuccimarra, F., Millican, M. J., Matsos, V. G., et al. (2026). "Programmable quantum simulation of anharmonic dynamics." arXiv:2603.04744v1. https://arxiv.org/abs/2603.04744v1

[166] Moore, R. E., Kearfott, R. B., and Cloud, M. J. (2009). *Introduction to Interval Analysis*. SIAM. https://doi.org/10.1137/1.9780898717716

[167] Murphy, A. H. (1973). "A new vector partition of the probability score." *Journal of Applied Meteorology* **12**, 595–600. https://doi.org/10.1175/1520-0450(1973)012<0595:ANVPOT>2.0.CO;2

[168] Nelson, E. (1959). "Analytic vectors." *Annals of Mathematics* **70**, 572. https://doi.org/10.2307/1970331

[169] Nielsen, M. A., and Chuang, I. L. (2010). *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press. https://doi.org/10.1017/CBO9780511976667

[170] Norris, J. R. (1997). *Markov Chains*. Cambridge University Press. https://doi.org/10.1017/CBO9780511810633

[171] Olver, F. W. J., Lozier, D. W., Boisvert, R. F., and Clark, C. W. (2010). *NIST Handbook of Mathematical Functions*. Cambridge University Press. ISBN 978-0-521-19225-5.

[172] Ozawa, M. (1984). "Quantum measuring processes of continuous observables." *Journal of Mathematical Physics* **25**, 79–87. https://doi.org/10.1063/1.526000

[173] Paulsen, V. (2002). *Completely Bounded Maps and Operator Algebras*. Cambridge University Press. https://doi.org/10.1017/CBO9780511546631

[174] Pazy, A. (1983). *Semigroups of Linear Operators and Applications to Partial Differential Equations*. Springer. https://doi.org/10.1007/978-1-4612-5561-1

[175] Poulin, D. (2010). "Lieb-Robinson Bound and Locality for General Markovian Quantum Dynamics." *Physical Review Letters* **104**, 190401. https://doi.org/10.1103/PhysRevLett.104.190401

[176] Ramsey, N. F. (1950). "A Molecular Beam Resonance Method with Separated Oscillating Fields." *Physical Review* **78**, 695–699. https://doi.org/10.1103/PhysRev.78.695

[177] Reed, M., and Simon, B. (1972). "Unbounded operators." In *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*, 249–317. Academic Press. https://doi.org/10.1016/B978-0-12-585001-8.50014-3

[178] Rivera, A. (2018). "Weighted local Weyl laws for elliptic operators." *arXiv preprint arXiv:1801.07598*. https://doi.org/10.48550/arXiv.1801.07598

[179] Rump, S. M. (2010). "Verification methods: rigorous results using floating-point arithmetic." *Acta Numerica* **19**, 287–449. https://doi.org/10.1017/S096249291000005X

[180] Ruskai, M. B. (1994). "Beyond strong subadditivity? Improved bounds on the contraction of generalized relative entropy." *Reviews in Mathematical Physics* **6**, 1147–1161. https://doi.org/10.1142/S0129055X94000407

[181] Shannon, C. (1956). "The zero error capacity of a noisy channel." *IRE Transactions on Information Theory* **2**, 8–19. https://doi.org/10.1109/TIT.1956.1056798

[182] Sheffield, S. (2007). "Gaussian free fields for mathematicians." *Probability Theory and Related Fields* **139**, 521–541. https://doi.org/10.1007/s00440-006-0050-1

[183] Simon, B. (1978). "A Canonical Decomposition for Quadratic Forms with Applications to Monotone Convergence Theorems." *Journal of Functional Analysis* **28**, 377–385. https://doi.org/10.1016/0022-1236(78)90094-0

[184] Stinespring, W. F. (1955). "Positive functions on C*-algebras." *Proceedings of the American Mathematical Society* **6**, 211–216. https://doi.org/10.1090/S0002-9939-1955-0069403-4

[185] Stinga, P. R., and Torrea, J. L. (2009). "Extension problem and Harnack's inequality for some fractional operators." arXiv:0910.2569. https://arxiv.org/abs/0910.2569

[186] Stoer, J., and Bulirsch, R. (2002). *Introduction to Numerical Analysis*. Springer, New York (Texts in Applied Mathematics). https://doi.org/10.1007/978-0-387-21738-3

[187] Suzuki, K., and Uemura, T. (2014). "On instability of global path properties of symmetric Dirichlet forms under Mosco-convergence." arXiv:1412.0725. https://arxiv.org/abs/1412.0725

[188] Szegedy, M. (2004). "Spectra of Quantized Walks and a sqrt(delta epsilon) rule." arXiv preprint. arXiv:quant-ph/0401053. https://arxiv.org/abs/quant-ph/0401053

[189] Szegő, G. (1939). *Orthogonal Polynomials*. American Mathematical Society (Colloquium Publications 23). https://doi.org/10.1090/coll/023

[190] Takesaki, M. (1970). *Tomita's Theory of Modular Hilbert Algebras and its Applications*. Springer. https://doi.org/10.1007/BFb0065832

[191] Tomiyama, J. (1957). "On the projection of norm one in W*-algebras." *Proceedings of the Japan Academy* **33**, 608–612. https://doi.org/10.3792/pja/1195524885

[192] Umegaki, H. (1954). "Conditional expectation in an operator algebra, I." *Tohoku Mathematical Journal* **6**, 177–181. https://doi.org/10.2748/tmj/1178245177

[193] Visser, M. (1998). "Acoustic black holes: horizons, ergospheres, and Hawking radiation." *Classical and Quantum Gravity* **15**, 1767–1791. https://doi.org/10.1088/0264-9381/15/6/024

[194] Watrous, J. (2018). *The Theory of Quantum Information*. Cambridge University Press. https://doi.org/10.1017/9781316848142

[195] Weedbrook, C., Pirandola, S., García-Patrón, R., Cerf, N. J., Ralph, T. C., Shapiro, J. H., et al. (2012). "Gaussian quantum information." *Reviews of Modern Physics* **84**, 621–669. https://doi.org/10.1103/RevModPhys.84.621

[196] Welch, J., Greenbaum, D., Mostame, S., and Aspuru-Guzik, A. (2014). "Efficient quantum circuits for diagonal unitaries without ancillas." *New Journal of Physics* **16**, 033040. https://doi.org/10.1088/1367-2630/16/3/033040

[197] Whitt, W. (2007). "Proofs of the martingale FCLT." *Probability Surveys* **4**, 268–302. https://doi.org/10.1214/07-PS122

[198] Zhang, F. (2005). *The Schur Complement and Its Applications*. Springer. https://doi.org/10.1007/b105056

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.